Swarm Search and Rescue with Differing Scales of Autonomy

Abstract

[To write. Draft facts for it: 33 operators, two 8-minute sessions each (Strategic Heavy and Tactical Heavy scenarios, counterbalanced), a heterogeneous 33-drone fleet, two optional assistants at two decision tiers. Headline results: reliance is a stable personal style; hand-over rises with mission size on both tiers; momentary load does not push operators onto the assistants; the Tactical Assistant is trusted more; a visible flaw on the strategic tier and an invisible one on the tactical tier.]

1 Introduction

The deployment of large numbers of robots has advanced from speculation from practice in a short time frame. Present applications of drones and similar uncrewed vehicles are increasingly requiring either large numbers of well-organised operators [23], or else provide early demonstrations of semi-autonomous command-and-control software [3]. There is significant appetite for the incorporation of autonomy for reasons including but not limited to the impracticalities of supporting one pilot for each asset, and a desire for a single operator to maintain a complete picture of the entire operation [29]. As swarms scale further, the one-to-one or many-to-one models of human control must give way to a one-to-many paradigm, and some degree of autonomy becomes not merely desirable but unavoidable.

How that autonomy is best integrated, however, remains an open question. Autonomy is far from new; defence operations have long been organised around autonomy at different levels of a hierarchy. Auftragstaktik (mission command) requires strategic command to communicate high-level goals to subordinates and grants them latitude over low-level execution, and indeed further delegation. A similar logic can be applied to robot swarms: tactical decisions may be delegated to individual assets or sub-groups, while strategic decisions remain with a human commander. Yet it is not obvious that the best configuration places the tactical under full autonomy while leaving the strategic entirely manual. Lower-level interventions may benefit heavily from human oversight; robots still struggle with physical execution, while autonomous planners are near-perfect at many allocation problems. Of course, higher-level decisions carry greater consequence and arguably warrant closer human control; it is natural to prefer an autonomy-at-tactical approach, however the best possible combination of autonomy levels may not be so simple. Each of these judgements also depends on operator workload and situational context. The need for interfaces and control paradigms which allow autonomy is therefore high [29], but the question of *where* in a command hierarchy that autonomy should sit, and whether this should be fixed or operator-configurable, has received little attention. Previous works have explored many forms of autonomy [18], but these have typically been treated at the level of a single decision layer.

This work represents a first step toward a longer-term vision: We aim to understand how autonomy effects both user and mission when deployed at each level of a multi-layered command hierarchy, and move towards a system where each level of concern from strategic down to tactical details, can be automated to varying degrees. Finally, we propose a "co-pilot" that, instead of attempting to fully automate the mission, identifies where the user is most needed, and suggest to them an autonomy profile which adapts on-the-fly.

We develop a simulated search-and-rescue scenario using a heterogeneous drone swarm, in which a two-tier assistive system (a strategic tier and a tactical tier) is available to the operator throughout.

[Step 4 note (kept from the old draft): this is the natural place to make the *ORCHID lineage* explicit — name `jennings2014human` (human-agent collectives / flexible autonomy) and `ramchurn2021trustworthy` alongside the already-cited `ramchurn2016disaster`. See cluster C1 in the Related Work notes box. Not drafted.] Neither assistant has errors injected: the manipulation is not the assistants' reliability but *where the operational demand sits*. In a within-subjects design each participant plays two 8-minute sessions — **Strategic Heavy** (frequent, small

missions, placing demand on cross-mission allocation) and **Tactical Heavy** (fewer, larger missions, placing demand on within-mission planning) — with the fleet and the assistants identical in both. Using either assistant is always optional, so the study observes when operators choose to hand a decision over, at which tier, and how that relates to the mission, the moment, the person and their trust.

[RQ block — REVISED to match what the analysis found (the old block assumed a perfect-agent "value of intervention" RQ4 that the data cannot answer). Candidate framings, each mapped to a results section; merge or cut as you like:

- **RQ1 (who).** How much of an operator's use of the assistants is a stable personal style, and is it predicted by attitudes measured beforehand? → Section 7.2.
- **RQ2 (when).** What makes an operator hand a decision over — the mission (its size), the moment (load, the queue), or practice — and does the answer differ between the strategic and tactical tiers? → Sections 7.3–7.7.
- **RQ3 (flaws).** How do operators respond to an assistant's flaw that is visible (the illogical strategy card) versus one that is not (the tactical assistant's wrong refusal)? → Section 7.8.
- **RQ4 (trust by tier).** Is one tier's assistant trusted more, where does the difference come from, and does stated trust match use? → Section 7.9.

Note: "failure" in this study means hardware drone failure during execution, plus the two side effects of the assistants' rules (Section 4.3) — not injected agent error. Operator-built deadlocks were auto-repaired and never surfaced.]

2 Related Work

Essential to any autonomous system is the notion of trust. Trust in this context is defined by Lee and See as "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability" [11]. Such qualities are abundant in swarm contexts, where the positions, motions, and goals of the robots are rife with uncertainty, and there is a significant risk of failure stemming from the attritable nature of many such assets and the high levels of risk in their environment [27]. It is an important consideration, as a system which is not trusted will likely not be used by the operator, and a system which is trusted too much will be relied on without appropriate checking and oversight. A system which balances under- and over-trust is said to be well calibrated [13].

It has been observed that once a user falls into the trap of under- or over-trust, they have a tendency to remain there unless actively coaxed back into a well-calibrated relationship [20]. The standard tool for this is explanations, however these are not always given due attention [17, 31], and often-times users are biased in favour of action over inaction so accept suggestions readily [15]. In the domain of HSI the notion of neglect benevolence describes the observed advantage from leaving the swarm to resolve its own behaviour instead of intervening overly often [16], additionally supporting the notion of autonomy which manages itself under human supervision instead of constant interaction with the recommender. Equally important to explanations, then, may be a design pattern where users work in tandem with autonomous tools and can elect to deploy them selectively, with varying levels of autonomy on varying scales of a swarm or sub-swarm.

Traditionally swarm robotic systems are characterised by autonomy, however with the importance of human oversight a question arises as to what level of autonomy should be given, and how the human's actions interface with some level of autonomy. Traditional scales such as Sheridan and Verplank's LOAs are a common metric [26], as well as more swarm-focussed scales of a similar form [7]. An important note in more developed robotic systems is that the degree of autonomy can vary at different parts of the robot's role (for example information acquisition may be more autonomous than decision-making) [21].

An important note is that the terms "automation" and "autonomy" are often used interchangeably but are in fact different things, with the former describing well-defined processes of tasks deferred to an *automated system*, and the latter describing *autonomous systems* which can make "decisions independent of human control" [12]. As we are referring to swarm robots in this work, the terms are somewhat more interchangeable, as to automate a swarm on a particular task requires the granting of some degree of autonomy (for example performing decentralised allocation, where an automated algorithm runs across robots who then autonomously resolve their tasks [4]). We tend to prefer

the term autonomy as it is more relevant to classical swarm paradigms, but remind the reader that there can be some distinction dependent on use case.

Similar to calibrating trust, it is important that any assistive autonomous system is sufficiently performant to merit use. Human performance in the presence of such an agent is highly sensitive to the reliability and performance of that agent [25]. Some degree of imperfection is an expectation (otherwise there would be no need for a human), and humans are typically willing to tolerate occasional failure if it represents an adequate time- and effort-save. In fact, the most trusted systems are often those which sit in between 'too poor to trust' and 'too reliable to bother checking' [10].

Another important factor is the level of workload. Prior works have found that under high task load, more reliable automation increases operator performance, however under low task load, more reliable automation actually *decreases* operator performance. This is likely due to the over-trust issue discussed above, and presents a difficult challenge to designers, as users may defer to the agent even when not overloaded [22].

The level of chosen autonomy will necessarily represent a trade-off between different desirable qualities as alluded to above, and no single solution is likely to best suit every scenario [14]. This motivates a flexible approach where the level of autonomy is not fixed, rather it can be chosen by the operator. Such flexible approaches have been explored to the extent where agents can be granted autonomy where desirable [5], or else autonomous agents can be overridden at any level while they attempt to perform autonomously [30]. So-called mixed-initiative approaches to swarm control allow a human to switch the entire swarm between manual and autonomous modes [18]. To our knowledge, however, no prior work has examined the behaviour of an operator when autonomous assistance is structured into a two-tier hierarchy and the operator is free to use it selectively. Such a design allows us to ask not only whether the assistive system improves performance, but *when* operators choose to engage it and what advantage that engagement confers.

We term the two structural dimensions of selective autonomy the **Depth** and **Breadth** of autonomy. **Depth** refers to how autonomous some agent is, pertaining directly to Sheridan and Verplank's LOAs [26]. **Breadth** refers to the scale of autonomy, whether that be an individual asset, a mission's worth of assets, or the entire reserve. The present work extends this framework by stacking two agents vertically in a hierarchy—one operating at the within-mission (tactical) level and one at the cross-mission (strategic) level—and asking how the operator engages with, and elects to intervene in, autonomy at each level.

Metrics exist to assess the difficulty of tasks in the context of swarms, such as efficiency and effectiveness, and it has been suggested that autonomy should vary depending on these [8]. As well as the complexity of the task, a more direct assessment of cognitive complexity via operator state metrics such as EEG has been employed [1], and such signals could in principle inform a meta-level recommendation system.

[Drafted gap paragraph — my pass, for your check; rewrite freely. Intended as the closing paragraph of Related Work. Requires the supplement bib for the four keys.]

Most recently, the one-operator, many-robot paradigm has moved from concept to field demonstration. Adams et al. showed that a single trained commander can supervise a swarm of one hundred heterogeneous ground and aerial robots across real urban field exercises, albeit with frequent workload peaks [2], while the MOSAIC framework demonstrated a single operator coordinating a heterogeneous five-robot team in a lunar-analogue mission by shifting individual platforms between driver-, task-, and mission-level autonomy [19]. In parallel, variable autonomy has consolidated into a recognised subfield in its own right, with dedicated venues [28] and emerging design guidelines and research agendas [24]. Across this line of work, however, autonomy is treated as a single dial — a *level* to which a robot or team is set — and the evidence base consists of demonstrations and design agendas rather than controlled operator studies. What remains unexamined is the setting in which assistance is available simultaneously at *distinct decision tiers* — cross-mission resource allocation and within-mission planning — and the operator chooses, moment to moment, how far to engage each. The present study addresses exactly this gap: a controlled, within-subjects examination of how the locus of operational demand shifts operator engagement, workload, and trust across a two-tier assistive hierarchy.

Step 4 — Reference integration notes (proposals, not drafted text). New keys live in `refs_supplement_new.bib` (add it via `\bibliography{refs,refs_supplement_new}`). Each cluster gives *where* to place it and *what* it buys; the wording is yours. Key nuance for the whole trust literature below: our agent is now **perfect** — cite the trust-in-*imperfect*-autonomy work as calibration/per-tier-trust *backdrop and motivation*, not as error-detection, since we inject no agent error.

- **C1 — ORCHID lineage** (Intro near `ramchurn2016disaster`, and/or opening Related Work). Keys: `jennings2014human` (HAC vision; "flexible autonomy"/"agile teaming"), `ramchurn2021trustworthy` (successor framing — its UAV disaster-response pipeline mirrors our two-tier setup and its "errors enter at every pipeline stage" point motivates the per-tier question), `soorati2022intelligent` (group agenda). *Angle:* frame the study as building on the ORCHID human-agent-collectives programme; `ramchurn2016disaster` (already cited) is its flagship.
- **C2 — task-allocation substrate** (short new paragraph in Related Work, by the automation/autonomy discussion). Keys: `ramchurn2010cfstp` (CFSTP: heterogeneous agents, deadline/workload tasks — the formal root of our strategic/tactical allocation), `capezzuto2020anytime` & `capezzuto2021largescale/capezzuto2021ar` (modern scalable extensions), `macarthur2011distributed` (already in bib, uncited), `ramchurn2010robocup`, `wu2015agile`, `baker2016planning` (UAV SAR planning). *Angle:* one or two sentences establishing the agent's allocation task as a well-studied MAS problem; cite the lineage, don't dwell.
- **C3 — single-operator / heterogeneous-swarm HSI** (near the LOAs / one-to-many discussion). Keys: `soorati2021designi` (in-group HSI interface; HARIS precursor), `abioye2024adaptive` (your own group; workload-adaptive HSI → RQ2/RQ3), `adams2025supervise` (single operator, 100 heterogeneous robots; now also in the gap paragraph — best used as an Intro feasibility anchor, an RQ2 workload-at-scale *contrast* (their physiological multi-dimensional estimation vs. our NASA-TLX), and qualitative motivation for the tactical tier: operators were "red-lined" precisely when tactical allocation autonomy was unavailable. Not a design comparator — no tiers, no trust, n=2 field observation), `chandarana2018determining` (multi-job-type swarm sizing). *Angle:* situate the one-operator + heterogeneous-asset setup and the workload stakes.
- **C4 — variable/adjustable autonomy** (attach to the Depth/Breadth paragraph and the mixed-initiative cites). Keys: `methnani2024charge` (survey — anchor), `theodorou2024editorial`, `ramesh2023experimental`, `verhagen2024meaningful` (firefighting; SAR-flavoured), `reinmund2024variable`, `oberacker2026mosaic` (MOSAIC; now also in the gap paragraph — the depth axis operating in the field: one operator slides a heterogeneous five-robot team across driver/task/mission autonomy levels, but allocation is folded into mission-level autonomy with no separately-operable strategic tier; arXiv preprint, verify status at submission); `tecuci_seven_2007` (already in bib, mixed-initiative). *Angle:* the RQ3 grounding ("where should autonomy sit across levels"); links Depth/Breadth to the variable-autonomy subfield.
- **C5 — trust across tiers** (extend the trust-calibration paragraph; motivates RQ3 per-tier trust and RQ4). Keys: `wischnewski2023measuring` (subsystem-failure → whole-system trust spillover — the key precedent for measuring trust *per tier*), `wildman2024trust` (multilevel trust in 3+-member HATs), `rittenberg2024trust` (between-subjects reliability manipulation), `naiseh2023explanation`, `centeiojorge2023exploring`; `parasuraman2010complacency` (already in bib). *Angle:* frame as system-wide vs. component-specific trust; justify measuring trust separately for each tier. NB the closest analogue — a multilevel "agent reliability pairing" study (Am. J. Psychology) — I could not fully verify; see the supplement TODO before citing.
- **C6 — selective use / recommendation acceptance** (the "elect to deploy them selectively" sentence). Key: `fink_let_2024` (already in bib) — giving users control over whether to accept a recommendation raises acceptance; direct support for the selective-deployment design and RQ2.
- **Optional (thesis-level, not this paper):** the AtomicOrchid platform papers and Fischer's "in-the-loop vs on-the-loop" work sit better with your HARIS/platform chapter (RQ1) than here; omitted from the supplement to keep scope tight.

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3 Scenario and Task Design

3.1 Operational Scenario

We built a simulated search-and-rescue operation in which a single operator directs a heterogeneous fleet of 33 drones from one hub. Missions keep arriving at positions on a map; each consists of two to six tasks, and each task needs a

fixed team of drone types. For every mission the operator makes two decisions in sequence:

- a **strategic** decision — *how many* drones of each type to commit to the mission, taken from the shared reserve that every other mission also draws on; and
- a **tactical** decision — *which* committed drone does *which* task, and in what order.

An assistant is available for each decision (Section 4); using either is always optional. Drones can fail mid-task, in which case the operator repairs the plan or abandons the mission. The score rewards completed tasks and penalises the time missions spend unfinished, so both getting it right and getting it done quickly matter.

3.2 Asset Types

The fleet comprises 11 drones of each of three types (Table 4). The types are named to the operator by role rather than colour — **Fast** (blue; 11 u/s), **Lifter** (red; 10 u/s) and **Camera** (green; 9 u/s) — and drones are labelled accordingly (e.g. `Fast-7`). Speeds are not shown to participants; the tutorial describes the types only as fastest, standard and slowest. Drones committed to a mission leave the reserve at once and return to it when their task chain is complete. The mission generator's archetype weights (below) were tuned in simulation so that aggregate demand, in drone-seconds, is approximately equal across the three types in both scenarios, so no single type is the systematic bottleneck.

3.3 Task Types

There are five task types of increasing complexity and value (Table 1). Each has a primary composition; three have a substitute composition with one fewer drone of the paired type, which takes about 2.5 times as long. The planner shows the substitute as an "OR" line on the task badge.

Table 1: Task types. Execution times are for the primary composition / the substitute.

Type	Name	Primary	Substitute	Time (s)	Points
T1	Recce	1 Fast	—	10 / —	20
T2	Recon	2 Fast	1 Fast	15 / 38	40
T3	Supply Drop	2 Lifter + 1 Camera	1 Lifter + 1 Camera	25 / 62	60
T4	Precision Supply Drop	1 Lifter + 2 Camera	1 Lifter + 1 Camera	30 / 75	80
T5	Search & Service	1 Fast + 1 Lifter + 1 Camera	—	45 / —	100

3.4 Missions

Missions arrive by a Poisson process. Each has a **category** that sets its number of tasks and how quickly its penalty grows — A Routine (2 tasks), B Moderate (3), C Significant (4), D Critical (5), E Mass Casualty (6) — and an **archetype** drawn at generation (Fast-, Lifter-, Camera-leaning or Mixed, weights 38/26/21/15) that sets which task types make up that count (Table 2). A mission therefore needs one drone type (e.g. an all-Recce Fast mission), or all three; only 12 of 606 strategic decisions concerned a two-type mission. Mission zones are circles of radius 80 units at least 150 units from the hub and 200 units from any other active zone. Where the findings speak of **mission size** they mean the number of drones the mission needs: the sum of its tasks' primary requirements, which the task badges display and which is the same for every operator who met that mission.

3.5 Score

The score S is completion reward minus an accrued time penalty, floored at zero:

$$S = \max\left(0,\ \sum_{m\in\mathcal{M}}\sum_{k\in\mathcal{T}_m} w_k\,\mathbf{1}[k \text{ completed}]\ -\ \sum_{m\in\mathcal{M}}\sum_{k\in\mathcal{T}_m} r_m\,\frac{w_k}{\sum_{k'\in\mathcal{T}_m} w_{k'}}\left(t_k^{\text{end}} - t_m^{a}\right)\right) \tag{1}$$

where w_k is the points value of task k (Table 1), r_m the penalty rate of the mission's category (0.05–0.40 points/s; Table 4), t_m^a the mission's arrival time and t_k^{end} the task's completion time, or the current time while it is unfinished. The penalty is thus charged per task, weighted by the task's share of its mission's points, from the mission's arrival

Table 2: Task list for each mission (category × archetype), as generated. Task counts grow with category; the archetype fixes the mix.

Cat.	#	Fast-lean	Lifter-lean	Camera-lean	Mixed
A	2	T1, T2	T3, T5	T4, T5	T1, T3
B	3	T1, T2, T2	T3, T3, T5	T1, T4, T4	T1, T3, T5
C	4	T1, T1, T2, T2	T3, T3, T5, T5	T1, T4, T4, T4	T1, T2, T3, T5
D	5	T1, T1, T2, T2, T5	T3, T3, T3, T5, T5	T1, T4, T4, T4, T5	T1, T2, T3, T3, T5
E	6	T1, T1, T2, T2, T2, T5	T3, T3, T3, T5, T5, T5	T4, T4, T4, T1, T5, T5	T1, T2, T3, T3, T4, T5

until that task completes; it is displayed in 15 s steps. If a mission is abandoned, its parent's penalty clock stops and its unfinished tasks are re-queued as a new *residual* mission (id suffix `-R`) that accrues from that moment, so abandoning transfers work rather than losing it. Because the four mission streams (Section 6.1) differ in how many points they offer, raw scores are compared only within a game.

3.6 Drone Failures

Any deployed drone can fail. In the main build (`study-v1.4` onwards; Appendix B) failures are a live hazard of $1/900$ per drone per second, rolled on a fixed one-second simulated grid, identical for every drone type and mission. A drone that fails after its part of a task is done (e.g. the Lifters of a Supply Drop after the drop) exits gracefully. Otherwise the task reverts to pending and the mission enters **recovery**: its card turns red and the tactical planner opens in recovery mode (Figure 5). The operator may reassign the drones still on that mission (never the hub reserve, which is what makes abandoning a real option), press Suggest for a repair, or abandon the mission. A mission with an open recovery takes no further failures until 30 s after it is resolved (`v1.5` onwards). On average a session had about four drone failures (Table 3).

4 The Two Assistants

Both assistants are deliberately simple, transparent rules rather than learned models, and in every session analysed neither had errors injected (Section 4.3). In the interface they are called the *Strategic Assistant* and the *Tactical Assistant*; the words "AI" and "agent" are not used on screen.

4.1 Strategic Assistant: Strategy Cards

When the operator clicks *Allocate* on a queued mission, a panel opens beneath its card and two strategy cards load, each after an independent delay of about 4–5 s (Figure 3b). Each card shows the drones it would commit per type, the projected mission completion time (ETA), the reserve that would remain, and three bars: Speed (the best ETA of the two divided by this card's ETA), Reserve (share of the current reserve left behind) and Resilience (spare drones above the minimum needed).

- **Aggressive** — enough drones of each type for every task to run at once (the parallel sum of the primary compositions), with at least one spare of each type the mission uses; capped by the reserve.
- **Conservative** — the smallest team that completes every task in series (primary compositions, or substitutes where the reserve cannot cover a primary), plus one spare of each type the mission uses, plus 15% of the remaining reserve of *every* type. Tasks run mostly in series.

The 15% top-up is applied to every drone type, including types the mission does not use. On a one-type mission the Conservative card therefore often includes, say, a Camera drone on an all-Fast mission. We call this the **illogical card**: it was on screen for 132 of the 606 strategic decisions, almost all on one-type missions, and every participant met it. It was discovered during data collection and deliberately left unchanged so as not to split the cohort; Section 7.8 studies it.

The operator can deploy a card as offered (*Deploy Selected*), choose *Edit this allocation* (a counter per type, pre-filled

from the selected card), or choose *Set manually instead* (the same counters, empty), which is possible before the cards have loaded (Figure 3c). On deployment the committed drones leave the reserve at once and fly to the mission zone, where they wait at its edge for the tactical plan.

4.2 Tactical Assistant: Suggest

Once a mission has its drones, it appears as pending in the tactical planner on the second screen (Figure 4). The planner opens **empty**: no plan is shown until the operator asks for one. The operator can build the plan by hand — drag a drone onto a task; shift-drag chains a drone on to a further task after its first, and the sequence numbers update live — or press **Suggest**. Suggest builds a plan with a virtual-timeline scheduler: tasks are taken most-constrained first (T5 before T1) and each is given the earliest-free drones of the required types from the committed team, chaining drones across tasks where that finishes sooner. The plan appears one drone→task entry per second, and *Deploy* unlocks when it is complete. The operator can edit any part of it before deploying. *Deploy* is enabled only when every task has at least its substitute composition. The planner works in "plan-all" mode: the whole sequence is confirmed once.

Because Suggest routes every drone in one global task order, its plans cannot deadlock. An operator-built plan can (a cycle of chained drones each waiting on the other's task); in every session analysed, such deadlocks were detected and silently re-routed onto one order by the system, so a participant could never encounter one.

In recovery mode Suggest proposes a repair from the drones on the mission. It considers only drones that are not currently executing a task. When the only fix needs a busy drone — chaining it on after its current task, which the operator could do by hand — Suggest answers "No workable plan". We call this a **wrong refusal**. To the operator it is indistinguishable from a correct refusal. It was found after data collection and also left unchanged.

4.3 Assistant Reliability

The platform can inject errors into either assistant (a per-session error rate per tier). This was switched off ($\varepsilon_S = \varepsilon_T = 0$) for every session analysed; the one participant who ran with errors switched on (P09) is excluded. Neither assistant was therefore ever deliberately wrong. The assistants' only flaws are the two side effects of their rules described above: the illogical card on the strategic tier, which is visible on the card face, and the wrong refusal on the tactical tier, which is not. [The old draft's RQ4 ("value of intervention into a near-optimal agent") and the per-tier error manipulation do not apply to these data; see the RQ note in the Introduction.]

5 Platform and Interface

The platform is a browser application (React/TypeScript; no server) designed for two monitors side by side (Figure 1). All randomness is seeded, so a seed and a configuration reproduce a session exactly, and every operator action and assistant output is logged with a millisecond timestamp.

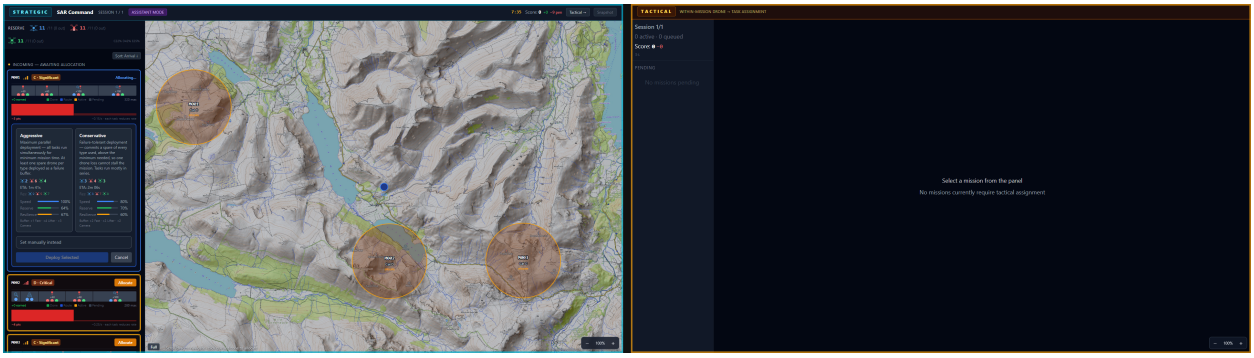


Figure 1: The two screens side by side. **Left, strategic screen:** reserve (top left), the queue of missions awaiting allocation (left column; the first is being allocated and shows its two strategy cards), and the strategic map with each mission's zone and the hub. **Right, tactical screen:** the tactical planner, here with no mission pending.

Strategic screen. The strategic screen (Figure 2) holds the reserve count per drone type, the mission queue and the strategic map. Each mission card (Figure 3a) shows the mission's id and category, a badge per task with its required drones and points, a progress legend (done, en route, active, pending) and a red penalty bar that grows while the mission is unfinished. The queue can be sorted by arrival or by criticality. About three and a half cards are visible at once; further waiting missions appear only on the map. The map can show zones only or every drone's path.

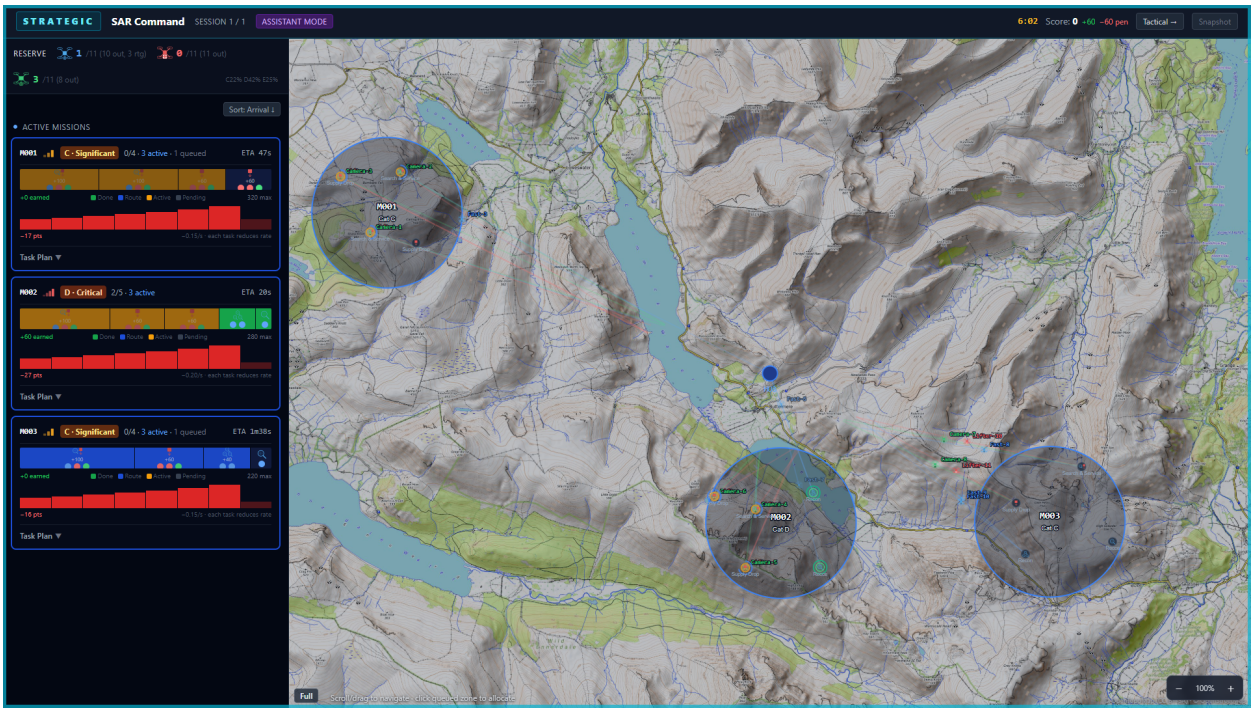


Figure 2: Strategic screen mid-session: missions in progress, drones in flight between the hub and the mission zones, and the penalty bars on the queued and active mission cards.

Tactical screen. The tactical screen lists missions awaiting a plan; selecting one opens the planner (Figure 4): the mission zone with each task as a waypoint labelled with its type and staffing (e.g. `1/2L 0/1C`, with its substitute below), the committed drones waiting at the zone edge, and a right-hand panel listing unassigned drones and the task schedule with each task's assigned drones and projected start and finish. Hovering a drone shows its full route.

Failures. When a drone fails, the mission's card turns red with a banner naming the lost drone, and the mission is listed as *reassign pending* on the tactical screen. The recovery planner (Figure 5) marks executing and travelling tasks, shows the failed drone crossed out, and offers *Suggest*, *Reassign* (enabled once every pending task is covered) and *Abandon Mission*.

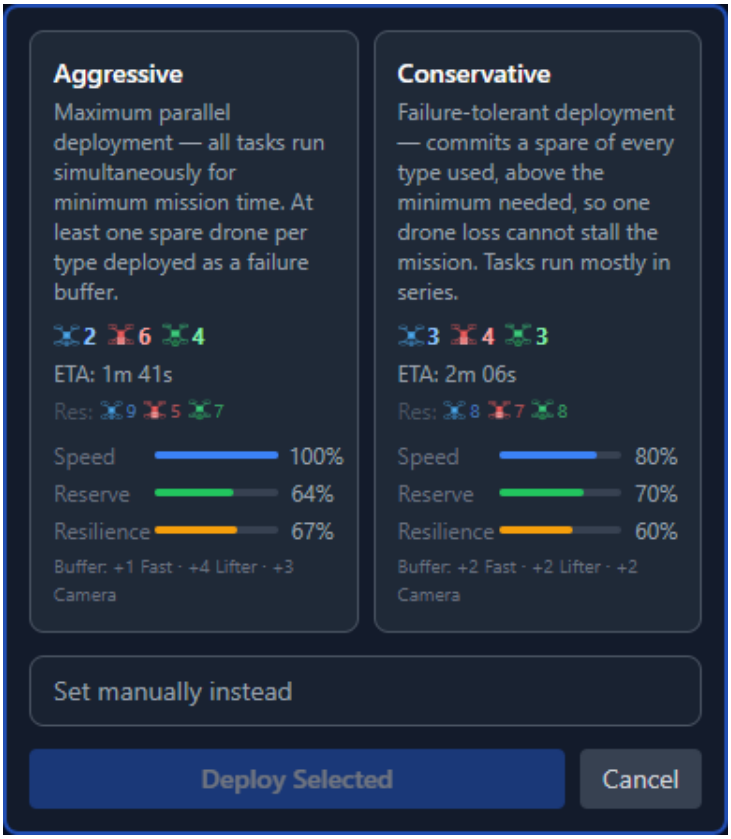
Tutorial. Before session 1 every participant completed a guided tutorial of 48 steps on a smaller fleet (6/6/6), with its own clock. It teaches the manual workflow first — allocate, plan, chain, deploy, recover from a failure by hand, and abandon a mission that cannot be recovered (two scripted failures) — and only then introduces the two assistants; the Suggest button is hidden until that lesson. The tutorial runs with error-free assistants.

6 Study Design

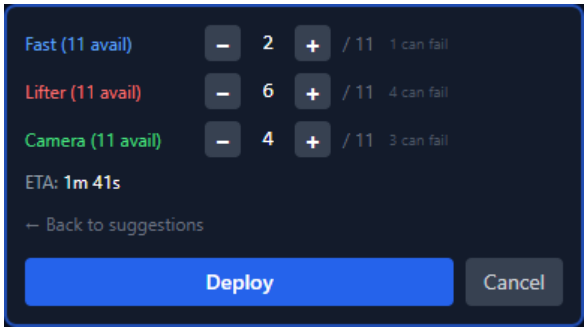
6.1 Design

The study used a within-subjects design with one factor, the **scenario**, i.e. where the operational demand sits. Each participant played one 8-minute (480 s) session of each scenario, in counterbalanced order (17 Strategic Heavy first,

(a) Mission queue: three missions awaiting allocation. Each badge shows a task's required drones and its points; the red bar is the accruing penalty.

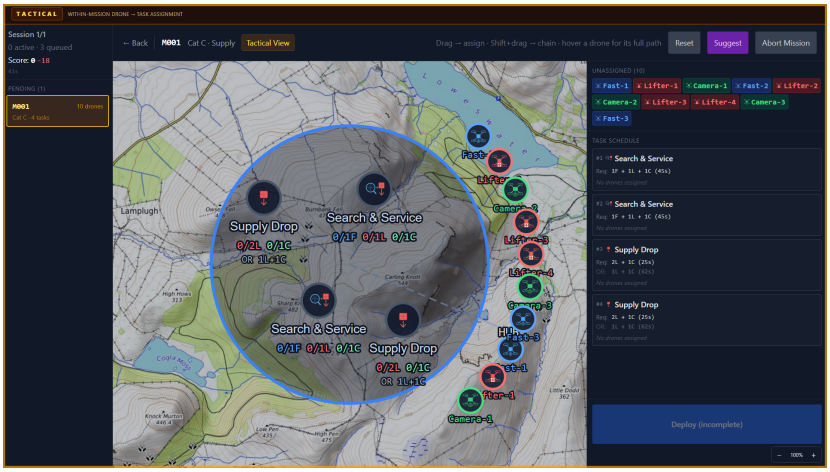


(b) The Strategic Assistant's two strategy cards for a Category C mission: drones per type, ETA, reserve left ("Res"), and the Speed, Reserve and Resilience bars.

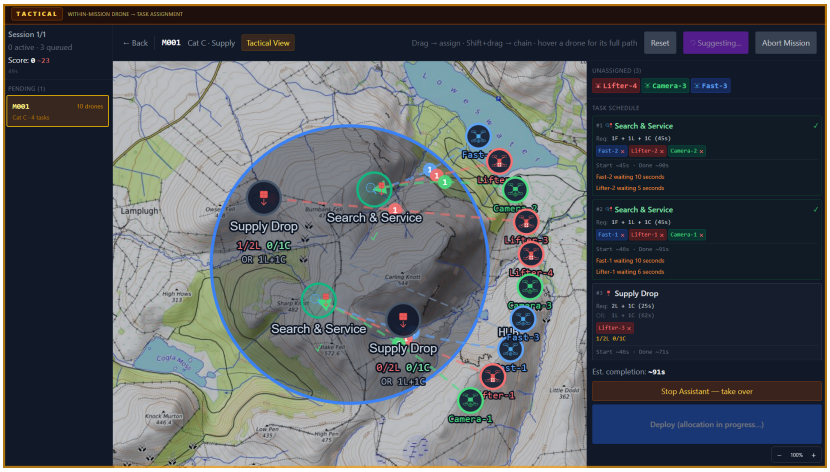


(c) Manual allocation (*Edit this allocation* or *Set manually instead*): a counter per type, how many of each could fail without stalling the mission, and the live ETA.

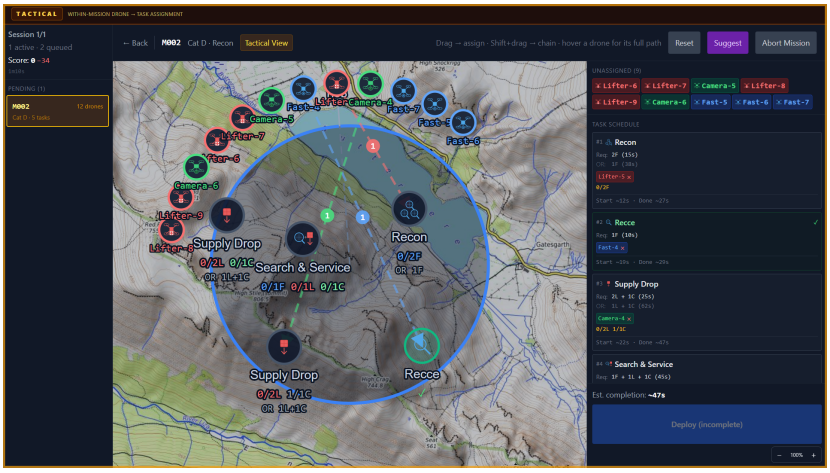
Figure 3: The strategic decision.



(a) The planner as it opens: every committed drone unassigned.



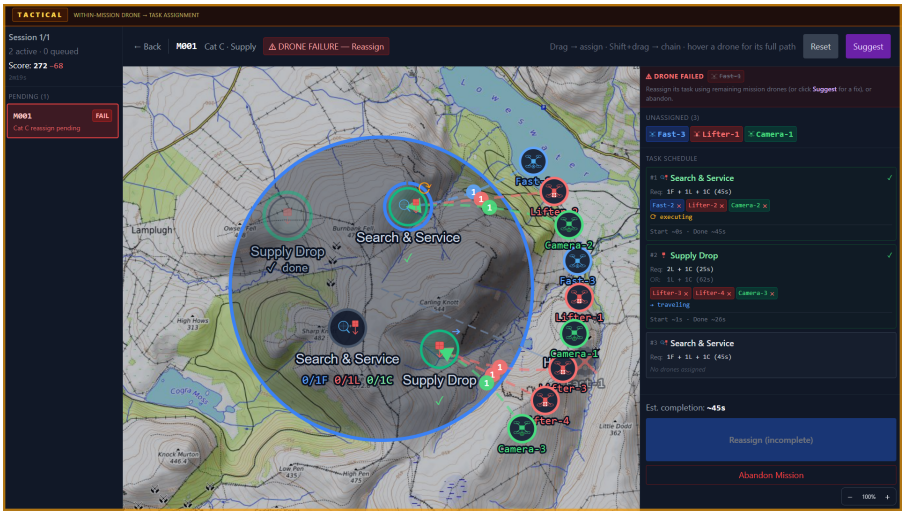
(b) Suggest filling in the plan, one drone→task entry per second; dashed lines and numbers show each drone's assigned task and order. *Deploy* stays locked until the fill completes.



(c) A plan being built by hand by dragging drones onto tasks.

Figure 4: The tactical decision in the tactical planner.

(a) Failure flagged in the queue.



(b) Recovery planner after Fast-1 failed: one Search & Service task is unstaffed; the remaining mission drones are listed as unassigned.

Figure 5: Drone failure and recovery.

16 Tactical Heavy first). The fleet, the assistants and every other parameter were the same in both.

- **Strategic Heavy** — missions arrive often (mean inter-arrival 37 s) and are mostly small (category weights A–E 40/38/16/5/1): about 13.5 missions per session, loading the cross-mission allocation decision.
- **Tactical Heavy** — missions arrive less often (75 s) and are mostly large (5/13/28/38/16): about 7.5 missions per session, loading the within-mission planning decision.

The two scenarios were tuned in simulation to comparable difficulty: a scripted near-optimal operator completes about 83% of tasks in both at the final failure rate.

Four games, not two. Every session used the same seed (42). A session's mission stream is seeded from the seed and a plan index, so there are exactly four fixed mission streams ("games"): Strategic Heavy 1 and 2 and Tactical Heavy 1 and 2. Up to P30 the plan index was the session number, so a participant's session-1 game was always the index-1 game of its scenario; from P31 (`study-v1.13`) the order was swapped, so later participants played the same games in the other position. The games differ in workload (tasks completed per session: SH1 25.1, SH2 28.9, TH1 21.5, TH2 32.8; Table 6), so every between-person comparison in the results holds the game fixed (z-scores within game, game means removed, or game as a fixed effect).

Figure 6: Four games, not two scenarios. Workload per session in each of the four fixed mission streams (Strategic Heavy 1 and 2, Tactical Heavy 1 and 2).

Build versions. The platform was updated during data collection (builds `study-v1.0` to `v1.13`); 24 of 33 participants played `v1.12` or `v1.13`. The changes that bear on the data are listed in Appendix B; the main one is that P01 and P02 played with half the later task points, which we rescale wherever a raw score is used.

6.2 Participants

Thirty-four people took part; 33 are analysed. P09 was run with assistant errors switched on (a pilot of the error manipulation) and did not follow the task, and is excluded throughout. The 33 were aged 21–44 (mean 28.7, SD 6.0); 25 men and 8 women. 28 held a postgraduate degree (23 master's, 5 doctorate) and 5 an undergraduate degree; 32 of 33 were from computing, AI, robotics or electronic engineering, and one from the armed forces. Self-reported experience was mostly moderate with simulations (20 moderate or extensive) and autonomous systems (24), and low with drones (21 none) and command or emergency management (24 none). [Recruitment route, ethics approval number, payment, and dates (sessions ran 2026-09).]

6.3 Procedure

1. **Consent and pre-study questionnaire.** Background and experience; a short comprehension check; and, from `study-v1.3`, three attitude scales: the AI Attitude Scale (AIAS-4, 10-point), a bespoke verification-propensity scale and a bespoke delegation-boundary scale (both 7-point).
2. **Tutorial** (Section 5).
3. **Session 1** (8 minutes; one scenario).
4. **Post-session questionnaires**: NASA-TLX and trust in each assistant (Section 6.4).
5. **A 30-second break screen.**
6. **Session 2** (8 minutes; the other scenario), then the same questionnaires.
7. **Debrief interview**, audio-recorded, with the researcher's written notes.

Participants were asked to think aloud while playing, and the researcher occasionally asked a short question during play; quotes that answer such a question are marked as following a leading question. Both screens, the cursor and the participant's voice were recorded throughout. [Total duration per participant; setting (lab, monitors); the debrief question list.]

6.4 Measures

Event log. Every session's first event dumps the full parameter set, and every operator action and assistant output is logged as it happens (606 strategic decisions and 630 tactical plans in all). The main behavioural measures are defined as follows.

- **Took a card**: a strategic decision deployed from a strategy card as offered. A card that was edited before deploying counts as a hand (manual) allocation.
- **Pre-emptive**: a manual allocation committed before the strategy cards had loaded.
- **Pressed Suggest**: Suggest was used at least once on that tactical plan (the planner starts empty, so this is always a deliberate request). **Kept unmodified**: a Suggest plan deployed with no change.
- **Handed over**: either of the two.
- **Decision time**: from opening the strategic panel (or the tactical planner) to the choice, minus the forced wait (the card reveal; one second per Suggest entry), unless stated otherwise.
- **Queue depth**: other missions that had arrived and had not yet received a strategic decision when the panel opened, rebuilt from the log.
- **Mission size**: drones needed (Section 3.4); also tasks, drone types, drones committed.

Questionnaires. After each session: **NASA-TLX** [6], six sliders from 0 to 20, reported as the unweighted mean (performance runs from 0 = perfect to 20 = failure; from `v1.12` the slider carried a note that it runs the other way, after two participants were confused by it). **Trust** in each assistant separately: six statements rated 1–7 ("The [Strategic/Tactical] Assistant is reliable", "I trust the … Assistant's recommendations", "… performs well", "I feel confident using …", "… provides useful guidance", "I would follow … suggestions without hesitation"), averaged.

[These six items are bespoke (adapted in spirit from [9], but not that scale) — describe and justify, or cite a source.] Technology-acceptance scales exist in the platform but are only given after a third session, so none were collected.

Qualitative material. Think-aloud and debrief audio was transcribed automatically (faster-whisper, word timestamps) and aligned to the event log by the on-screen session clock. Debriefs and the written interview notes were coded against a codebook of 28 codes written before coding (C01–C28; e.g. C03 "small myself, big to the assistant", C11 "hands over when overloaded", C12 "noticed over-allocation or unneeded types"). Each code is present or absent per participant, with every supporting excerpt checked to occur verbatim in the source. Quotes in the results were selected from these sources and checked verbatim. [Coding was done by a single coder, an AI model (Claude), with a second AI pass ($\kappa = .96$); the only independent check (stated preference, round 1) gave $\kappa = .27$. A human second coder is needed before the "X of 33 said" counts are reported as more than descriptive.]

6.5 Analysis

The analysis is exploratory, with a pre-specified core: ten hypotheses (H1–H10) were written and signed off before any test was run, each with one primary test, Holm-corrected together; everything else is reported as exploratory. The findings were then refined over five review passes in which every result was checked and re-specified. [Describe the analysis process honestly here: the analyses were scripted with the help of AI coding agents and every result re-run and reviewed by the researcher (register: `docs/analysis/register`). Decide how much of this belongs in the paper.] Common elements:

- **Decisions within people.** Decision-level outcomes are modelled with generalised estimating equations (GEE; logistic for yes/no, Gaussian for times on the log scale) clustered on participant, with small-sample bias-reduced standard errors. Predictors of interest are split into a *within-person* part (the decision's value minus the person's own mean) and a *between-person* part (the person's mean), so "within person" estimates compare each operator only with themselves.
- **Game held fixed**: game (one of the four) and session as fixed effects, or z-scores within game.
- **Expected and adjusted rates.** Where a raw rate could reflect who reaches a situation rather than the situation itself, a logistic model predicts each decision's chance of handing over from the mission (drones, tasks, types), the minute of the session, the game and the person (their rate on all their *other* decisions, so a decision never predicts itself). *Adjusted* = the overall rate plus the mean of (observed − expected) in a bin.
- **Intervals**: 95% intervals by resampling people (whole participants, with replacement; 400–2,000 draws), so one heavy user cannot make an interval look tight.
- **Nulls** are reported with the effect size the data could have detected.

7 Results

[The results below are generated from the analysis register by `scripts/paper/build_paper.py` (`results.tex`). Grey text is the register's own wording, accurate but written for the review page, to be rewritten. Each subsection is one finding (its summary, then each analysis as figure, key numbers, method, and what participants said, with the quotes selected on the review page). Sidelined analyses are in Appendix C. Findings are ordered: who the operator is; when operators hand over (mission size, manual builds, practice); workload and the queue; the assistants' flaws; trust; what operators wanted.]

7.1 Overview

Table 3 summarises each scenario, and Table 6 each game. Strategic Heavy sessions had nearly twice as many missions and decisions as Tactical Heavy ones, but about the same number of tasks completed; Tactical Heavy offered more points and accrued more penalty. Operators took strategy cards on 60% of decisions in Strategic Heavy and 73% in Tactical Heavy, and pressed Suggest on 72% and 79% of plans. Workload (NASA-TLX) did not differ between the scenarios.

Table 3: Descriptive overview by scenario: mean (SD) over the 33 sessions of each scenario. Every participant played one session of each, so p is a paired Wilcoxon signed-rank test (33 pairs; fewer where a measure is undefined, e.g. no manual decisions). Behavioural measures use the definitions of Section 6.4. [a]P01 and P02 ran at half the later task weights; their points are doubled here.

Measure (per session)	Strategic Heavy	Tactical Heavy	p
Missions arrived (non-residual)	13.5 (0.5)	7.5 (1.5)	$<.001$
Strategic decisions made	11.4 (2.3)	6.9 (1.5)	$<.001$
Tactical plans confirmed	11.2 (2.5)	6.8 (1.5)	$<.001$
Tasks completed	26.5 (6.0)	27.2 (9.0)	.902
Completion points[a]	1276 (316)	1627 (608)	.003
Time penalty (points)	276 (59)	395 (182)	.001
Score (points − penalty)[a]	1000 (364)	1233 (564)	.042
Strategy card taken (% of decisions)	60 (27)	73 (28)	.004
Manual before cards loaded (%)	19 (23)	11 (20)	.039
Card edited, then deployed (%)	6 (16)	6 (14)	.508
Suggest pressed (% of plans)	72 (26)	79 (29)	.029
Suggest plan kept unmodified (% of plans)	69 (28)	75 (32)	.084
Drone failures	4.0 (2.6)	3.8 (2.6)	.636
Missions abandoned	1.2 (1.6)	0.8 (0.9)	.165
NASA-TLX, raw mean (0–20)	8.8 (3.1)	8.6 (3.5)	.607
Mental demand	11.7 (4.8)	11.0 (6.1)	.456
Temporal demand	13.6 (3.9)	12.4 (5.0)	.226
Performance (0 = perfect)	7.5 (4.9)	8.8 (4.9)	.152
Effort	9.6 (4.7)	8.9 (4.7)	.352
Frustration	6.0 (5.0)	6.2 (5.8)	.467
Trust, Strategic Assistant (1–7)	4.38 (1.52)	4.84 (1.33)	.098
Trust, Tactical Assistant (1–7)	5.52 (0.88)	5.36 (0.93)	.537

7.2 Each operator has a personal style — individual, not a type

Summary. Operators differ a great deal in how they use the assistants, and the differences are personal. We describe each session by ten habits (how often a strategy card is taken, how often Suggest is pressed, how long a decision takes, and so on). From those habits alone, a person's second session resembles their own first (median profile r = .48) and essentially nobody else's (r = −.01): their own session is in the closest three of 33 four times as often as chance (41% vs 9%), although the second session is a different scenario, and who the operator is explains about six times more of card-taking than the mission does. Six of the ten habits carry over clearly; one (topping up drones on single-type missions) does not at all. Between people, use of the two assistants is only loosely linked ($\rho \approx .35$), so leaning on one is not a fixed part of one overall "delegate" strategy (though within a person the two do rise and fall together from session to session). The styles do not fall into types, and nothing measured beforehand (attitude to AI, views on human authority, experience) predicts them. Strong evidence for a stable personal style; good evidence against types; "no evidence" (not "evidence of no effect") for disposition.

Strength of evidence (register rating): strong.

7.2.1 The habits we grade them on

Figure 7. To compare styles we describe every 8-minute session by ten habits, each something the logs record at every decision. Seven concern the strategic tier (the strategy cards): how often a card is taken as offered; how often cards are previewed before choosing; how long a decision takes; whether, when building by hand, the operator does it before the cards have even appeared; whether a card is edited before committing; and how many drones are committed compared with the cautious Conservative card, on single-type and on multi-type missions. Three concern the tactical tier (Suggest): how often Suggest is pressed on a plan, whether a suggested plan is deployed unchanged, and whether Suggest is pressed when recovering from a failure. The chart shows each habit's spread over all 66 sessions in its own units. Most habits vary enormously between sessions (card uptake runs from 0% to 100% of a session's decisions); two are nearly universal (most sessions keep every suggested plan unchanged, and most never edit a card). The right-hand

Figure 7: The habits we grade them on

column previews the next results: how strongly each habit carries over from a person's first session to their second.

Key numbers.

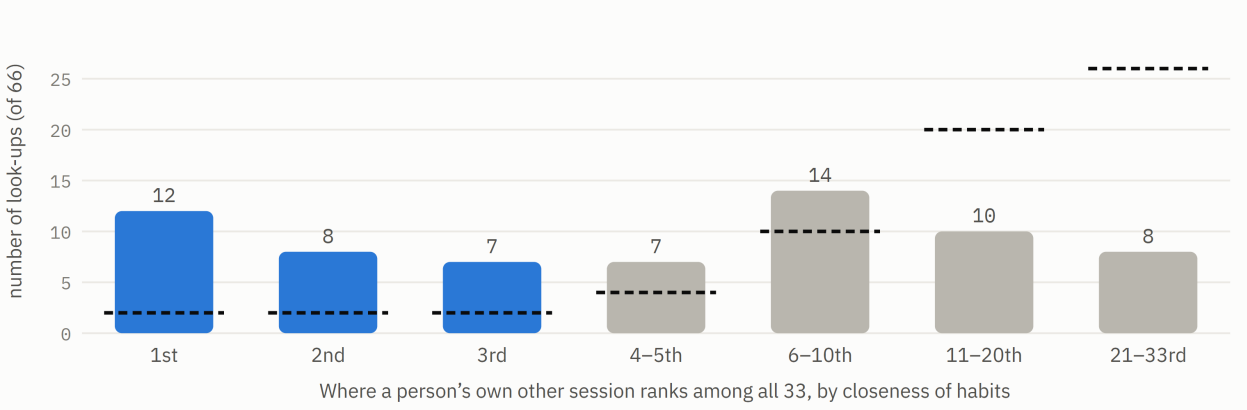
- Card taken as offered: median 67% of a session's strategic decisions (middle half 47–91%, range 0–100%)
- Suggest pressed: median 86% of tactical plans (55–100%); suggested plan kept unchanged: median 100% (middle half 100–100%)
- Decision time beyond the card wait: median 6.5 s (5.4–10.3 s, range 1.6–43 s)
- Card edited before committing: median 0% (most sessions never); going manual before the cards appear: median 50% of hand-made decisions (0–75%)
- Drones vs the Conservative card: single-type -0.5 (-1.25 to 0); multi-type +0.6 (-0.1 to +1.3)

Method. Rows: 66 sessions (33 people × 2; P09 excluded). Each habit is a share or an average over that session's decisions. A habit with nothing to count in a session is undefined there: going manual before the cards (no hand-made decisions) in 15 sessions, drones on single-type missions (no single-type mission opened) in 19, Suggest in recovery (no failures) in 4, keeping the Suggest plan (Suggest never pressed) in 2. For the profiles these are filled with the median of that game. Decision time is analysed as log(1 + seconds) and shown in seconds. For the comparisons that follow, each habit is turned into a z-score among the sessions of the

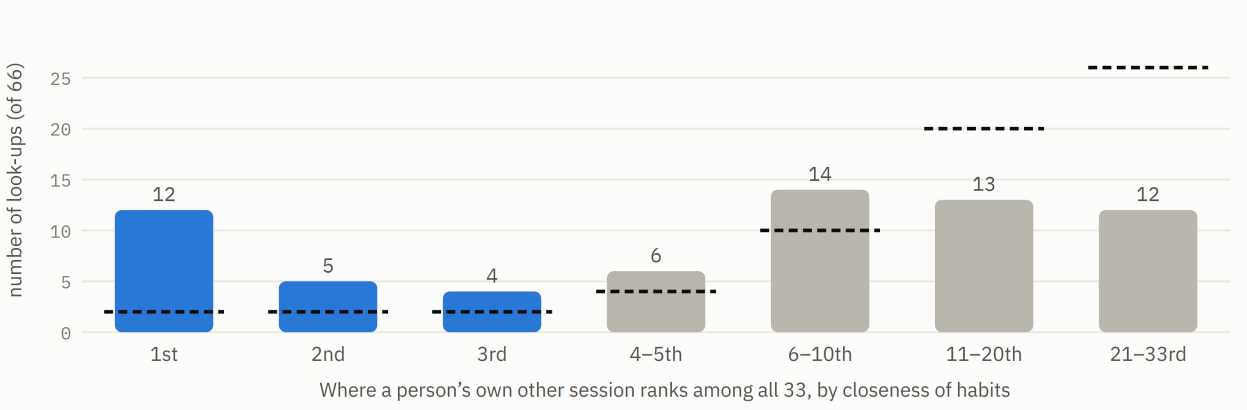
same game (each of the four fixed games), so a profile says how a person compares with others who faced the same missions. The ten habits were fixed in round 1 (H1) before any grouping was tried; outcomes (score, trust, questionnaires) are never among them. ρ = Spearman correlation between a person's session-1 and session-2 values, game-adjusted (see "How far each habit holds").

What participants said. Asked to describe their approach, 15 of 33 named a consistent strategy of their own (C01). In their own words it is mostly about the habits graded here: whether they wait for the cards or go manual before they load (P21, P32), whether they trim or edit a card, whether they start from Suggest and touch it up (P31, P21), and how they check (P14, P17). Few described a rule for when to press Suggest. Most said they pressed it on nearly everything.

7.2.2 Where each person's own session ranks



(a) All ten habits.



(b) With the reliance axis removed.

Figure 8: Where each person's own session ranks

Figure 8. Each person's ten-habit profile from session 1 is compared with all 33 session-2 profiles, and the 33 are ranked from most to least alike (and the same the other way round, session 2 among the session 1s). If habits were not personal, a person's own other session would be equally likely to land anywhere in that ranking (dashed line). Instead it is the single closest match 12 times out of 66 (2 expected by chance) and in the closest three 27 times (6 expected), even though the second session is a different scenario.

Key numbers.

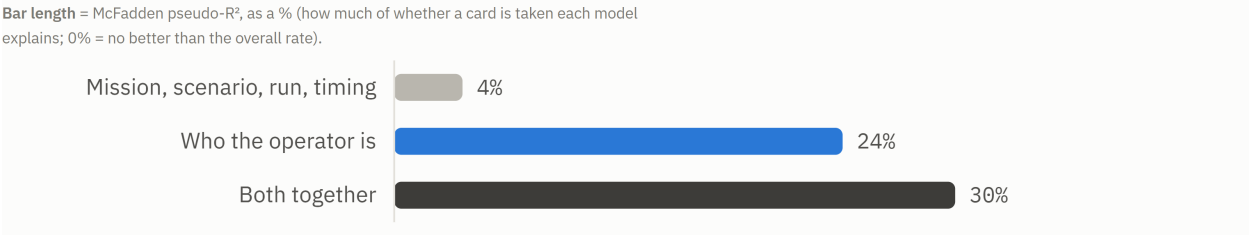
- Own session ranked 1st: 12 of 66 look-ups vs 2 expected; in the closest three: 27 vs 6 (41% vs 9%); permutation $p < .001$
- Average rank of own session 8.2 of 33 (chance 17.0); with the reliance axis removed 10.1 ($p < .001$)

- Profile correlation with own other session: median r = .48; with other people's: median r = −.01; own closer for 29 of 33

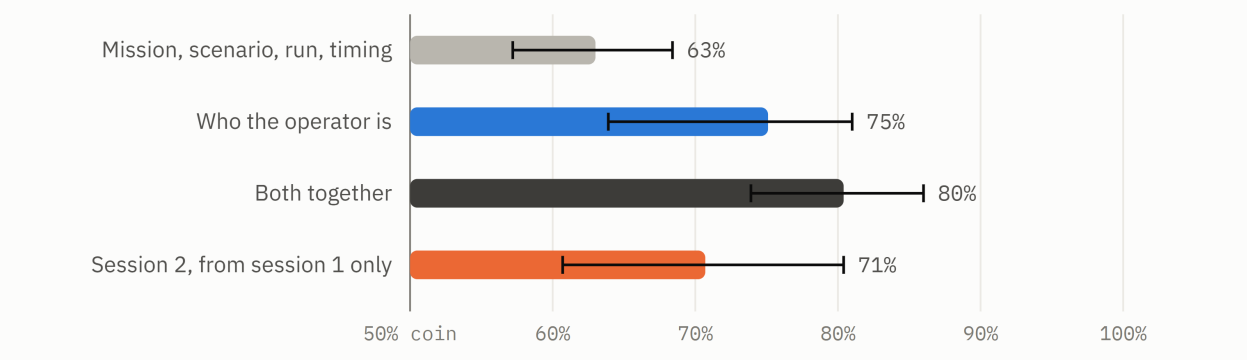
Method. Rows: 66 sessions (33 people × 2). Each session is the ten habits of the first section, each turned into a z-score among the sessions of the same game, so a profile says how this person compares with others who played that game. "Alike" = straight-line (Euclidean) distance between two ten-habit profiles. 66 look-ups (33 people, both directions); under chance every rank 1–33 is equally likely, so the dashed line is 66 × (ranks in the bin) / 33. The p-value shuffles who-is-who among the session-2 profiles 10 000 times. "Reliance axis removed" subtracts each profile's score on the first principal component (see the axis numbers) before comparing.

What participants said. Nobody was asked to compare their two sessions directly. In the think-aloud, though, several people open session 2 with the same move and say the same thing as in session 1: P29 'start conservative' and does single-colour missions by hand, P11 builds all-blue missions themself, P21 hand-builds in parallel. That fits a recognisable personal style. The people who said they changed are in the retest section.

7.2.3 Who decides matters more than what is being decided



(a) McFadden pseudo-R^2.



(b) The same comparison as a ranking test (AUC); whiskers are 95% intervals (people resampled).

Figure 9: Who decides matters more than what is being decided

Figure 9. Card uptake runs from 8% to 100% across people (median 61%). Knowing *who* is deciding predicts whether a card is taken about six times better than knowing everything about the mission and when it arrived. In plain terms: shown two decisions, one where a card was taken and one where it was not, and told only who made each, you would pick the right one 75% of the time (a coin: 50%; knowing only the mission: 63%). A person's session-1 rate on its own still does it 71% of the time for their session-2 decisions.

Key numbers.

- Pseudo-R^2: mission + timing 4%; person 24%; both 30%
- Card uptake per person 8%–100%, median 61%; 6 people ≥ 90%, 3 people ≤ 25%

- Telling a taken from a not-taken decision (AUC): mission and timing 63% [57%, 68%]; the person, from their other decisions, 75% [64%, 81%]; both 80% [74%, 86%]; session-2 decisions from the session-1 rate alone 71% [61%, 80%] (321 decisions).
- Right-or-wrong guessing: always 'card' 66%; mission 65%; person 70%; both 77%.
- Person-only pseudo-R^2 with the person labels shuffled: 4% on average (95th percentile 6%), against 24% observed.

Method. 606 strategic decisions. Three logistic models of "card taken": (a) mission and timing only — drone types needed, drones needed, scenario, session 1 or 2, decision number; (b) the person only (one term each); (c) both. McFadden pseudo-R^2 = how much each improves on predicting the overall rate for everyone (0 = no better, 1 = perfect). Descriptive, not a test: with ~18 decisions each, the person-only model flatters itself a little; refitted 200 times with the person labels shuffled it scores 4% on average, so about 20 of its 24 points are real. Retest of raw card uptake session 1 vs 2: ρ = .57 (.62 once game-adjusted, next section). **In plain terms** (the second view): AUC = the chance a model ranks a random card-taken decision above a random not-taken one. The person enters as their rate on all their OTHER decisions (on the log-odds scale), so no decision predicts itself; 'session 2 from session 1' uses only each person's session-1 rate. Intervals: people resampled 1,000 times, predictions held fixed.

What participants said. People often described their use of the assistants as part of who they are: 'naturally conservative' (P32), 'I'm more conservative' (P26), 'I always let the agent do it first' (P13). At the extremes, P14 used the assistants 'for almost every allocation' and P16 'manually overrode most of the time'. Against this, several gave the situation as the reason: workload (P32, 9 of 33 coded C11), being resource-constrained (P22), mission size (C03). P32 calls themself 'naturally conservative' and also says their use was 'very much about workload', so a person's style and the situation are both in play.

7.2.4 How far each habit holds from one session to the next

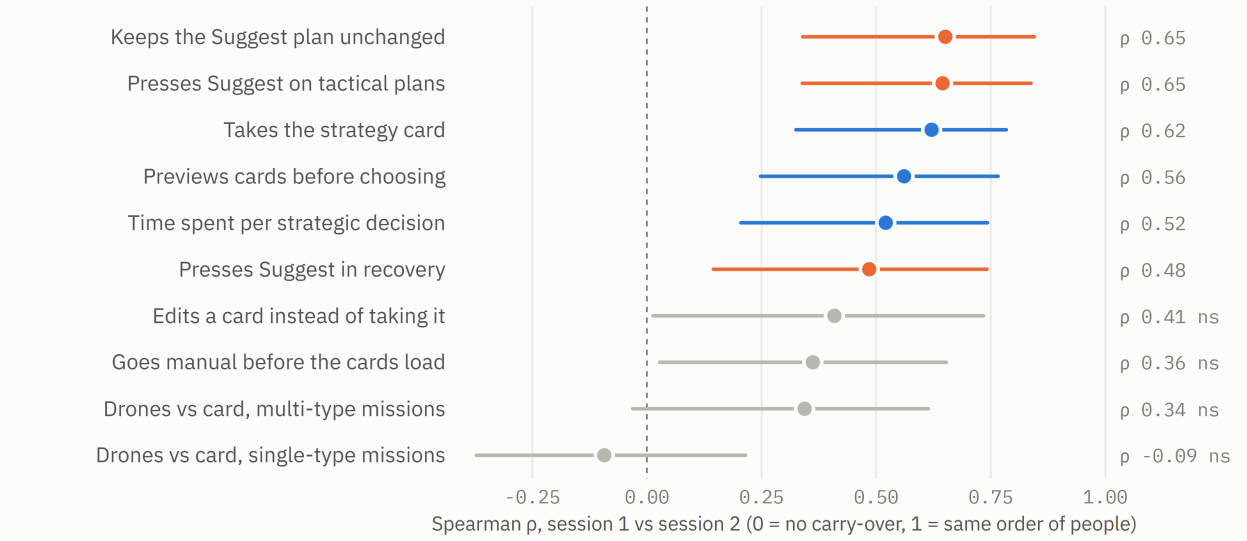


Figure 10: How far each habit holds from one session to the next

Figure 10. One line per habit: how strongly a person's place on that habit in session 1 predicts their place in session 2 (ρ; 0 = no carry-over, 1 = same order of people). Six of the ten carry over clearly and survive correction for testing ten (coloured). Editing cards, going manual before the cards load and topping up drones on multi-type missions carry over weakly, and topping up drones on single-type missions does not carry over at all (ρ −.09). "Keeps the Suggest plan" is high partly because 24 of 33 people kept every Suggest plan in both sessions.

Method. 33 people. For each habit, Spearman ρ between session-1 and session-2 values, each first z-scored within the game played; 95% intervals from resampling people 2000 times; Holm correction across the ten. The same ρ values are the right-hand column of the habits chart in the first section.

What participants said. Some people say they kept their approach and some say they changed it. P18 'sort of changed from the first one' (led) and P30 'used more personalized' in session 2. Four people describe a real shift in the think-aloud: P06 and P31 came to rely on the assistants more and check less, while P35 and P15 went back to deciding for themselves after the strategy cards made errors. That fits habits that carry over clearly but not completely.

7.2.5 Strategic and tactical reliance go together only loosely between people

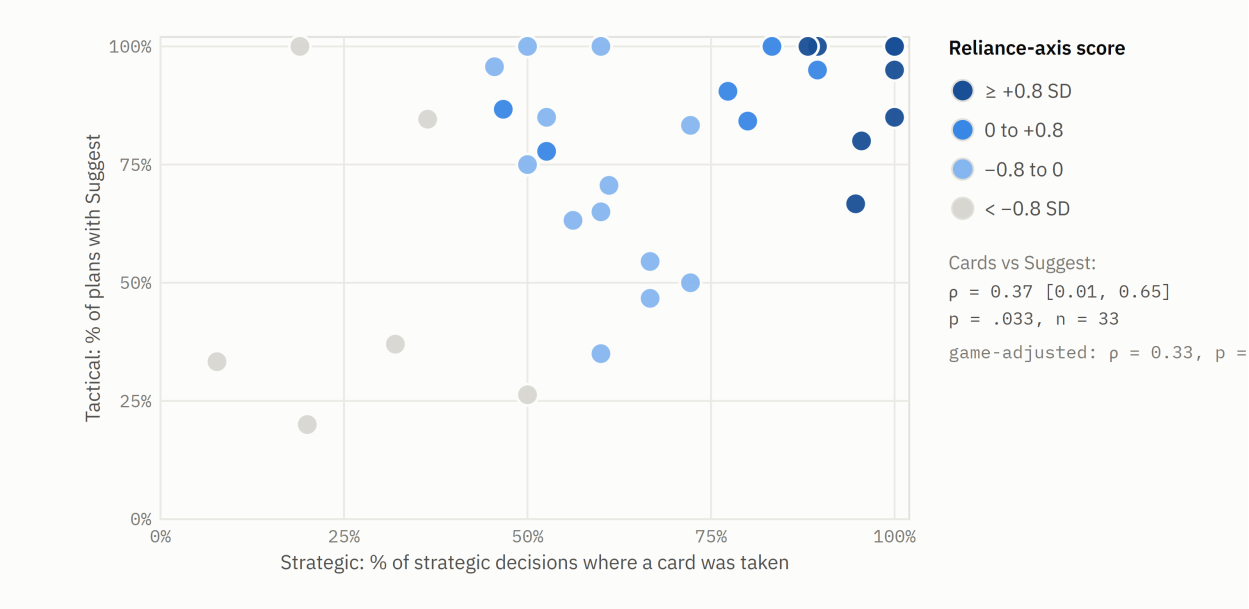
Figure 11. Each dot is a person: x = share of strategy cards taken, y = share of tactical plans where Suggest was pressed. Between people the two are only weakly linked (ρ = .37, and .33 once each is measured against others who played the same game): plenty of people press Suggest on almost every plan while taking only half the cards, so knowing how much someone relies on one assistant tells you little about the other. Part of the reason is a ceiling: Suggest is pressed on a median 86% of plans. The second view asks a different question: when the *same* person takes more cards in one session than the other, do they also press Suggest more? Yes (ρ = .51). So use of the assistants is not one fixed trait spanning both tiers, but within a person there is a session-level "leaning on the assistants today" that moves both together.

Key numbers.

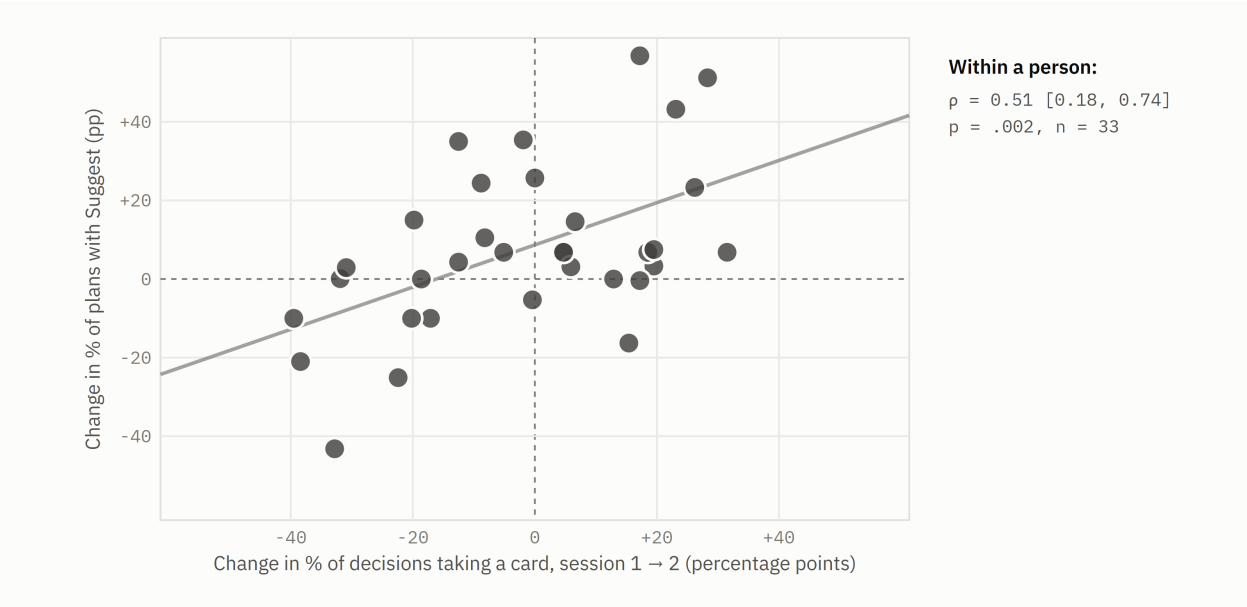
- Between people, card uptake vs Suggest use: ρ = .37 [.01, .65], p = .033 (raw); ρ = .33 [−.03, .62], p = .058 (game-adjusted)
- Within a person, change in card uptake vs change in Suggest use, session 1 → 2: ρ = .51 [.18, .74], p = .002
- Card uptake vs Suggest in recovery: ρ = .42, p = .014; vs keeping the Suggest plan unchanged: ρ = .20, p = .26
- Card taking and previewing: ρ = .79 (largely mechanical: choosing a card means clicking it)

Method. Between people: one value per person, both sessions pooled (n = 33), Spearman ρ with a Fisher interval; the game-adjusted version first subtracts the mean of everyone who played the same game from each session, then averages the person's two sessions. Within a person: the session-2 minus session-1 change in card uptake against the change in Suggest use, both game-adjusted (n = 33). Checked: the within-person link holds without game adjustment (ρ = .56), and points the same way whichever scenario came first (ρ = .47, p = .058 with Strategic Heavy first, n = 17; ρ = .63, p = .010 with Tactical Heavy first, n = 16).

What participants said. Many describe relying on the two assistants differently, which fits the loose link between people. The most common split is to keep the strategic decision and hand over the tactical one: P22, P28, P30 and P32 say so, and P14 checked the tactical tier harder. A few did the reverse: P11 said they would use the strategic one more and built small plans themself, and P16 overrode the cards but 'more or less fully relied' on Suggest. P10 describes the two moving together: taking the Aggressive card makes Suggest the obvious next step.



(a) Between people.



(b) Within a person: change from session 1 to session 2.

Figure 11: Strategic and tactical reliance go together only loosely between people

7.2.6 Individual, not types

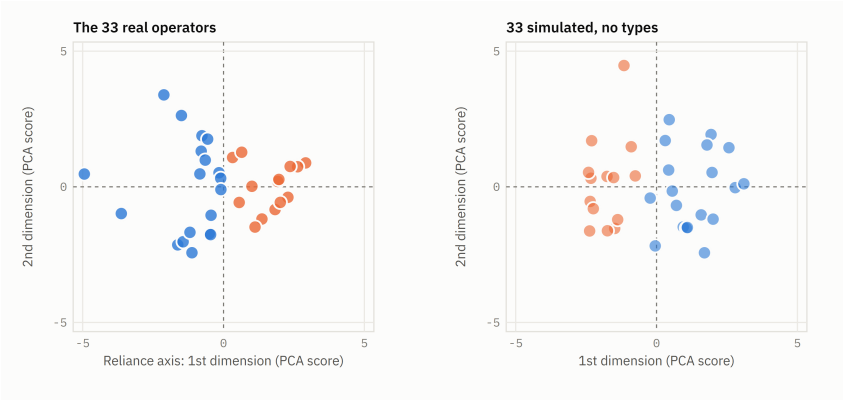
Figure 12. Distinct people, but not distinct kinds of people. If there were types (say "delegators" and "hand-builders"), people would sit in clumps with gaps between them, and splitting them into groups would find clean, well-separated groups. Three plain checks say they do not. **(1) The map:** placed by their two biggest habit dimensions, the 33 operators form one continuous cloud; it looks just like 33 simulated operators that have no types by construction, and the best two-group split simply cuts the cloud in half. **(2) Separation:** however many groups we ask for (2, 3 or 4), the real people separate no better than the no-types simulations do, and every value is below the usual threshold for real group structure. **(3) Recurrence:** grouping each session separately, the same people barely end up together twice, no more than in a smooth cloud where people keep their place. The archetype names (Delegator, Deliberate builder, Fast tinkerer) are best kept as labels for the ends of the main dimensions.

Key numbers.

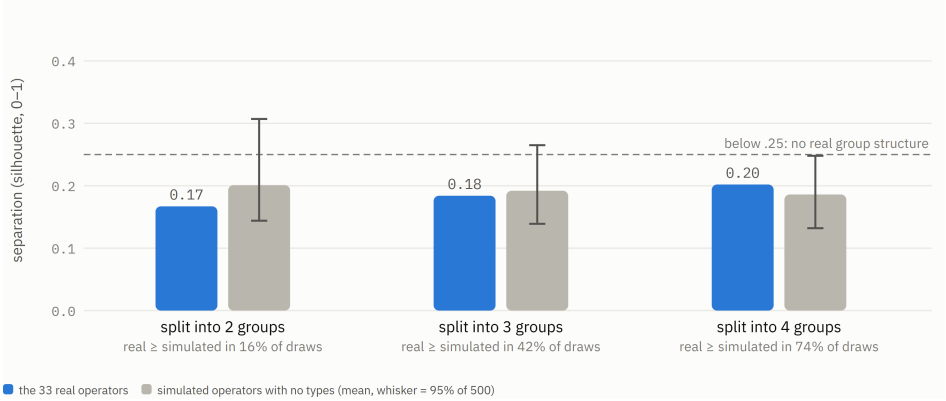
- Separation (silhouette), real vs no-types simulations: 2 groups .17 vs .20 [.14, .31]; 3 groups .18 vs .19 [.14, .27]; 4 groups .20 vs .19 [.13, .25]; a no-types sample separates at least as well 84%, 58% and 26% of the time
- Groups in session 1 vs session 2 (ARI): 2 groups .08, 3 groups .09; a stable no-types cloud gives .13 on average (95% of simulations $-.03$ to .47 for 2 groups, $-.02$ to .40 for 3), at least as high as the real value about 55% of the time; shuffled labels give up to .12–.13
- Round 1: two groups vs one, ΔBIC 1.9 (needed > 10; 42% of no-type simulations did at least as well)
- Archetypes (extras2): three explain 46% vs 43–59% in the no-types continuum (p = .83); same archetype both sessions 61% vs 30–79% (p = .32)

Method. Rows: the 33 people, each described by the ten habits pooled over both sessions and standardised. Grouping = k-means (the computer splits people into k groups so that each sits as close as possible to its group's centre; best of 20 starts). Separation = silhouette: for each person, how much closer they are to their own group than to the nearest other group, averaged (1 = tight clumps far apart; 0 = boundaries arbitrary; below .25 is conventionally "no substantial structure", Kaufman & Rousseeuw). Benchmark = 500 simulated samples of 33 operators drawn from one smooth cloud with the same distribution of each habit and the same correlations between habits as the real people, so they have no types by construction. (We use these rather than simply shuffling each habit, because shuffling also destroys the links between habits, and a cloud of linked habits separates slightly better than shuffled data without having any types: shuffled data give about .14.) Recurrence: each session's profiles (z-scored within game) are grouped separately and compared with the adjusted Rand index (1 = the same groupings, 0 = chance), against (a) shuffled labels and (b) 500 simulated pairs of sessions with no types but the same session-to-session stability of every habit as the real people. The map uses the first two principal components (50% of the differences). Round 1's pre-registered test (mixture models compared by BIC) and the archetype simulations (extras2) reach the same conclusion; their numbers are kept below.

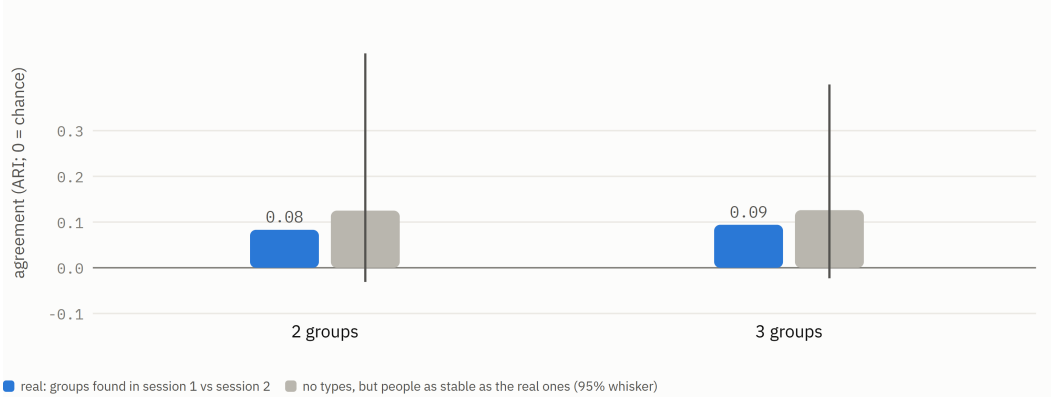
What participants said. Nobody described themselves or others as a 'type' of operator. People talked about their own rules, and several, such as P29, were still unsure of them partway through a session ('I haven't actually decided ... which ones are best to handle myself').



(a) The people, mapped on the first two principal components.



(b) Cluster separation (silhouette) against no-types simulations.



(c) Recurrence of groupings across sessions (adjusted Rand index).

Figure 12: Individual, not types

7.2.7 Nothing measured beforehand predicts the style

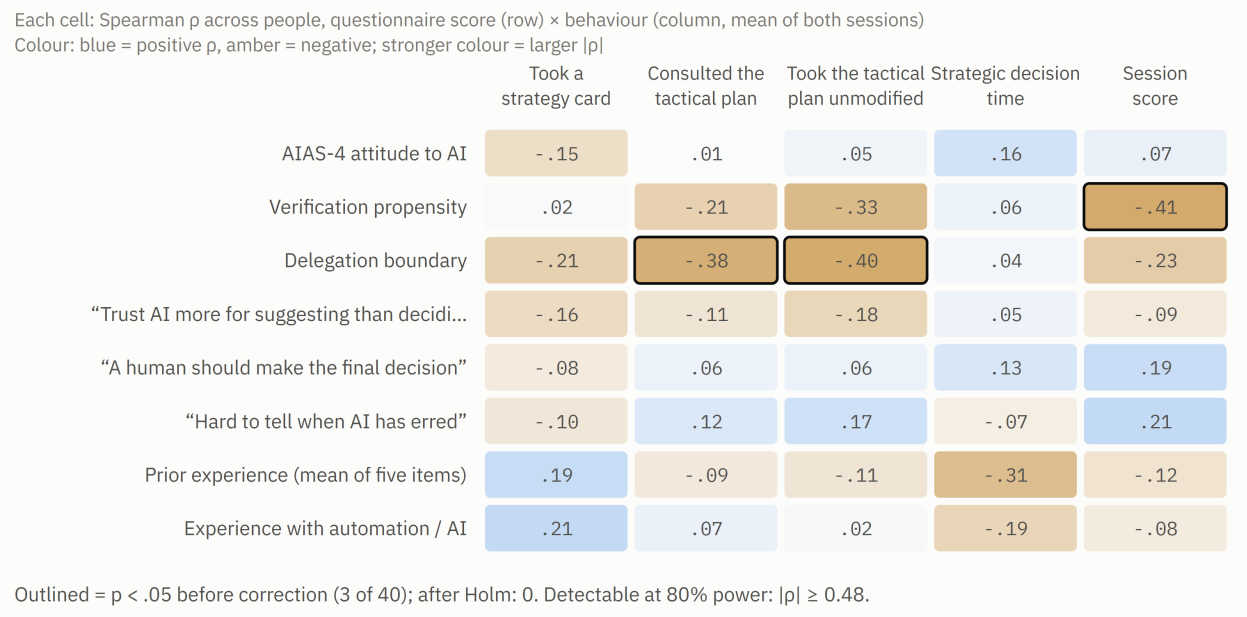


Figure 13: Nothing measured beforehand predicts the style

Figure 13. Attitude to AI (AIAS-4), verification propensity, views on human authority over AI (delegation scale and its single items, including "a human should make the final decision"), and prior experience: 40 tests against five behaviours. Three reach p < .05 before correction, and <b>none survives Holm correction for the 40 tests. Three nominal hits is what chance alone produces: shuffling which person's questionnaire goes with which person's behaviour gives 2.0 such hits on average, and 3 or more in a third of shuffles. This is "no evidence", not proof of no link: the sample could only have detected fairly strong correlations ($|\rho| \geq .48$), and it is almost uniformly AI-positive (AIAS-4 averages 8.0 / 10) and from computing or engineering (32 of 33). Within this sample, the large differences in style are not explained by what people say they think about AI.

Key numbers.

- 0 of 40 tests survive Holm; 3 nominal (verification × score ρ −.41; delegation × Suggest consulted −.38 and × plan kept −.40)
- Chance alone gives 2.0 nominal hits on average; 3 or more in 34% of 2000 shuffles of who-goes-with-whom; a p as small as the smallest one (.021) turns up somewhere in the grid in 51% of shuffles
- AIAS-4 × card uptake ρ −.15; "A human should make the final decision" × card uptake ρ −.08
- AIAS-4 odds ratio for card taking 0.80 per SD [0.50, 1.27]; detectable only at about ×2
- AIAS-4 mean 8.0 / 10 (SD 1.2); 32 of 33 from computing / engineering; detectable $|\rho| \geq .48$

Method. Per person: each questionnaire score (taken before session 1) against the average over both sessions of: card uptake, Suggest consulted, Suggest plan kept unmodified, strategic decision time, session score. Spearman ρ per cell; Holm correction across all 40 together (build_findings_synthesis `analyse_disposition`). Detectable effect = the correlation the sample had an 80% chance to find at α = .05 (n = 32 with questionnaires). Round 1 H7's decision-level model gives the same answer for AIAS-4: odds of taking a card × 0.80 per SD [0.50, 1.27].

What participants said. When people talked about AI in general, it did not line up neatly with what they did. P04 does not trust the AI to be better than them but uses it because it is quicker. P24 says they really trust AI yet trims the

cards to save drones. P15 calls the AI 'a little bit stupid' but hands it complex plans. P35 relies on it because they are 'not trained'. Most people explained their use by time pressure and job size rather than by a view about AI, which fits the null.

7.3 Small jobs by hand, big jobs to the assistant — on both tiers, and it is where the assistant saves time

Summary. The bigger the job, the more likely an operator hands it to an assistant, on both tiers. Measured the same way on both (drones the mission needs), the strategy card is taken on 56% of the smallest missions vs 75% of the largest, and Suggest is pressed on 54% vs 91%. Comparing each person only with themselves, every doubling of the mission multiplies the odds of handing over by about 1.6 (strategic) and 2.9 (tactical); 22 of 33 people hand over more on their biggest missions than their smallest on each tier. It survives every check we tried, including dropping the illogical Conservative cards and allowing for everyone having played the same missions. Three qualifications: on the strategic tier most of the effect is operators deciding small missions before the cards load; the strategic curve is lumpy (a dip at 5 drones, missions that mostly arrived in bursts in the Strategic Heavy games), so the dependable strategic contrast is one-type vs three-type missions; and handing over on one tier goes with handing over on the other (a mission allocated from a card gets Suggest at about twice the odds). The rule is sensible: the card is much quicker than building by hand on three-type missions and no quicker on one-type ones, and a Suggest plan is not slower: compared on missions of the same size it finishes about 15 s sooner, mostly because it is carried out faster.

Strength of evidence (register rating): strong.

7.3.1 Both tiers on one axis: the bigger the mission, the more it is handed over

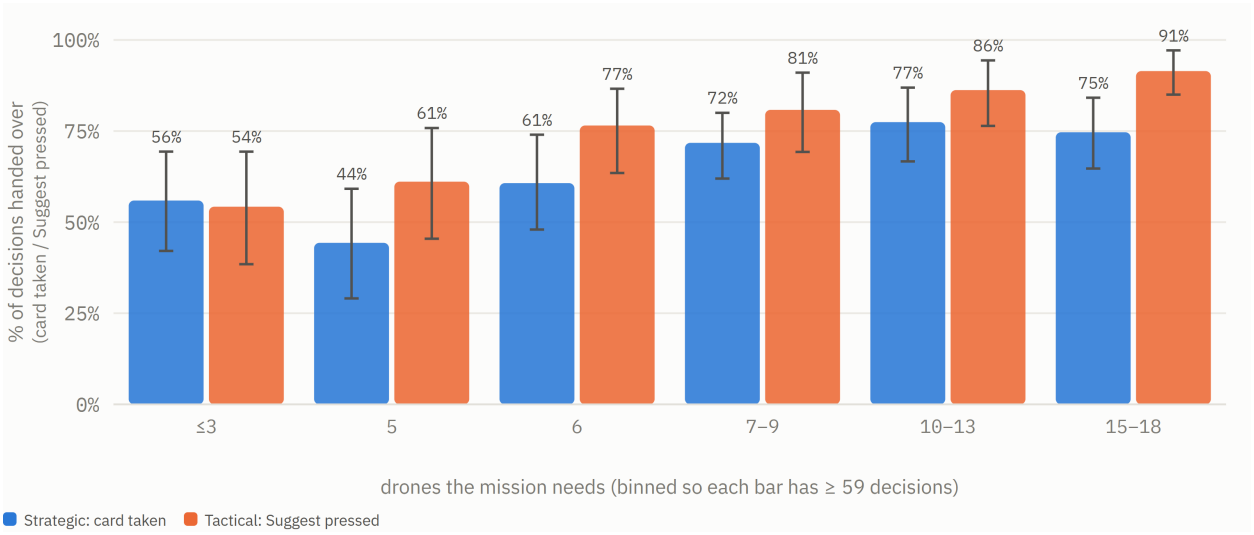


Figure 14: Both tiers on one axis: the bigger the mission, the more it is handed over

Figure 14. For every mission, the share of decisions where the operator handed over (took a strategy card as offered, blue, or pressed Suggest in the tactical planner, orange) by how many drones the mission needs. Both rise. The tactical tier rises more steeply and keeps rising, from 54% to 91%. The strategic tier climbs from about half on small missions to about three quarters, with a dip at 5 drones. That dip is seven particular all-Fast missions, all in the two Strategic Heavy games, and six of them arrived in bursts: in one game two of them arrived 11 s apart in a run of seven missions inside one minute (minutes 1.6 to 2.6), in the other four of them arrived within 80 s (minutes 3.7 to 5.0) alongside three other missions. So by the time operators reached them 3–4 other missions were waiting, and they decided fast and often before the cards loaded. Busy arrival is likely part of it, but not all: missions of other sizes in the same bursts got the card more often (a 3-drone and a 6-drone mission in those bursts got it 67% and 64% of the time), and holding the queue fixed barely changes the 5-vs-3 comparison (see "Why the 5-drone missions dip").

Key numbers.

- Strategic, per doubling of drones needed: odds $\times 1.62$ [1.23, 2.15], $p < .001$ (606 decisions, 33 people)
- Tactical, per doubling of drones needed: odds $\times 2.91$ [2.03, 4.17], $p < .001$ (627 plans)
- Both tiers, one shared slope: $\times 2.07$ [1.64, 2.61], $p < .001$ (1,233 decisions); the tactical slope is steeper than the strategic one by $\times 1.38$ [0.99, 1.91], $p = .056$ — suggestive, not shown
- Raw rates, smallest to largest bin: strategic 56% $\rightarrow$ 75%, tactical 54% $\rightarrow$ 91%
- Arrival bursts (Strategic Heavy games, fixed per game): the seven 5-drone missions had on average 4.4 other missions arrive within 60 s of them, against 3.6 for the 3- and 6-drone one-type missions and 3.1 for every other mission; 6 of 7 arrived inside or straight after a burst; the seventh arrived 10 s into a game and got the card 69% of the time

Method. **Rows.** Strategic: all 606 strategy-card decisions (one per strategic choice). Tactical: all 627 non-recovery tactical plans whose mission size is known (630 minus 3 residual missions with no recorded need). **Size** = drones the mission needs: the sum of the primary requirements over its tasks — the number an operator reads off the task badges, the same for both tiers. Binned so each bin has ≥ 59 decisions. **Intervals**: we resample whole people (2,000 times) and recompute the rate, so a person's decisions travel together and one heavy user cannot make an interval look tight. **Model**: a logistic GEE (a regression for yes/no outcomes that allows each person's decisions to be correlated) with size split into two parts — this decision's size minus the person's own average size (the *within-person* part) and the person's average (the between part); game and session are held fixed. The within-person odds ratio answers: "when the same person meets a mission twice as big, how much do the odds that they hand it over change?" Odds $\times 2$ means, for example, a 60% chance becomes 75%. Both tiers in one model: the two tables stacked, one shared within-person slope plus a tier term; the tier $\times$ size term tests whether the slopes differ.

What participants said. 10 of 33 state the size rule in debrief or interview (C03). The think-aloud adds at least six more who give the same reason while playing: P04, P06, P12, P20, P32 and P36. They do single-colour or few-drone missions themselves and hand over large, multi-colour, many-chain ones, usually because it is quicker. There is dissent. P30 thinks simple tasks are the ones the AI 'should be able to do even easier'. P29 switched direction mid-session ('if it's easy, then it's hard for it to do it wrong'). P20 kept complex orderings for themself, and P10 hands trivial one-option plans to Suggest.

Quotes selected for this section.

> *"I think if it's just one colour, I'm gonna do it, like, assign it myself, but if it's multiple colours, I think I'm gonna let, um, the AI choose for me, for the, the strategic side." — P29 (think-aloud s2 0:17:33)*

> *"either it's a really big mission, and it's like, oh, I can trust it with a large mission. [. . .] Or it's a simple mission, and I'll be like, oh, I can just bang it up myself." — P11 (debrief 0:28:41)*

> *"the simple task you probably want to do yourself if you're going to get the solution quicker and the more complex tasks you're going to want somebody to give you that solution" — P36 (think-aloud s1 0:09:32)*

> *"But when it was just one color, I could just kind of count how many is necessary and then add, like, one extra more. Mm-hmm. And that was usually enough. So, it was fast for me to just click manual" — P29 (debrief 0:24:29)*

> *"I did feel like some of them I should do myself and I trusted it more with the complicated tasks" — P29 (debrief 0:24:54)*

> *"Automated larger, more complex tasks more, but smaller tasks could do himself" — P15 (written interview)*

> *"When they only needed a single colour of drone I tended to do it manually" — P34 (written interview)*

> *"If I look at a mission with 15 drones and I need to move two or three chains, that's just going to take me about a minute, and it's just no point. It's just a computation problem." — P11 (debrief 0:29:41)*

> *"I'll use the suggest, because there's a lot of drones here, so because there's a lot of drones to allocate, the AI is going to provide me a nice map" — P05 (think-aloud s1 0:09:36)*

7.3.2 Person-adjusted: every size measure, both tiers, one picture

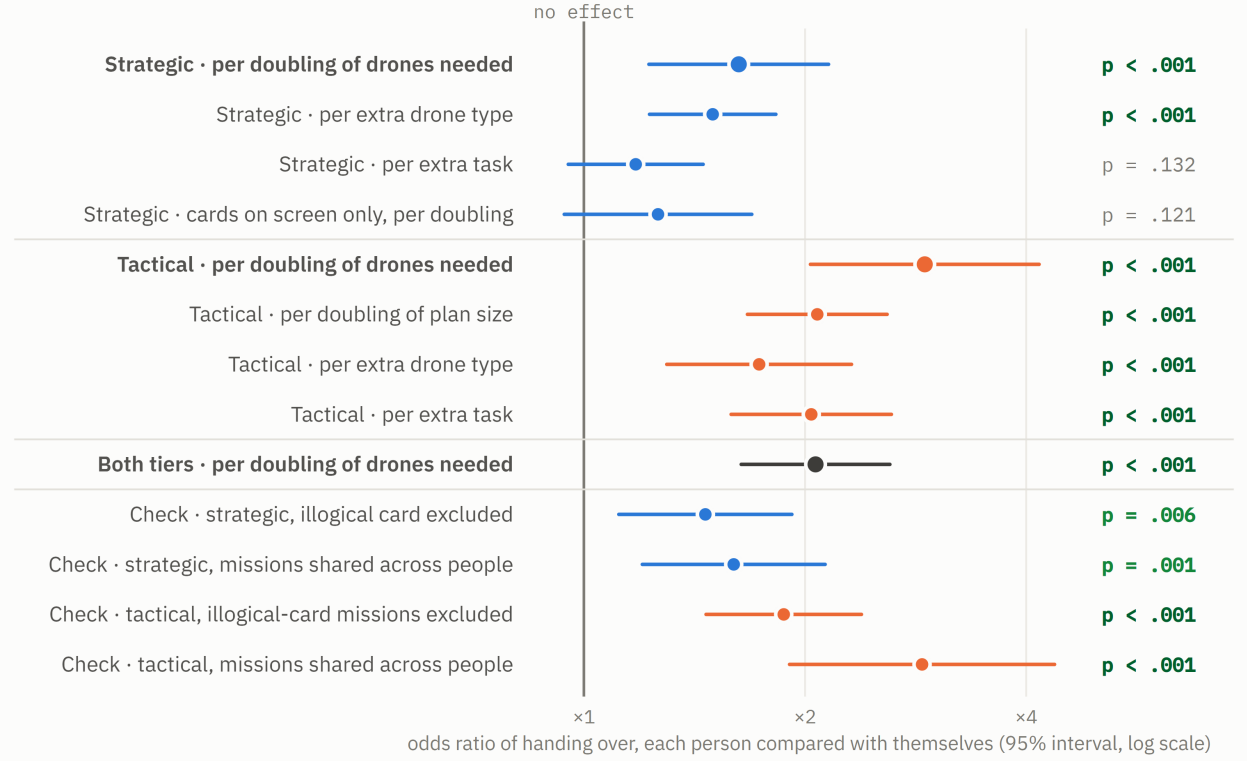


Figure 15: Person-adjusted: every size measure, both tiers, one picture

Figure 15. Each row is one within-person estimate: how the odds of handing over change when the same person meets a bigger job. Rows right of the line mean "bigger → hand over more". All size measures point the same way on both tiers; the two that do not clear the line are strategic tasks and the strategic slope once the cards are on screen. The bottom block holds the artefact checks. Dropping every illogical Conservative card leaves the effect in place. So does allowing for the fact that every operator played the same fixed missions, so one odd mission could otherwise make the interval look tight. With each mission as one point, the share handed over still rises with its size on both tiers.

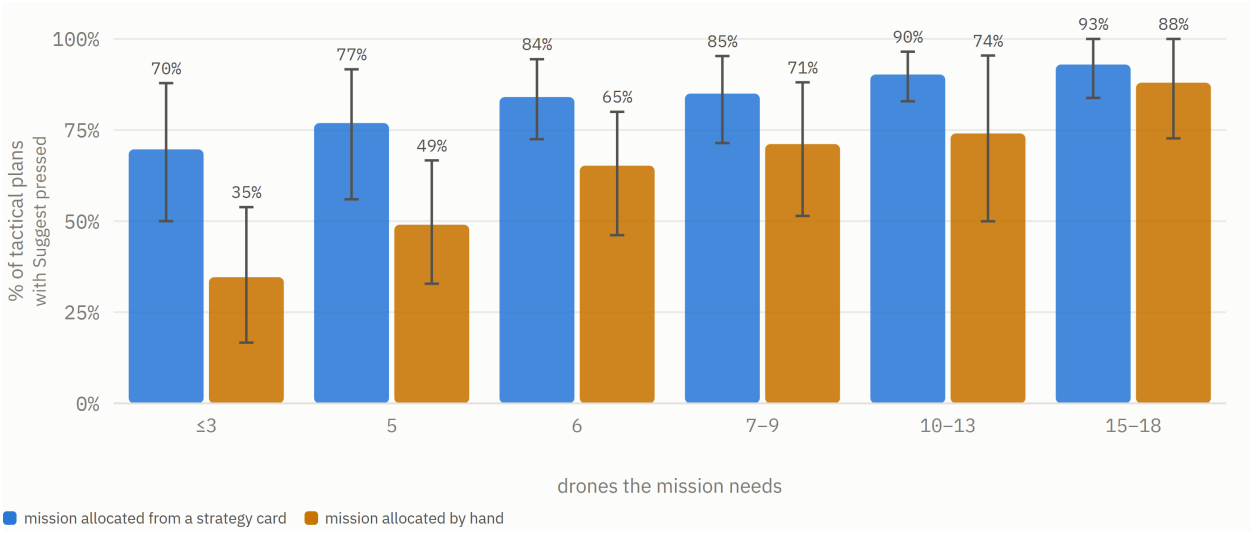
Key numbers.

- Strategic, illogical Conservative card excluded: $\times 1.46$ [1.12, 1.92], p = .006 (474 decisions)
- Missions shared across people (interval clustered on person and mission): strategic $\times 1.60$ [1.20, 2.13], p = .001; tactical $\times 2.89$ [1.91, 4.37], p < .001 (65 missions)
- Each mission as one point (missions with ≥ 5 decisions): strategic $\rho = .63$ over 41 missions, tactical $\rho = .69$ over 41 (both p < .001)

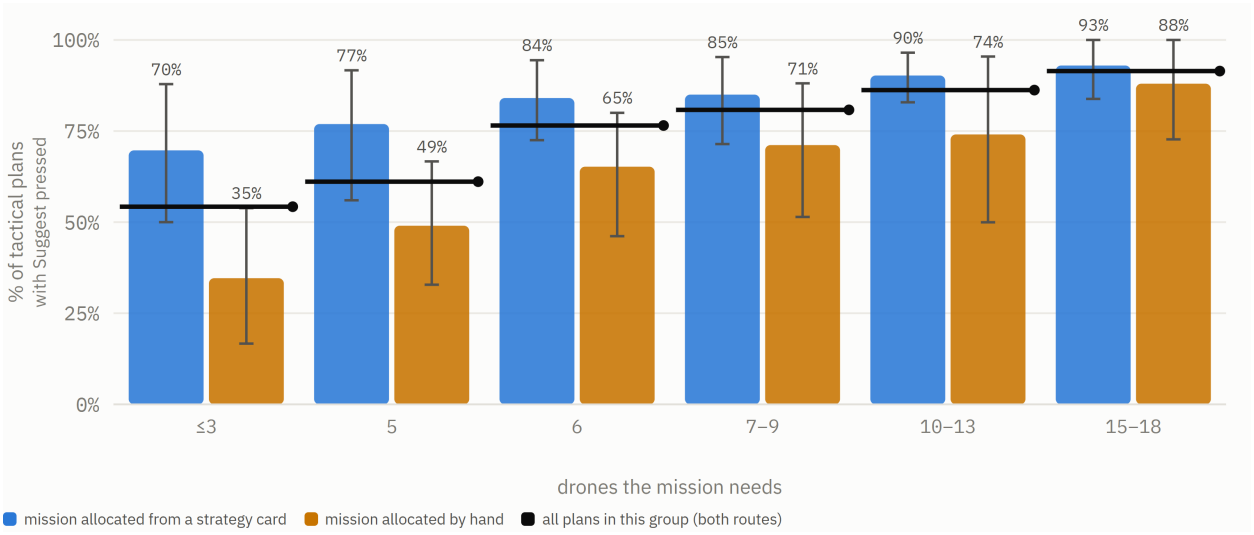
Method. GEE logistic models clustered on person (exchangeable correlation, small-sample "bias-reduced" standard errors), size split into within- and between-person parts, game and session held fixed. Odds ratio = how many times the odds of handing over multiply per unit of size. Read $\times 1.6$ per doubling as: a person who takes the card on 60% of 6-drone missions would be expected to take it on about 71% of 12-drone ones.

What participants said. No one spoke to the model checks themselves. One check bears on what people said, though. Several said part of their hand-building on one-type missions was to avoid the Conservative card's spare drones of unneeded types (P19, P29, P21, P13). The finding's check that drops those illogical cards is the right control for that stated reason.

7.3.3 Tactical tier: Suggest rises with mission size, and goes with the strategic choice



(a) Size measured as drones needed.



(b) With a marker on each pair at the rate for all plans in that size group.

Figure 16: Tactical tier: Suggest rises with mission size, and goes with the strategic choice

Figure 16. In the tactical planner the dose-response is smooth whichever way size is measured: drones the mission needs, drones committed, tasks, drone types or plan size. The two bar colours split plans by how the mission was allocated strategically. A mission allocated from a strategy card gets Suggest more often than one allocated by hand, at every size and on every measure: **handing over at one tier goes with handing over at the other**. Comparing the same person's missions of the same size, a card-allocated mission gets Suggest at about twice the odds ($\times$2.10). The gap is wide on small missions (70% vs 35% at $\leq$3 drones) and small on the biggest (93% vs 88%), but that narrowing is both rates approaching 100%; tested properly, the gap does not reliably shrink with size. Operators say this too: one participant (P10) described, while playing and again afterwards, that once they had taken the Aggressive card "anything but Suggest" made no sense, because a generous allocation leaves only one sensible way to assign the drones; two others (P05, P30) talked through taking a card and then pressing Suggest on the same mission. P10's reason, that it is the generous card that makes Suggest the obvious next step, does not show up across everyone: Suggest follows the Aggressive and the Conservative card at the same rate. The link is card vs hand, not which card. We use drones needed as the main measure: it is fixed by the mission, the same for every operator and what the task

badges show. Plan size and team size also contain the operator's own strategic choice: for the same mission, a card commits about 11% more drones than a hand allocation, so part of the strategic route is built into those x axes.

Key numbers.

- Within person, per doubling of drones needed: odds $\times$2.91 [2.03, 4.17]; per extra task $\times$2.04 [1.59, 2.62]; per extra drone type $\times$1.73 [1.30, 2.31]; per doubling of drones committed $\times$2.51 [1.83, 3.45]; per doubling of plan size $\times$2.08 [1.67, 2.59] (all $p < .001$)
- Allocated from a card vs by hand, same person, size held: $\times$2.10 [1.30, 3.40], $p = .003$ (drones needed); $\times$1.78 to $\times$2.12 on the other four measures, all $p < .05$
- Aggressive card vs Conservative card, same person, drones needed held: $\times$0.88 [0.45, 1.75], $p = .72$ (400 plans); raw Suggest rate after Aggressive 81% (200), after Conservative 89% (203), after a hand allocation 63% (227)
- Does the gap shrink with size? Route $\times$ size: $\times$0.76 [0.30, 1.92], $p = .56$ (drones needed); $p = .28$ to .98 on the other measures: no reliable narrowing
- Suggest pressed, card-allocated vs hand-allocated: $\leq$3 drones 70% vs 35%; 7–9 drones 85% vs 71%; 15–18 drones 93% vs 88%
- Same mission, card vs hand: $\times$1.11 more drones committed and $\times$1.12 larger plan (both $p < .001$); drones needed, tasks and types identical by construction
- Person level: people who take more cards also press Suggest more (Spearman $\rho = .37$, $p = .033$, 33 people)
- P10 in the log: Suggest on 8 of 9 Aggressive-allocated missions (never took the Conservative card) vs 6 of 9 hand-allocated

Method. **Rows**: the 630 non-recovery tactical plans that were confirmed (the planner starts empty, so pressing Suggest is a deliberate choice); 627 have a known mission size. **Measures**: drones needed (sum of the task badges' requirements), drones committed (the team the strategic choice sent), tasks (residual missions left out), drone types, plan size (team $\times$ tasks). **Size model**: logistic GEE clustered on person, size split into this plan's size minus the person's own average (within) and that average (between), game and session fixed. **Route**: the same model plus "allocated from a card" (also split within/between person), so its odds ratio compares the same person's card-allocated and hand-allocated missions of the same size. **Narrowing test**: a route $\times$ size term (ratio of odds ratios; below 1 = the gap narrows as size grows). **Same-mission check**: regression of each measure on route with one dummy per mission, so only operators who met the same mission are compared.

What participants said. Three people tie the two tiers together the way the logs do (card-allocated missions get Suggest more). P10 says repeatedly that once they have taken the Aggressive card there is 'only really one way' to assign, so Suggest is the obvious next step. P05 and P30 take a card and then press Suggest on the same mission. The same continuity runs the other way for P21, who lays out a plan in their head while building the allocation by hand and says it would 'disrupt my thought process' to switch to Suggest for the tactical step; P30 likewise finds it easier to give the assistant resources and let it optimise. The opposite pairing is said too. P18 takes a card and then plans a routine mission by hand, and P26 and P28 build strategic allocations by hand but press Suggest. On size, the reasons given are drag-and-drop effort and the number of combinations or chains.

Quotes selected for this section.

> *"Now I'm using the suggestion because I took the aggressive approach which means I know that I have enough drones." — P10 (think-aloud s2 0:13:08)*

> *"Better to use tac agent when using the aggressive strat agent" — P10 (written interview) (paraphrase)*

> *"If I look at a mission with 15 drones and I need to move two or three chains, that's just going to take me about a minute, and it's just no point. It's just a computation problem." — P11 (debrief 0:29:41)*

> *"Dragging 14 drones is going to take way more time than hitting the assign and waiting for eight seconds." — P10 (think-aloud s2 0:13:43)*

7.3.4 Card then Suggest, person by person: a broad tilt, but a few people carry much of it

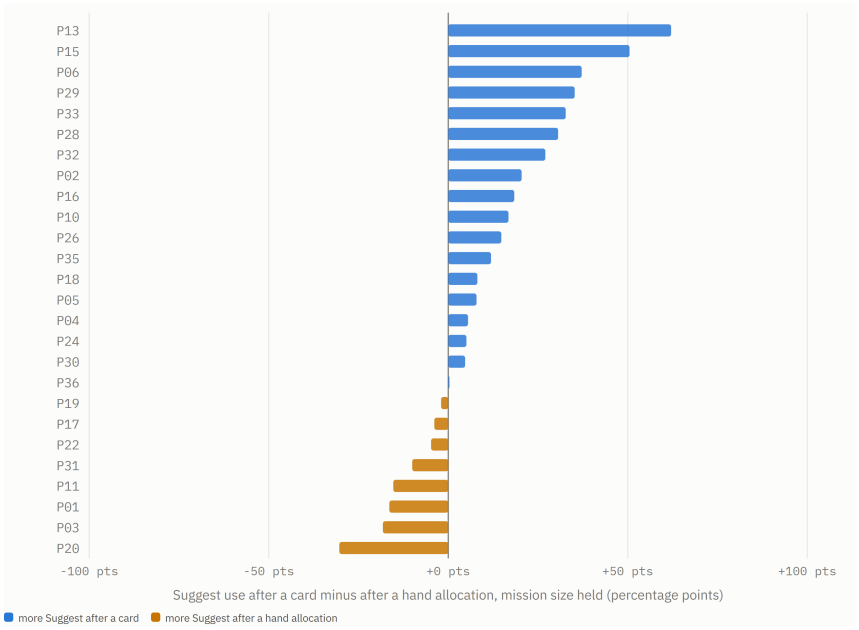
Figure 17. Is the card → Suggest link down to a few people? Partly. Each bar is one person: how much more often they pressed Suggest on missions they had allocated from a card than on ones they allocated by hand, with mission size held (the second view shows the two raw rates). Of the 26 people who did both often enough to compare, **17 lean towards Suggest after a card, 8 the other way, 1 neither**, so it is not one or two people. But it is uneven: the three biggest gaps (P13, P06, P15) make up about half of the total, and the person-adjusted odds ratio (×2.10 for everyone) falls to ×1.63 without those three (still p = .03) and to ×1.46 without the top five (p = .08). Leaving out any single person never moves it outside ×1.90–×2.40. So: a general tendency, strongest in a handful of people. Seven people cannot show a gap at all, six because they nearly always took a card before planning.

Key numbers.

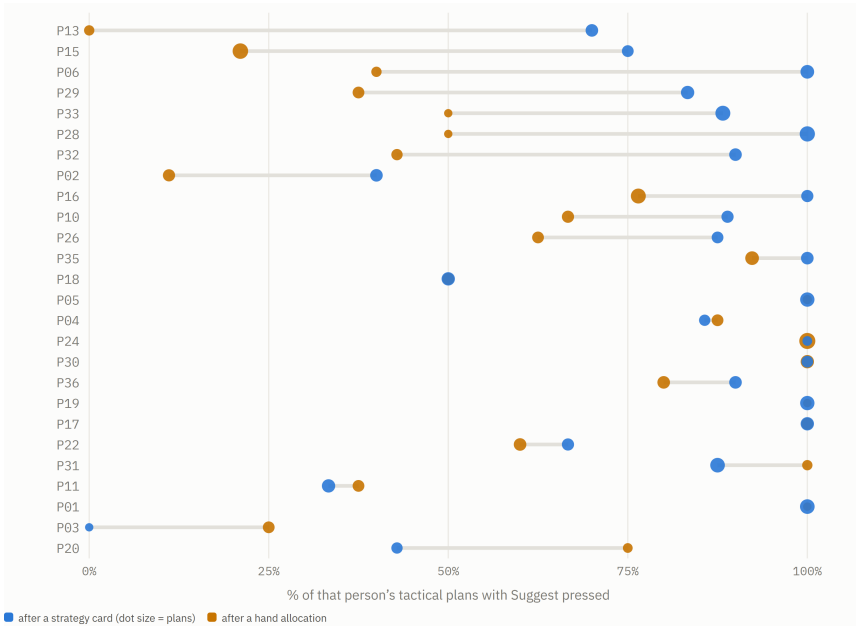
- People with ≥ 2 plans each way: 26. Gap (size held) positive 17, negative 8, none 1; sign test p = .11; mean gap +11 points.
- Card vs hand odds ratio, everyone ×2.10; leaving out any one person ×1.90 (without P15) to ×2.40 (without P11), always p < .05.
- Without the largest gaps: top 1 (P13) ×1.97, p = .006; top 3 (P06, P13, P15) ×1.63 [1.05, 2.52], p = .03; top 5 ×1.46 [0.96, 2.22], p = .08; top 8 ×1.23, p = .34.
- The three largest gaps are 52% of the summed gap. Not comparable: 6 people almost only planned after a card, 1 almost only after a hand allocation.
- Also holding fixed whether the mission needs one drone type: ×2.13 [1.30, 3.49]; and the number of tasks too: ×2.06 [1.24, 3.43], p = .005.

Method. Rows: the 627 tactical plans with a known mission size, split by how the mission was allocated (strategy card or by hand). For each person with at least two plans each way, the gap is the mean of (pressed Suggest − expected) after a card minus the same after a hand allocation, where 'expected' comes from a person-clustered model of Suggest on mission size (drones needed, within and between person), game and session, without the route; so the gap compares missions of the same size. The odds ratio is the section above's route model (card vs hand, same person, size held), refitted leaving out one person at a time, and then leaving out the people with the 1, 3, 5 and 8 largest gaps. The sign test asks whether more people lean one way than chance.

What participants said. The people at the ends of the chart, in their own words. **Strongest tilt** (P13, P15, P06, then P29): none of them talks about linking the two steps. They describe a rule for the whole mission: P15 hands over when there are 'too many decisions', P29 does one-colour missions themself and lets the assistant take multi-colour ones, and P06 drags and drops less as the work piles up. P13's hand allocations are disagreements with over-stocked cards ('why would I spend more resources'), and they say 'I always let the agent do it first' on the tactical step, yet none of their five plans after a hand allocation used Suggest. Their words do not explain their gap. **Leaning the other way** (P20, P03, P11, P31): their hand allocations are of a different kind. P20 builds by hand when the card cannot be met ('I've run out of blues') and then presses Suggest because the mission has many tasks. P03 works every allocation out first and checks the card against it. P11 rarely presses Suggest at all and does not want 'to be reduced to pressing allocate, suggest, deploy'. P31 lets Suggest get 'the majority of the way' whatever happened at the strategic step. **Only one route**: P21, who almost always allocated by hand, gives the clearest account of carrying a plan from one step to the next ('it's going to disrupt my thought process'). The card-only people (P07, P12, P14, P25, P27, P34) mostly give speed and urgency as the reason; P14 used the assistants 'for almost every allocation'.



(a) Each person's gap, mission size held.



(b) Each person's two raw rates.

Figure 17: Card then Suggest, person by person: a broad tilt, but a few people carry much of it

7.3.5 Why the rule makes sense: the card pays only on big strategic jobs; Suggest is not slower

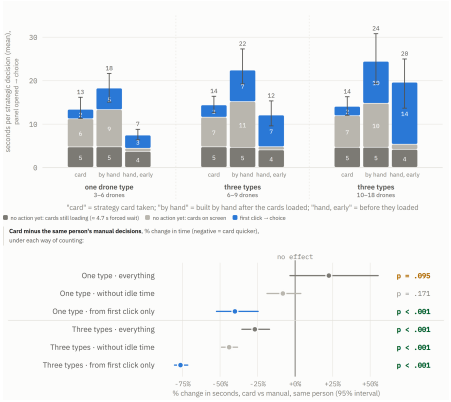
Figure 18. Each decision's time split into the time before the operator's first action and the time after it, by mission size, as stacked bars. **First action** = the first click on that decision: highlighting a card or changing a manual count (strategic; taking a card always needs a highlight first), pressing Suggest or dragging a drone (tactical). Before it, the strategic panel has two parts: the cards still loading (a forced wait of about 4.7 s; deciding by hand early can cut it short) and **idle time** with the cards on screen. Strategic: counting everything, a card is much quicker than a manual build on three-type missions (−27%) and no quicker on one-type ones (+22%, not reliable). Card-takers sit idle about as long as manual builders (7 s vs 9–11 s on average), so dropping idle time makes the card look better everywhere; on one-type missions it then ties with a manual build (−8%), and on three-type missions the card is 44% quicker. Deciding before the cards load (7 s) beats the card when everything is counted and is level with it once idle time is dropped, on one-type missions. Tactical, on all 627 plans: by hand and with Suggest take about the same planner time (29 vs 27 s on average), and the same with the idle time removed (15 vs 17 s); held to the same person and size the difference is under 1 s. On large missions hand-building takes longer (32 vs 22 s without idle time), but the size × Suggest term is not reliable. So the speed story is clear on the strategic tier, and on the tactical tier Suggest saves no planner time. Its time saving is in carrying out (next section).

Key numbers.

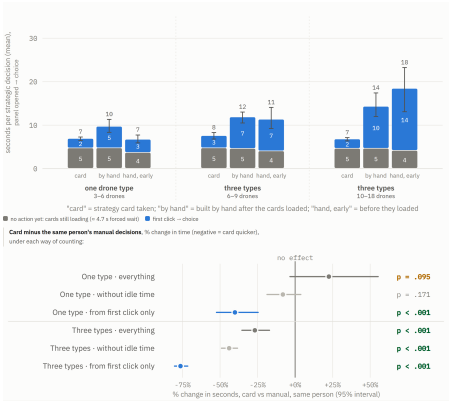
- Strategic, card vs manual, same person, % change in time: everything: one-type +22% [−3, +55], p = .09; three-type −27% [−35, −17], p < .001
- Without idle time: one-type −8% [−19, +4], p = .17; three-type −44% [−49, −38], p < .001. From the first click only: one-type −40%, three-type −76% (both p < .001)
- Strategic means (s), loading + idle + acting: one-type card 4.7 + 6.5 + 2.2 = 13.3; by hand after cards 4.7 + 8.6 + 5.0 = 18.2; before cards 3.7 + 0.7 + 3.0 = 7.4. Three-type 10–18 drones: card 4.6 + 7.3 + 2.1 = 14.0; by hand 4.6 + 10.1 + 9.7 = 24.4
- Tactical means (s), idle + building + ready → confirmed: all plans by hand 14.2 + 13.3 + 1.6 = 29.1 (144) vs Suggest 10.6 + 10.9 + 5.8 = 27.2 (465); large missions 17.9 + 30.0 + 1.9 = 49.8 (22) vs 10.0 + 15.3 + 6.2 = 31.5 (159)
- Tactical, Suggest minus by hand, same person, drones needed held: everything −2.8 s [−9.0, +3.5], p = .39; without idle time +0.2 s [−2.7, +3.0], p = .92; Suggest × size (without idle) −1.3 s per doubling, p = .47
- Tactical plans idle for more than 10 s: 54% had another mission's strategic panel opened before the first action (176 plans); all plans 22%

Method. **Rows**: strategic, the 606 card decisions (the last panel opening before the choice, as everywhere in this finding); tactical, the 627 non-recovery plans with a known mission size (kept Suggest plans vs hand-built; the 18 edited Suggest plans are in the table only). **Segments**, from the event log: strategic, panel opened → first click (split at the moment the cards appeared: before = "cards still loading", after = "idle, cards on screen") and first click → choice; tactical, planner opened → first Suggest press or drag ("idle"), → plan ready (last drag, or the end of the one-entry-per-second Suggest fill), → confirmed. These are the same phases as the next section, which adds carrying out. Bars are means (means add up, medians do not), whiskers 95% intervals of the bar's total from resampling people. Sizes: strategic, one-type missions (3–6 drones) and three-type missions of 6–9 and 10–18 drones; tactical, drones needed ≤ 6, 7–9, 10+. **Models**: strategic, Gaussian GEE on log seconds, card × three-type, all manual decisions pooled as the comparison, the person's own card rate, drones, game, session and minute held fixed, run once per way of counting; tactical, GEE on seconds, Suggest split within/between person, drones needed, game and session fixed. **Caveat**: idle time mixes reading the cards or plan with being busy elsewhere, and the log cannot tell them apart. For tactical plans idle for more than 10 s, 54% had another mission's strategic panel opened in that time, so long idle time is often attention elsewhere.

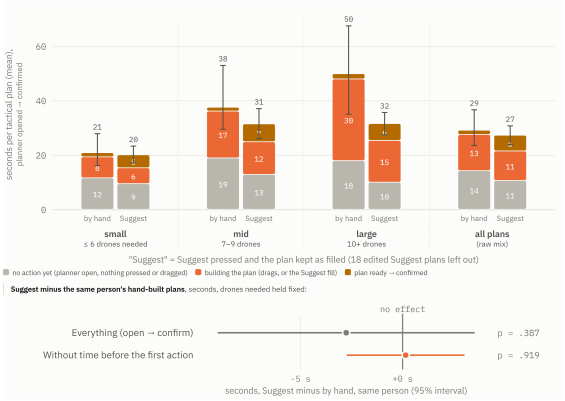
What participants said. Speed is the most common reason given for using the assistants: 15 of 33 coded C04. Many add that the plan is not better, just faster (P04, P17, P11). The card's loading wait is the main complaint. Six were coded C06, and in the think-aloud P18, P21, P29 and P32 go manual because they 'can't wait' for the cards,



(a) Strategic tier, all time counted.



(b) Strategic tier, idle time removed.



(c) Tactical tier, all time counted.

Figure 18: Why the rule makes sense: the card pays only on big strategic jobs; Suggest is not slower

mostly on small missions. That matches the finding that the card pays off only on big strategic jobs. P20 first said the strategic assistant saved them more time, then corrected themself on remembering its loading stage.

Quotes selected for this section.

> *"I think the key strength when I using the agent is it can help me to save the time and energy to those specific deployments" — P28 (debrief 0:32:10)*

> *"the point is, it takes me so much time to untangle that, that maybe it probably will not be faster if I do it myself manually. So as long as the AI makes sure that everything is covered once." — P17 (think-aloud s1 0:07:07)*

> *"Manually can't wait for the processing, to be honest." — P21 (think-aloud s1 0:09:32)*

> *"I think even if the tactical agent gave me a dodgy suggestion and the mission failed and I lost points, I probably would have gained more points from the time saved..." — P11 (debrief 0:30:11)*

> *"if you use human's knowledge to do it, it will take a longer time, much longer time. But in this situation, in some situations, the time is very limited. So the AI will give much quickly." — P26 (debrief 0:30:28)*

> *"Mainly speed as was faster than doing it manually" — P07 (written interview)*

> *"The top advantage in this situation was the speed of response." — P31 (debrief 0:29:10)*

> *"More about speed than accuracy (tactical is faster)" — P34 (written interview)*

> *"I think I was definitely a bit slower than the, than the algorithm there." — P22 (think-aloud s1 0:07:11)*

> *"You know, it's quicker to get a suggestion, but it is less clicking, and it seems to do what I would do." — P32 (think-aloud s1 0:10:07)*

> *"So, it just seems quicker to just send some myself than wait for the recommendation. I might just be impatient." — P32 (think-aloud s1 0:03:09)*

> *"So, when we are trying to do with the tactical screen, I find the agent is quite slow." — P27 (debrief 0:29:10)*

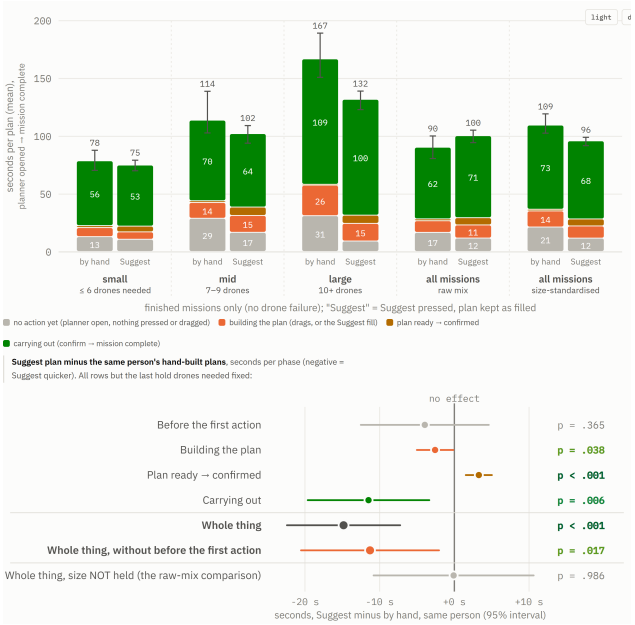
> *"Yeah, when I located, I think sometimes the AI will take a little longer. And I just mess up my strategy" — P26 (debrief 0:28:37)*

7.3.6 Where the tactical time goes: hand-built and Suggest plans are close, with a small edge to Suggest in carrying the plan out

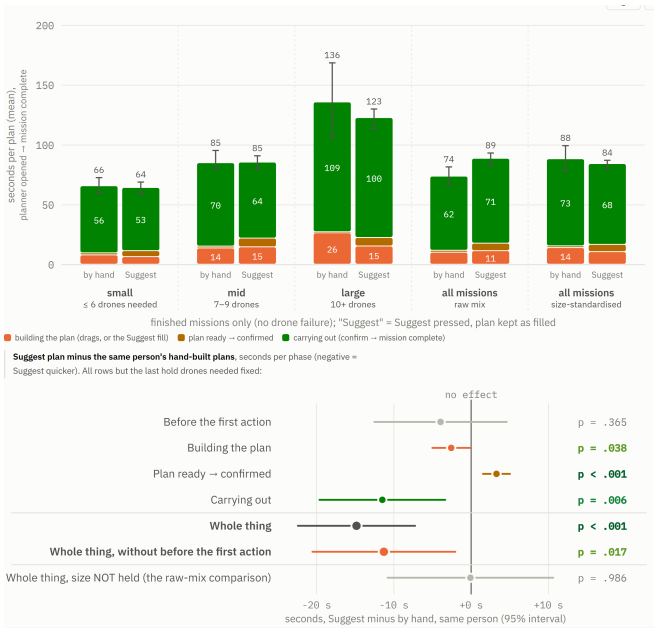
Figure 19. The headline is how close the bars are. The operator's own time on a plan is the same either way: Suggest saves about 3 s in building and gives it back waiting between ready and confirmed (net about +1 s). The only reliable difference is in carrying the plan out, about 11 s sooner for a Suggest plan (roughly a tenth of a mission's time), so the assistant's plans are very slightly better rather than much faster to make. Read the large group with care: only 6 hand-built large plans finished cleanly. The raw "all missions" bars look the other way (hand-built 90 s, Suggest 100 s); that is a mix effect, not a contradiction: 78% of finished hand-built plans are small missions, against 42% of Suggest plans. Given the same mix of sizes, hand-built plans take 109 s and Suggest 96 s. So the common feeling that the assistant is "faster" is only weakly supported: it does not save the operator time, and the plans it makes finish a little sooner. Each tier's use still looks sensible: Suggest is used where plans are big, which is where its small edge in execution adds up. And relying on the assistants more does not make an operator do better: within a person, card missions are completed in full no more often than their own hand-built ones (odds $\times 0.81$, p = .27; raw 60% vs 62%), and session score is flat in sessions where they rely more (-0.10 SD per 25 points of card uptake, p = .61). The ready $\rightarrow$ confirmed gap is when operators turn to other missions: 45% of Suggest plans had another mission's strategic panel opened while the planner was open, against 19% of hand-built ones.

Key numbers.

- Suggest minus hand-built, same person, drones needed held (315 finished plans): before the first action -4.0 s [-12.5, +4.6], p = .37; building -2.6 s [-5.0, -0.1], p = .038; ready $\rightarrow$ confirmed +3.3 s [+1.5, +5.0], p < .001; carrying out -11.5 s [-19.6, -3.4], p = .006; whole -14.8 s [-22.4, -7.3], p < .001



(a) All four phases.



(b) Without the time before the first action.

Figure 19: Where the tactical time goes: hand-built and Suggest plans are close, with a small edge to Suggest in carrying the plan out

- Whole thing without the time before the first action: −11.3 s [−20.5, −2.1], p = .017; whole thing with size not held: −0.1 s [−10.8, +10.6], p = .99
- Means (s), hand vs kept Suggest: small 78.5 vs 74.8 (63 vs 97 plans); mid 113.6 vs 102.0 (12 vs 59); large 166.6 vs 131.6 (6 vs 75); all, raw 90.2 vs 100.2; all, size-standardised 109.3 [102.7, 119.4] vs 95.7 [91.9, 99.1]
- Size mix of finished plans: hand-built 78% small, 15% mid, 7% large (mean 5.7 drones needed); Suggest 42%, 26%, 32% (mean 8.8)
- All 627 plans, first three phases: before −2.9 s, p = .29; building −3.2 s [−6.2, −0.1], p = .04; ready → confirmed +3.3 s [+2.0, +4.6], p < .001
- Carrying out, including missions with a drone failure (378): −8.1 s [−17.1, +0.9], p = .08: same direction, less certain
- Projected completion vs the assistant's own plan at open, median: hand-built +10.3 s [3.7, 28.3] (144 plans), kept Suggest +1.6 s [0.8, 4.0] (468), edited Suggest +38.7 s (18)
- Interleaving, round-1 specification re-run: odds ×4.83 [2.24, 10.40], p < .001 (630 plans, 33 people); raw 45% vs 19%; 28% vs 11% actually allocated another mission
- Operator time, building + ready → confirmed, Suggest minus hand-built, same person, size held: −2.6 s + 3.3 s ≈ +0.7 s (negligible)
- From the retired "who does better" finding: within a person, card missions vs own manual missions, completed in full odds ×0.81 [0.55, 1.18], p = .27 (raw 60% vs 62%); session score per +25 points of card uptake −0.10 SD [−0.47, 0.27], p = .61. Between people, heavier card users complete more (×1.32 per 25 points, p = .004): the difference is in the people, not the advice

Method. **Rows**: confirmed non-recovery tactical plans. **Phases** from the event log, the same as in section 4: planner opened (*tactical_opened*); first action = first Suggest press or first drag; plan ready = last drag, or for a Suggest plan the end of its fill (last press + 1 s per drone-task entry), whichever is later; confirmed (*tactical_confirmed*); mission complete (*mission_completed*). **Carrying out** can only be measured when the mission finished in full with no drone failure: 315 of 630 plans; the chart shows those (81 hand-built, 231 kept Suggest; edited Suggest plans left out, 3 survive). Size groups by drones needed: small ≤ 6, mid 7–9, large 10+. Bars are means, whiskers 95% intervals of the bar's total from resampling people. **Size-standardised**: each route's mean in each size group, weighted by the share of all finished plans in that group (small 51%, mid 23%, large 26%), so both routes get the same mix; this answers "how long would each route take on the same missions?". Only 6 large hand-built plans finished cleanly (5 people), so that bar and the standardised hand-built bar are the least certain. **Model**: Gaussian GEE of seconds in each phase on "Suggest pressed" (split within/between person), log drones needed, game and session. Seconds add up, so the phase differences sum to roughly the whole. The bottom row drops drones needed from the model: still within person, but with size not held the difference vanishes (−0.1 s), which shows the raw reversal comes from size, not from who uses which route. **Projected completion** and **interleaving** as before.

7.3.7 What operators say

Ten of 33 state the rule unprompted (C03: small/simple myself, big/complex to the assistant); 15 value the assistant for speed rather than accuracy (C04, the most common reason given); 8 describe working on something else while it runs (C05). The behaviour matches the words. People who state the rule press Suggest less overall (59% vs 86% of plans) and their size slope is, if anything, steeper on the tactical tier (×1.43 steeper, p = .10) — not reliably different from everyone else's, so stating the rule is not what makes someone follow it.

Key numbers.

- C03 10 of 33; C04 15 of 33; C05 8 of 33
- C03 vs others, size slope ratio: strategic ×1.28 [0.69, 2.38], p = .43; tactical ×1.43 [0.93, 2.20], p = .10

Method. Codes from the qualitative coding (docs/analysis/qual/codes_matrix.csv); excerpts are pulled by code on the page. Slope comparison: the within-person size model with a size × C03 term (ratio of the two groups' odds ratios).

What participants said. 10 of 33 state the size rule unprompted (C03), 15 value the assistant for speed rather than accuracy (C04) and 8 work in parallel (C05). The clearest statements tie all three together: small jobs are quicker by hand, big ones are a 'computation problem', and the wait frees attention for the next mission. Dissent from P30, P20 and P29 is in the first section. P32 said their use followed workload, not how hard a mission looked.

7.3.8 In their words: working in parallel while the assistant plans

To close the finding: what people said about pressing Suggest and turning to other work while it fills in. It sits here rather than under the time-phase chart because it is about how people used the assistant, not about the chart's numbers.

What participants said. 8 of 33 describe starting Suggest and working elsewhere while it fills in (C05). The think-aloud shows more of it (P19, P32, P36). Nobody said a Suggest plan finishes the mission sooner. Two people said the opposite about plan quality: P31 adjusted a Suggest plan they thought visited tasks in a poor order, and P21 found it lacked the cross-task redundancy they built themself. P36 names a cost of handing over: if you did not build the plan, you are playing catch-up when something fails.

7.4 Manual allocations are anchored on the cards — and the 'trimming small missions' was mostly dropping the illogical spare

Summary. When operators allocate by hand after seeing the cards, their numbers track the cards: one more drone on the Conservative card means about half a drone more in the manual build, beyond what the mission needs. Operators who decided before the cards appeared show no such link — a clean control, so this is anchoring, not both tracking the mission. Manual builds sit at or below the cards (44% below both, 33% equal to one, 14% between, 8% above). The earlier "trim small missions, pad big ones" mostly dissolves: on one-type missions the 'trim' is almost entirely dropping the Conservative card's spare drones of types the mission does not need.

Strength of evidence (register rating): moderate.

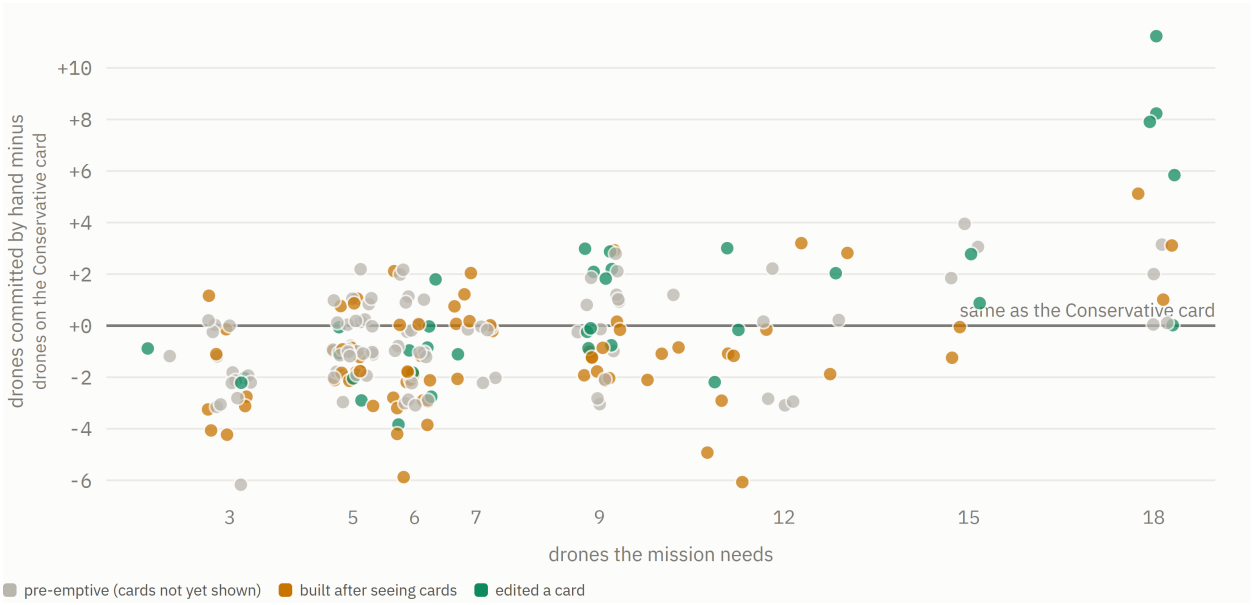
7.4.1 Each manual decision against the Conservative card

Figure 20. Every manual allocation (209) as one dot: drones committed minus drones on the Conservative card, against the drones the mission needs. Switch to "needed types only" and the band of one-type missions moves up to zero: on the types the mission actually needs, manual builds on small missions match the Conservative card (+0.05 drones on average); the −1.2 drones on all types is the card's spare of each unneeded type being left off. On big three-type missions manual builds sit about one drone above the Conservative card (and still far below the drones the mission lists, since drones are reused across tasks in series).

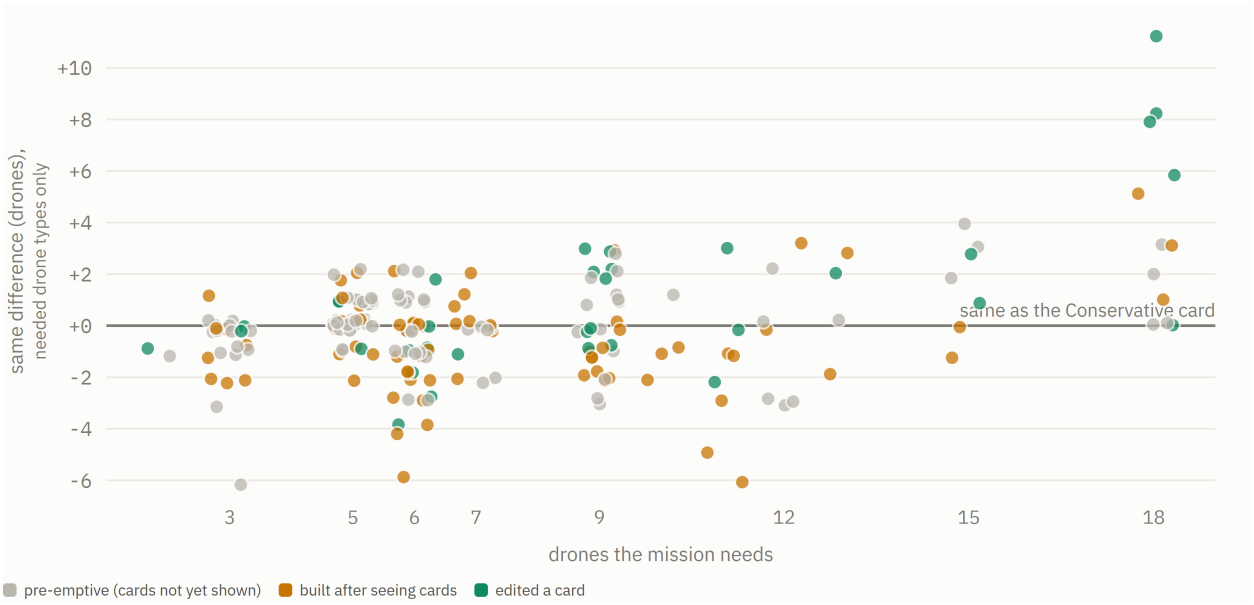
Key numbers.

- Mean manual − Conservative: −0.55 drones overall = −0.51 on unneeded types + −0.04 on needed types
- One-type missions: −1.17 overall, +0.05 on needed types, −1.22 on unneeded; three-type with 10+ drones: +0.95
- Per doubling of mission size the difference grows by +1.51 drones [0.53, 2.49] (after seeing the cards) and +1.09 [0.47, 1.72] (pre-emptive): the size pattern is just as strong in people who never saw the cards, so it is not a reaction to the card
- Needed types only, after seeing the cards: +0.93 per doubling [−0.18, 2.04], p = .10

Method. Rows: strategic choices of type manual. The cards' true asset counts per colour come from the last *strategic_modal_opened* before the choice; the operator's counts from *assetsChosen*. "Needed types" = colours with a primary requirement in the mission. Kinds: pre-emptive (decided before the cards loaded — the cards existed in the log but were never on screen), edited a card (started



(a) All drone types.



(b) Needed drone types only.

Figure 20: Each manual decision against the Conservative card

from a card, then changed it), built after seeing the cards. Size slope: Gaussian within-person GEE of the difference on log2 drones needed.

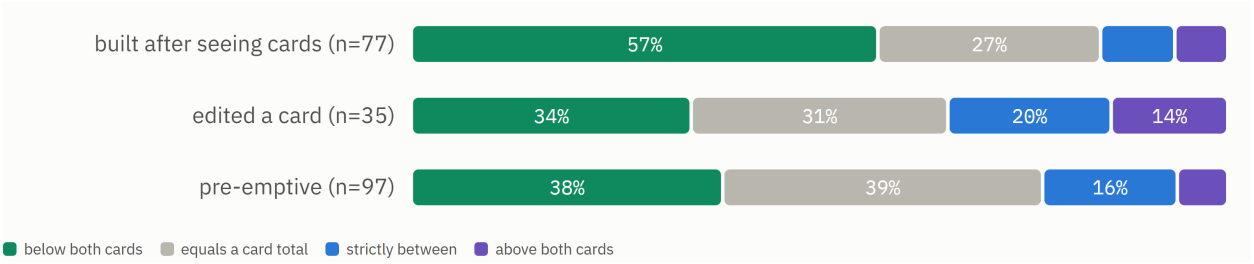
What participants said. The think-aloud shows what was happening on the cards. People trimming small missions mostly said they were removing the card's drones of types the mission did not need (P10, P19, P21, P29, P13). Others cut one or two drones to protect the reserve (P24, P28). P20 shows the anchoring directly: they take the Aggressive card's numbers and does 'one less than each of those'. On big missions short of drones, P10 'send[s] everything I have', above the cards. One dissent: P29 found their own leaner builds sometimes had to be aborted.

Quotes selected for this section.

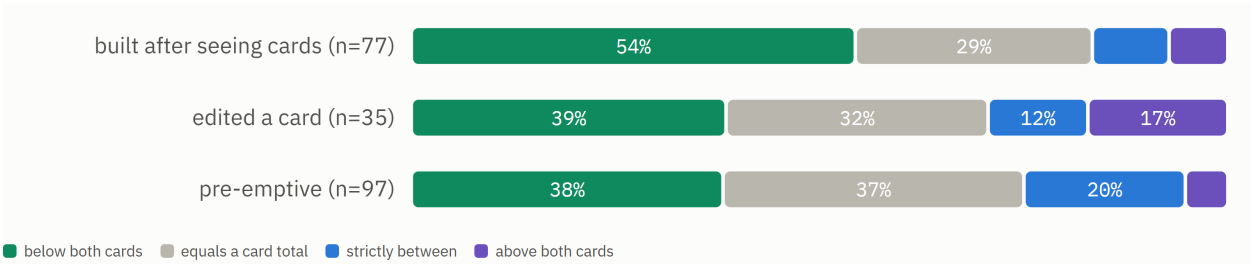
> *"All right. Let's just send everything I have. [. . .] Edit. Give me five red ones. Five green ones. Hit it." — P10 (think-aloud s2 0:17:46)*

> *"I'm trying to reserve more and more drones in the base, rather than send them out to finish the task quickly, because I cannot predict when the next task will appear." — P28 (debrief 0:28:05)*

7.4.2 Where manual builds sit relative to the two cards



(a) As built.



(b) Balanced for mission size, drone types, the gap between the cards, game, session and minute.

Figure 21: Where manual builds sit relative to the two cards

Figure 21. Mostly below both cards or equal to one of them, rarely above. Pre-emptive builds — made without seeing the cards — land equal to a card total more often than builds made after seeing them (39% vs 27%), so 'equal to a card' is partly just the obvious allocation, not copying. Part of that gap is the missions: pre-emptive builds were on smaller, more often one-type missions. Balanced for the mission (second view), it is 37% vs 29%, too close to tell apart.

Key numbers.

- All 209: below both 44%, equal to a card 33%, strictly between 14%, above both 8%
- Built after seeing (77): 57% below, 27% equal, 9% between, 6% above; pre-emptive (97): 38%, 39%, 16%, 6%
- Mission mix: pre-emptive builds needed 6.9 drones on average (after seeing 7.6, edited 9.3) and 52% were one drone type (38%, 23%).

- Balanced: after seeing 54% below, 29% equal, 9% between, 7% above; pre-emptive 38%, 37%, 20%, 5%; edited a card 39%, 32%, 12%, 17%.
- Pre-emptive minus after seeing, balanced: equal to a card +8 points [−10, +23] (raw +12); below both −16 [−33, +4] (raw −19).

Method. Each manual total compared with the two card totals (all types): below both, equal to either, strictly between, above both. Counts per kind of manual decision. **Balanced view**: one logistic model per position (below both / equal / between / above) on the kind of build + drones needed + number of drone types + the gap between the two card totals + game + session + minute. Every one of the 209 builds is predicted as each kind in turn and averaged, so the three kinds are compared on the same mix of missions; the four shares are rescaled to 100%. Intervals: people resampled 1,000 times, models refitted. The person is not held fixed (too few builds each).

What participants said. 6 of 33 wanted something between the two cards (C08), and P17 says in play that they aim 'between' them. Most of what people say describes trimming a card or matching it, which fits 'below or equal'. 'Equal to a card' was often their own count arrived at first: P30 and P03 work out a number while the cards load and see whether a card matches, and P32 stopped waiting once the cards matched their count.

Quotes selected for this section.

> *"But whenever I clicked manual, it's either been for the cause of setting too much, or I want it to be somewhere between those two levels." — P17 (debrief 0:23:38)*

> *"I felt like sometimes I could find like a middle ground that had like maybe a little bit of gambling in there, essentially, that you wouldn't have the necessary coverage." — P22 (debrief 0:28:57)*

> *"Sometimes the same, and often too extreme (far too aggressive and far too conservative). Manually overrode most of the time" — P16 (written interview)*

7.4.3 What operators say about building by hand

Four accounts of manual allocation sit alongside the numbers. Holding drones in reserve (C07, 6 people) fits the 'below both cards' mass — though in the logs these six are not lower than others on needed types (+0.36 vs the Conservative card). Wanting a middle ground or more options (C08, 6 people) fits the wish, not the behaviour: only 10% of their manual builds land strictly between the cards. Treating a card as a first draft (C09, 4 people) shows in the logs — 16 of their 33 manual decisions are edited cards. Testing the assistant against themselves (C10, 5 people) fits slower, deliberate manual builds (round 1: 20 s vs 12.6 s). All four describe manual as working *with* the advice, not rejecting it.

Key numbers.

- C07: 55 manual decisions from 6 people, mean +0.36 vs Conservative on needed types, 21 edited cards
- C08: 40 from 5 people, 10% strictly between the cards
- C09: 33 from 4 people, 16 edited cards
- C10: 40 from 5 people

Method. Codes from docs/analysis/qual/codes_matrix.csv; per-code manual numbers from the same rows as the dot chart (small groups, descriptive only).

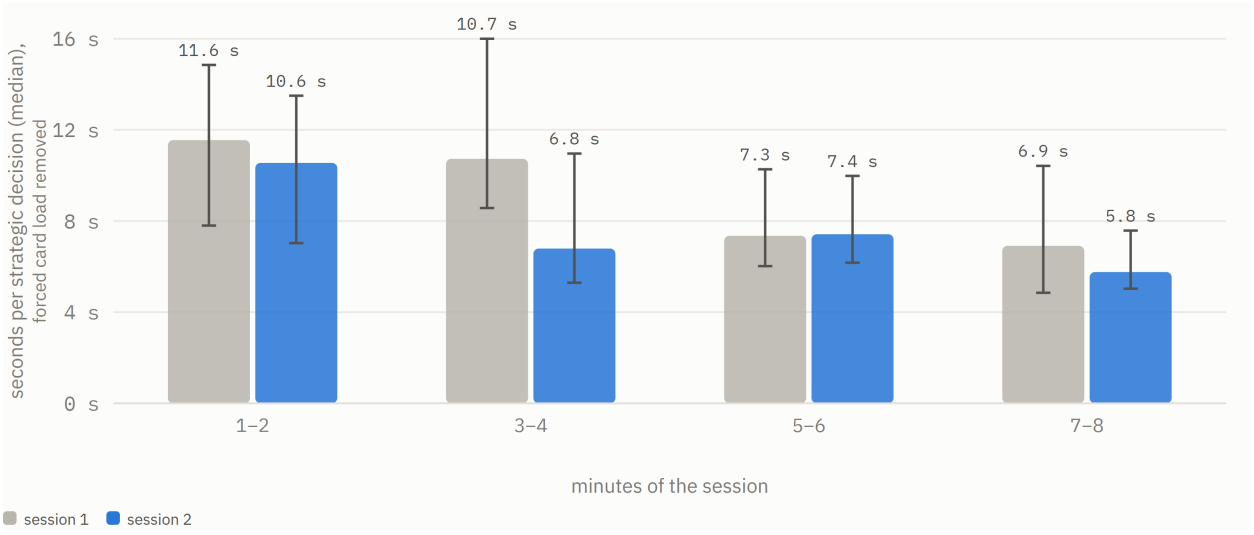
What participants said. The four accounts are all present in people's own words. They hold drones back (C07, 6), want a middle ground (C08, 6), treat a card as a draft (C09, 4: 'you can fix bad, you can't fix nothing') and test themselves against the assistant (C10, 5). The think-aloud shows the draft habit more widely than coded: P10, P14 and P31 edit cards, and P21 and P31 use Suggest as a first draft on the tactical tier. The one person who tested themself said it ended in favour of the assistant (P29).

7.5 Practice makes operators faster, not better at choosing when to hand over

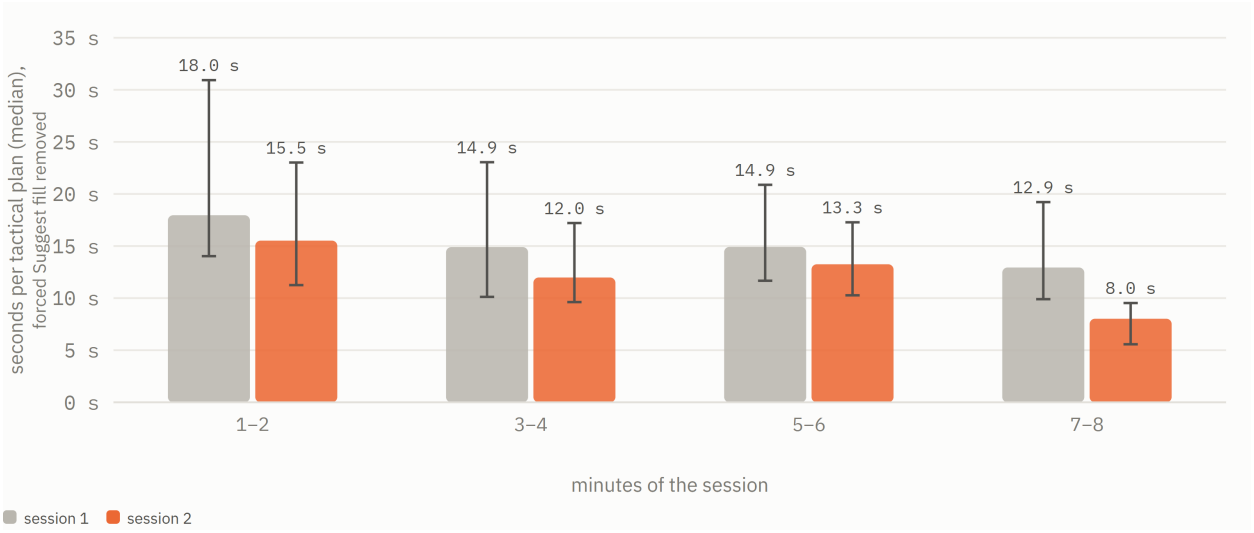
Summary. Decisions speed up within each session: about 6% faster per minute on both tiers, and not because the queue gets longer. On the strategic tier most of that gain is gone by the start of the next session (median 11.6 s at the start of session 1, 6.9 s at its end, 10.6 s at the start of session 2), so it reads as a within-session warm-up more than durable learning. On the tactical tier session 2 starts about where session 1 ended. The size rule does not sharpen with practice: the big-minus-small hand-over gap is the same or smaller in session 2. The six people who say they got better did improve their scores (all six, against 37% of others), but they did not come to use the assistants more, and across everyone a bigger score gain does not go with more use.

Strength of evidence (register rating): moderate.

7.5.1 Faster within a session; on the strategic tier the gain mostly resets by the next



(a) Strategic tier.



(b) Tactical tier.

Figure 22: Faster within a session; on the strategic tier the gain mostly resets by the next

Figure 22. Median net decision time through each session, by pair of minutes. On the strategic tier decisions get much faster through a session (11.6 → 6.9 s in session 1) and then start nearly as slow again (10.6 s), before speeding up

once more (5.8 s). The model keeps about half of the gain across the break, but uncertainly. On the tactical tier the first session's speed-up is weaker and session 2 starts about where session 1 ended, then speeds up again. Holding the number of waiting missions fixed does not remove the within-session speed-up, so it is not just time pressure from a longer queue.

Key numbers.

- Strategic, medians: session 1 minutes 1–2 11.6 s [7.8, 14.5] → minutes 7–8 6.9 s [4.9, 10.2]; session 2 10.6 s [7.0, 13.5] → 5.8 s [5.0, 7.6]
- Strategic, model: end of session 1 −39% [−51, −24]; start of session 2 −20% [−37, +2], p = .07; start of session 2 vs end of session 1 +31% [0, +72], p = .053 (slower again)
- Tactical, medians: 18.0 s [14.0, 30.9] → 12.9 s [10.1, 19.2]; session 2 15.5 s [11.3, 21.9] → 8.0 s [5.6, 9.6]
- Tactical, model: end of session 1 −24% [−45, +4], p = .09; start of session 2 vs end of session 1 +2% [−33, +56], p = .92; session 2 slope −7% per minute vs −4% in session 1 (difference p = .26)
- Within-session slope with the queue held fixed: strategic −6% per minute, tactical −8% per minute (both p < .001)
- Overall (unchanged): strategic −5.7% per minute [−7.8, −3.6], session 2 vs 1 −19% [−30, −6]; tactical −5.7% per minute [−9.1, −2.2], session 2 vs 1 −24% [−36, −9]

Method. **Net time** = seconds to decide minus the forced wait (the card reveal for card choices; 1 s per entry for Suggest plans). Bars: medians of all decisions made in minutes 1–2 and 7–8 of each session, with 95% intervals from resampling people. **Model**: Gaussian GEE on log net time, clustered on person, with minute of session, session 2 and their interaction, plus route and mission size; game held fixed. From it: the end of session 1, and the start and end of session 2, each relative to the start of session 1, and the start of session 2 relative to the end of session 1 (a positive value = slower again after the break). A log-scale coefficient converts to "% faster". The start of a session is also the start of a game (first missions, short queue), so part of a within-session speed-up may be the session's own rhythm rather than learning.

What participants said. People describe getting faster and finding their rhythm: P10 'find your stride', P30 'got into the rhythm', P35 'getting familiar', P11 'get the hang of it quite quickly'. Several state a lesson from session 1 at the start of session 2: P22 'don't send everything straight away', P06 'the task will get a lot pretty quickly'. P18 is a partial dissent: the second session felt better prepared but more stressful.

Quotes selected for this section.

> *"there was definitely an improvement further throughout session one, and in session two, you really find your stride" — P10 (debrief 0:22:57)*

> *"i think the second scenario i used more personalized um and why i think it was easier because i had to i sort of got into the rhythm of having to anticipate other problems" — P30 (debrief 0:23:52)*

> *"Right. So now we also know from the last one, don't send everything straight away. Cause otherwise we end up, we're waiting for drones." — P22 (think-aloud s2 0:17:11)*

> *"And I could definitely feel that there is absolutely a learning effect." — P10 (debrief 0:22:39)*

> *"I think with having some experience, yeah, I was able to quickly understand this" — P05 (think-aloud s1 0:11:14)*

> *"It was interesting. It was, you get the hang of it quite quickly, I think." — P11 (debrief 0:26:32)*

7.5.2 Does 'I got better with practice' match the logs? For score yes; not for speed or assistant use

Figure 23. Six people say they got better, faster or more confident (C22: P10, P11, P15, P21, P30, P32). All six raised their within-game score from session 1 to 2, against 10 of 27 others. Their tactical planning sped up more than others' on average (5 of 6 faster), but not reliably; their strategic decisions did not speed up more. They also **did not come to use the assistants more**: strategy cards taken changed by −1 point (others −4) and Suggest by +11 points (others

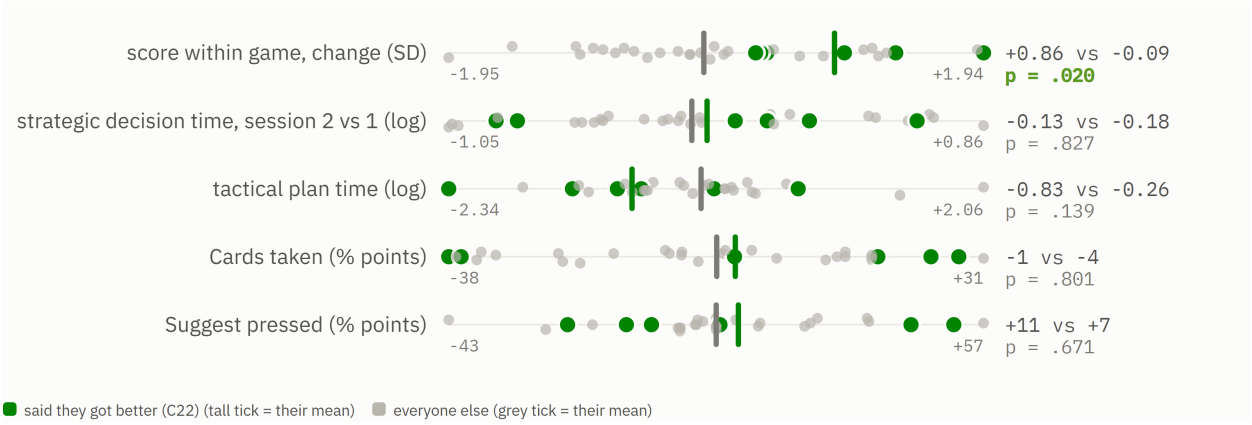


Figure 23: Does 'I got better with practice' match the logs? For score yes; not for speed or assistant use

+7), with 3 of the 6 using each assistant more in session 2. Across all 33 people, a bigger score gain does not go with a bigger rise in either kind of use. So "got better" is right about outcomes, but it is not "relied on the assistants more".

Key numbers.

- Score change: C22 +0.86 SD vs others −0.09, p = .020
- Tactical time change (log): −0.83 vs −0.26, p = .14; strategic: −0.13 vs −0.18, p = .83
- Change in cards taken (game-adjusted): C22 −1 point vs others −4, p = .80; Suggest pressed: +11 vs +7 points, p = .67
- Used more in session 2: cards 3 of 6 C22 vs 12 of 27 others; Suggest 3 of 6 vs 19 of 27
- Everyone: score change vs change in card-taking $\rho = -.07$ (p = .71); vs change in Suggest $\rho = -.18$ (p = .32)

Method. Per person: session 2 minus session 1 in (a) score standardised within game, (b) log median net strategic time, (c) log median net tactical time, (d) share of strategy cards taken and (e) share of plans with Suggest pressed. For (d) and (e) each session's rate is first taken relative to that game's average rate across everyone, so playing a different game in session 2 does not count as a change of habit (raw changes agree). Group mean vs the other 27, two-sided permutation p (20,000 reshuffles). The across-everyone check is a Spearman rank correlation between score change and change in use (33 people). With P01/P02 left out of the game means, P21's score change is slightly negative (five of six, not six).

What participants said. 6 of 33 say they got better (C22). Their reasons are understanding the task (P32), experience (P11, P15) and optimising how they worked with the assistants (P21), not leaning on the assistants more. Others outside C22 describe practice changing how much they use the assistants, in both directions. P06 and P31 relied more and checked less, P28 expected to rely less once familiar, and P35 used the cards less after spotting errors. That mixed picture fits the logs: no overall rise in use with practice.

7.6 A busier moment does not push people onto the assistants — even though they say it does

Summary. Comparing each operator with themselves, having more missions waiting does not change how often they take a strategy card or press Suggest: the line is flat once we account for what each decision was (mission size and drone types), when it came in the session, which game it was and who made it. This is "no evidence of an effect", with enough data to rule out large effects (strategic: at most about +15 points of card-taking going from 0 to 5 waiting). Nine of 33 said in the debrief that they hand over when overloaded; their own behaviour moved with the queue no more than anyone else's. In play they said overload-type things at slightly busier moments, but their reliance around those moments did not change.

Strength of evidence (register rating): a null result.

7.6.1 Queue depth against reliance — raw, and adjusted

Figure 24. Each point is the share of decisions where the operator took a strategy card (or pressed Suggest), grouped by how many OTHER missions were waiting for a strategic decision at that moment. Whiskers are 95% intervals. The dotted vertical line marks where the queue list runs out of room (about 3.5 cards visible at rest, counting the one being opened): beyond it, waiting missions are only on the strategic map. Nothing changes at that line either.

Key numbers.

- Strategic raw: 67% (0 waiting), 70%, 68%, 58%, 51%, 71%, 50% (6+); adjusted: 66%, 65%, 65%, 61%, 67%, 74%, 59% — all intervals overlap.
- Tactical raw: 73%, 79%, 83%, 79%, 76%, 73%, 67%; adjusted: 76%, 78%, 77%, 75%, 82%, 78%, 78%.
- The queue rises with the minute (Spearman $\rho = .49$) but hardly with mission size ($\rho = -.08$): the adjustment mostly removes WHO reaches deep queues and WHICH GAME they are in.
- Deep queues are a Strategic Heavy phenomenon: 90–97% of decisions at 4+ waiting are in those two games.

Method. **Rows:** 564 strategic decisions on original (non-residual) missions and 630 tactical plans, 33 people (P09 excluded). **Queue** = other missions that had arrived and not yet been given a strategic choice when the decision panel opened, rebuilt from the event log (agrees with the logged count for 100% of decisions). **Expected**: a logistic model predicts each decision's chance of a card (or Suggest) from four things only — the mission (drones needed, number of tasks, number of drone types), the minute of the session, the game (one of four fixed scenarios), and the person (their own rate on all their OTHER decisions, so a decision never predicts itself). The queue is deliberately left OUT. **Adjusted** for a bin = the overall rate + the average of (observed − expected) in that bin. In plain words: 'given who made these decisions, on which missions, at what point in which game, how much more or less did they rely on the assistant than we would expect?' If load mattered, the deep-queue bins would sit above what their missions and people predict; they do not. **Worked example (strategic, 4 waiting, 41 decisions):** raw 51% looks like a drop from 67% at 0 waiting. But 90% of those decisions were in the two Strategic Heavy games, on missions averaging 6.9 drones (vs 8.7) with fewer three-type missions (54% vs 66%), made by people whose usual rate is 56% (vs 66%). The model expects 50%; observed is 51%; adjusted $\approx$ 67%. **Intervals**: people resampled with replacement 400 times, the expected-model re-fitted each time, so the interval includes the uncertainty of the adjustment itself.

What participants said. 9 of 33 say they hand over more when busy (C11). Almost all of them mean the harder session or 'when workload gets higher', not the moment a queue builds. One clear voice goes against it in play: P18, asked during a stressful session, said 'I think less… I'm too focused to look at it'. P21 felt pressure pushed them past the strategy card rather than onto it. P29 and P05 describe pressure as the thing that tips them towards the assistant. That is the claim the adjusted data do not bear out.

Quotes selected for this section.

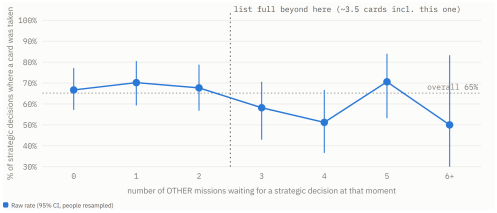
> *"When workload gets higher you rely on AI much more" — P05 (written interview)*

> *"You can have a lot of tasks to be overloaded and try and offload some of that work to AI and then for the very complex stuff, maybe you don't." — P20 (debrief 0:28:46)*

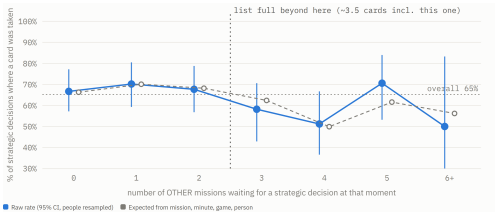
> *"So I think if, at first glance, I felt really overwhelmed by the amount of drones to be assigned. I would use the assistant to sort of, yeah." — P18 (debrief 0:26:17)*

> *"there was no more active missions coming in, so therefore I could take more time and ensure that it was correct. But perhaps if I was more time pressured, then I'd ask it to suggest." — P05 (think-aloud s1 7:31)*

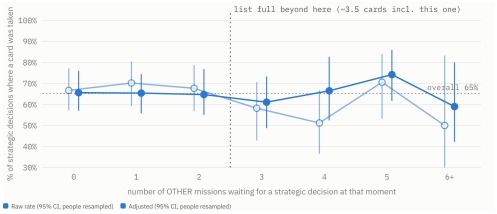
> *"But in an urgent situation and the number of the swarm is not enough, so I need to consider about the strategies and I think I need the help of AI assistant." — P35 (debrief 0:28:19)*



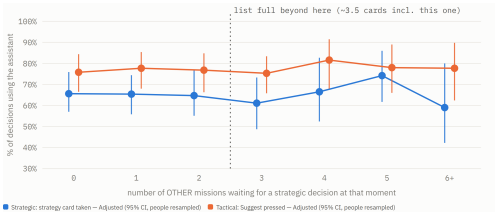
(a) Strategic tier, raw rates.



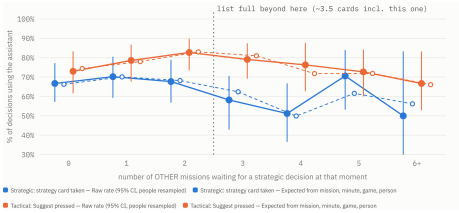
(b) Strategic tier, raw with the expected rate (dashed).



(c) Strategic tier, adjusted.



(d) Both tiers, adjusted.



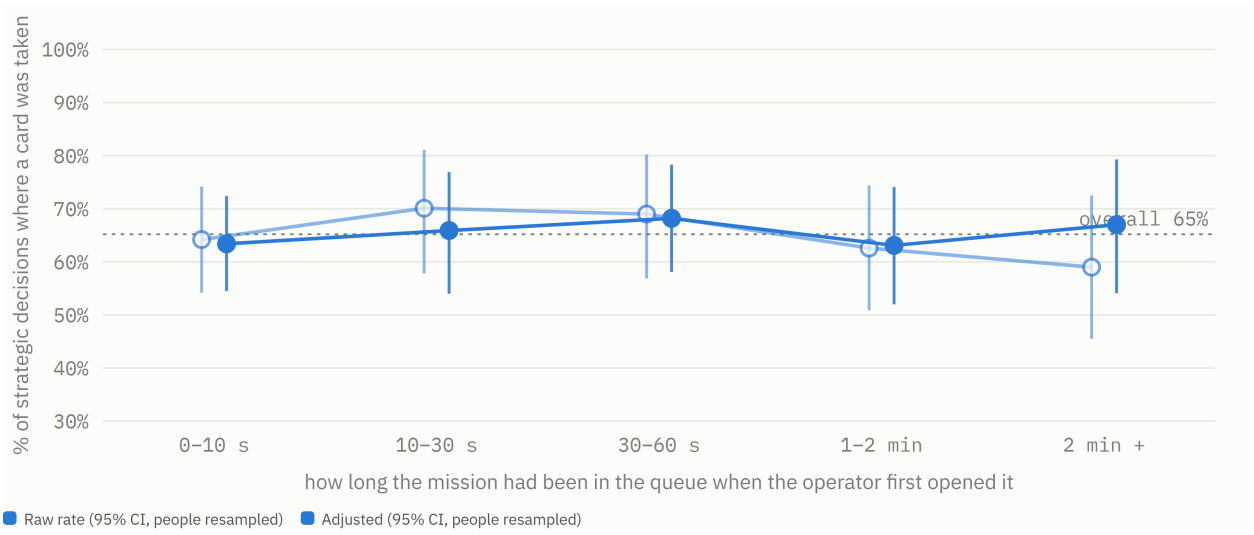
(e) Both tiers, raw with their expected rates (dashed).

Figure 24: Queue depth against reliance — raw, and adjusted

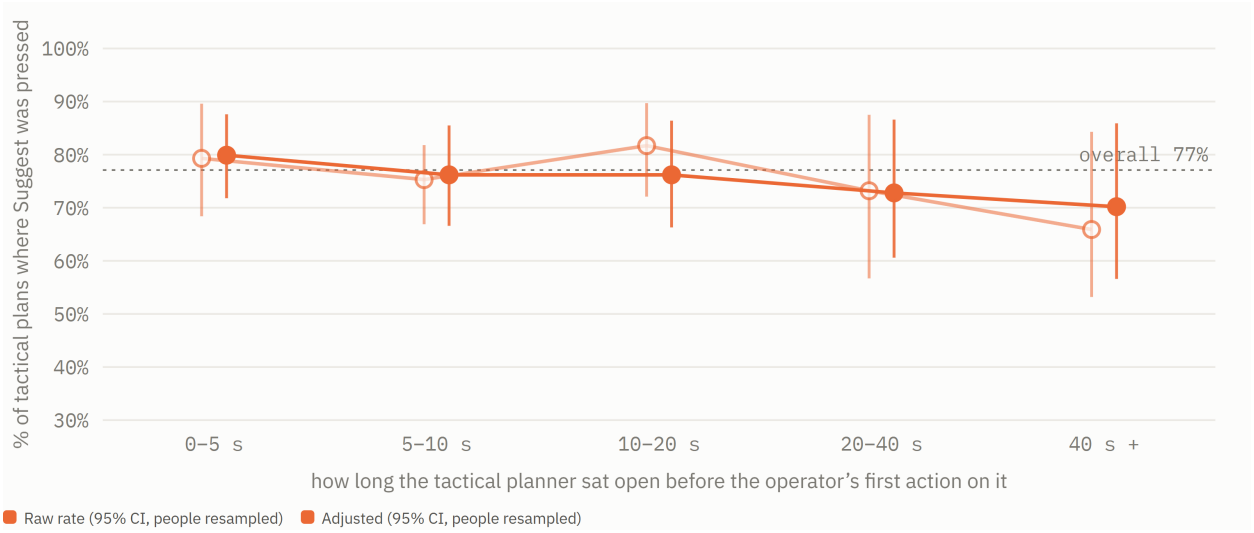
"I think the first one, I feel the pace is very, very fast. So, I can't do it very well in a very short time. So, I used a lot of AI suggestions." — P33 (debrief 0:27:33)

"Main reason to use AI was to save time under high workload" — P06 (written interview)

7.6.2 A mission left waiting is not handed over more often



(a) Strategic tier: how long the mission had waited in the queue.



(b) Tactical tier: how long the plan had been left open.

Figure 25: A mission left waiting is not handed over more often

Figure 25. Do people hand a decision over when they realise they have left it waiting? Here each decision is grouped by how long its mission had been waiting: on the strategic tier, from the mission's arrival to the first time the operator opened it; on the tactical tier, from the planner opening to the operator's first action on it (the plan the operator left sitting while they did something else). **The lines are flat.** Strategic card use is 63–68% (adjusted) whether the mission waited 10 seconds or more than two minutes. Tactical Suggest use, if anything, drifts down when a plan was left open longer (80% → 70%), the opposite of the idea, but not reliably.

Key numbers.

- Strategic (mission left in the queue): adjusted 63% at 0–10 s, 66% 10–30 s, 68% 30–60 s, 63% 1–2 min, 67% at 2 min+ (n 173, 117, 100, 91, 83). Within person, odds $\times 1.10$ per doubling of the wait [0.96, 1.28], p = .17; holding the queue fixed $\times 1.10$, p = .19.
- Tactical (plan left open): adjusted 80% at 0–5 s, 76% 5–10 s, 76% 10–20 s, 73% 20–40 s, 70% at 40 s+ (n 270, 182, 93, 41, 44). Within person, odds $\times 0.80$ per doubling [0.60, 1.06], p = .12; holding the queue fixed $\times 0.76$, p = .10.

Method. Rows: every strategic decision on a real (not re-queued) mission, 564 from 33 people, and every tactical plan, 630. Wait: strategic, mission arrival to the first time its strategy panel was opened; tactical, planner open to the first drag or Suggest. 'Adjusted' is the overall rate plus the average of (observed − expected) in that bin, where the expected value comes from mission size and drone types, minute of the session, the game and the person's own rate on their other decisions (the same model as section 1). Intervals resample people. The odds ratio compares each person with themselves: log2 of the wait (+5 s), centred within person, in a model with the same controls, people clustered; a second version also holds the queue depth fixed, since long waits come at busy moments.

7.7 Busy operators speed up a little but do not change what they pick

Summary. The earlier story — that with a long queue operators abandon order and pick smaller missions — does not survive a fair comparison. Long queues happen almost only in the two Strategic Heavy games, where every mission is small, so the opened mission is smaller simply because everything waiting is. Compared with what was actually waiting, operators keep the same priorities at any queue length: a strong pull towards the oldest mission, a mild preference for larger ones, and only falling 'oldest-first' rates because chance falls as more missions wait. What does change is pace: a bit more decisions per minute when more is waiting (suggestive).

Strength of evidence (register rating): mixed.

7.7.1 Order: still oldest-first well above chance

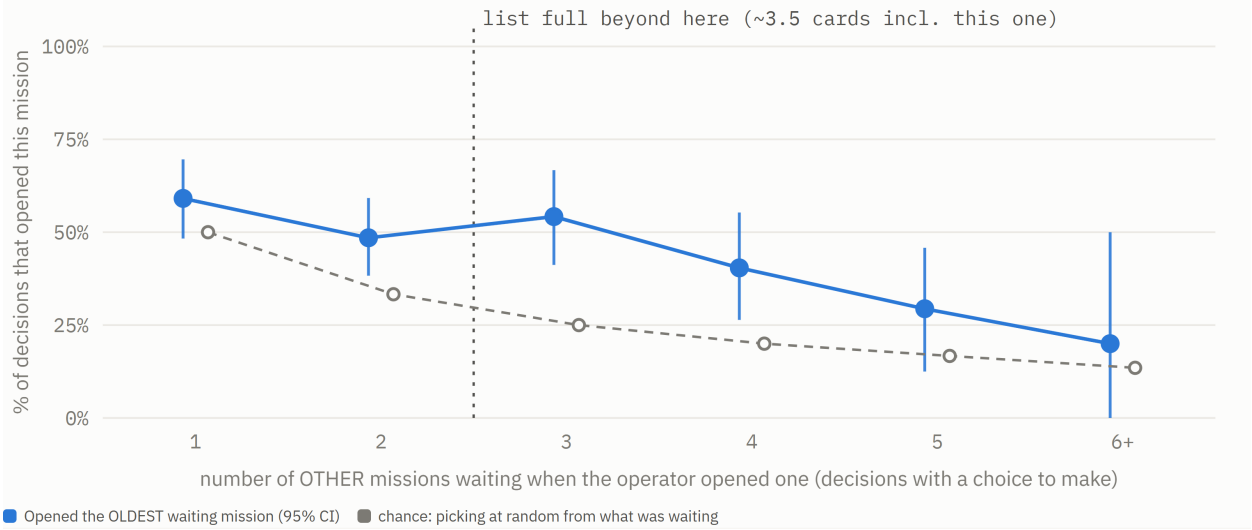


Figure 26: Order: still oldest-first well above chance

Figure 26. The share of decisions that open the oldest waiting mission falls as the queue grows (59% with one other waiting, about 30% with five) — but so does the chance of hitting the oldest by picking at random (50% → 17%). Measured against chance, the pull towards the oldest stays roughly the same. 'One of the top 3 in the list' is near chance at every depth, so the list position itself does little beyond 'oldest first'.

Key numbers.

- Opened the oldest: 59% (1 other waiting; chance 50%), 48% (33%), 54% (25%), 40% (20%), 29% (17%), 20% (14%).
- Within person, the excess over chance does not shrink with depth: +2.4 points per extra mission [−1.0, +5.7], p = .16.

Method. 414 strategic decisions where at least one other mission was waiting (33 people). For each, the set of missions waiting when the panel opened (arrival order from the log). Chance = 1 / (number waiting) for 'oldest'; 3 / (number waiting) for 'top 3'. Intervals: people resampled 400 times. Within-person test: (picked oldest − chance) against queue depth, person-centred, with minute and game.

What participants said. Nobody says they took missions oldest-first. Five say, in play or after, that criticality comes first ('I'll go with the most critical first'): P20, P22, P31, P19 and P27. P27 adds 'can it be completed' as the first filter. The order model has no criticality term, so this stated rule is untested. P19 wished for a sort control, and P27 for smart ordering, unaware that an order-by-criticality button existed.

7.7.2 What makes a waiting mission get opened, short vs long queues

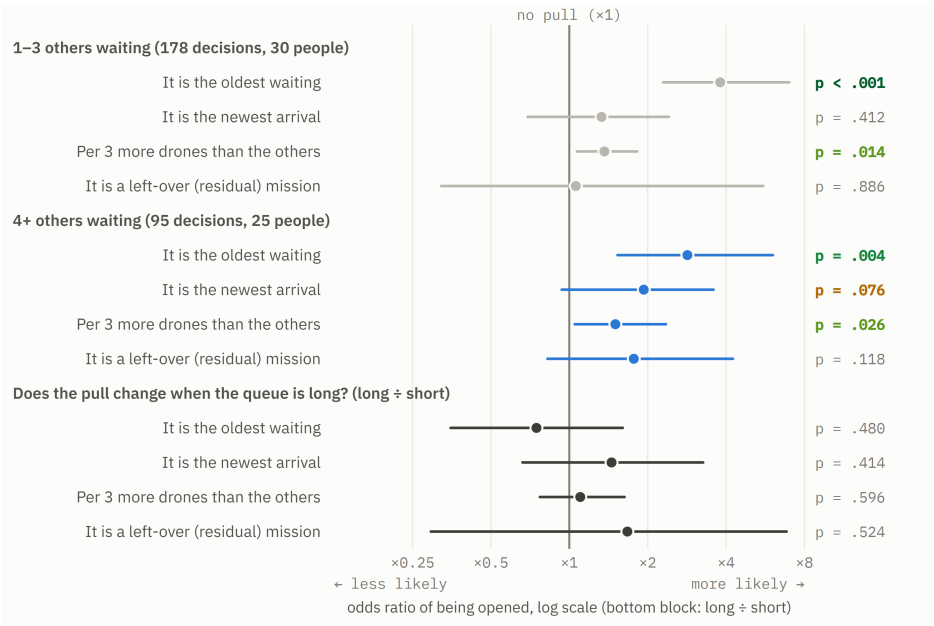
Figure 27. Putting it together in one model of the choice among what was waiting: being the oldest roughly triples a mission's chance of being opened, larger missions are mildly preferred, and being the newest arrival helps a little. These pulls are statistically clear where marked, and none of them changes significantly between short (1–3 others waiting) and long (4+) queues: the "long ÷ short" rows all have intervals spanning 1. Under pressure, priorities do not visibly change, though with 95 long-queue decisions only a large change could have been detected.

Key numbers.

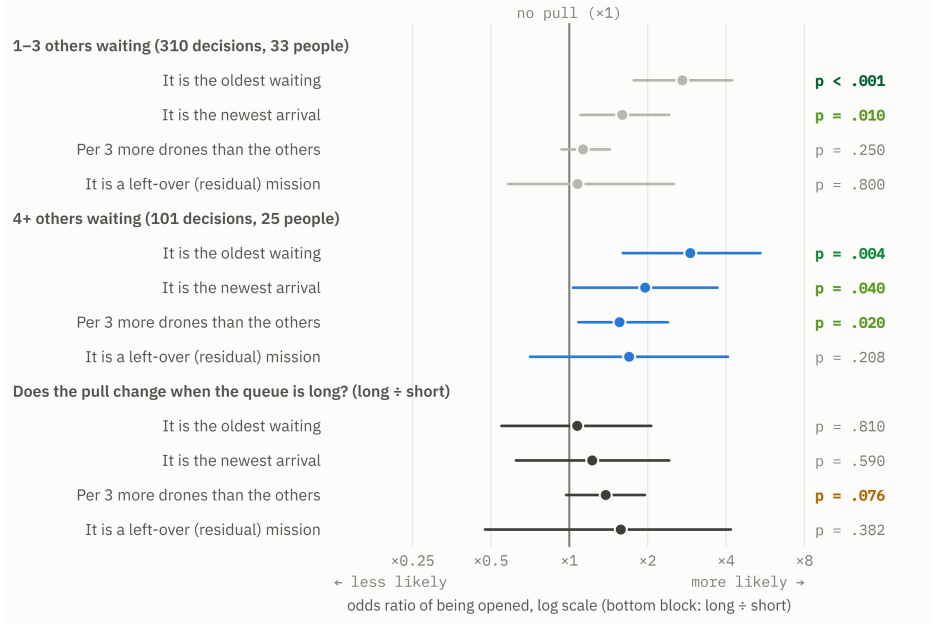
- Strategic Heavy games, short queues (178 decisions): oldest ×3.80 [2.29, 6.99], p < .001; newest ×1.33 [0.69, 2.41], p = .41; +3 drones ×1.36 [1.07, 1.82], p = .014; residual ×1.06, p = .89
- Strategic Heavy games, long queues (95 decisions): oldest ×2.84 [1.53, 6.06], p = .004; newest ×1.93 [0.94, 3.58], p = .076; +3 drones ×1.50 [1.05, 2.35], p = .026; residual ×1.77, p = .12
- Long ÷ short, Strategic Heavy games: oldest ×0.75 [0.35, 1.60], p = .48; newest ×1.45, p = .41; size ×1.10, p = .60; residual ×1.67, p = .52 (none differs)
- All four games: oldest ×2.72 (short) and ×2.92 (long), both p ≤ .004; newest ×1.60, p = .010 and ×1.96, p = .040; size ×1.13, p = .25 and ×1.56, p = .020; long ÷ short for size ×1.38 [0.97, 1.95], p = .076, all others p ≥ .38

Method. Conditional logit: each decision is a choice among the missions waiting at that moment; the model estimates how much each feature (oldest, newest, size relative to the set in units of 3 drones, residual) multiplies a mission's odds of being the one opened. Fitted separately for short (1–3 others) and long (4+) queues, in the Strategic Heavy games only (like with like) and in all four. Intervals and p-values: people resampled 1000 times (the short- and long-queue fits on the same resampled people), so they allow for each person making many decisions; p = twice the share of resamples on the far side of "no pull". The "long ÷ short" rows are the ratio of the two odds ratios with their own interval and p, i.e. a direct test of whether the pull changes with the queue.

What participants said. Besides criticality, people describe choosing by feasibility: they skip what the reserve cannot cover yet and come back (P17, P27, P30). P19 jumped to a newly arrived higher-priority mission, which matches the small 'newest' pull. Nobody mentions preferring larger missions, and nobody says their way of choosing changed when the queue was long.



(a) Strategic Heavy games only.



(b) All four games.

Figure 27: What makes a waiting mission get opened, short vs long queues

7.8 The dip in strategy-card use belongs to the people who later said they noticed the illogical card, early in the session; it shows up whether or not they had met that card yet

Summary. The Conservative card sometimes adds a drone type the mission does not use (22% of decisions, nearly all on one-type missions); all 33 people met it, usually on their 1st or 2nd decision. The 15 who said they noticed it take the cards less afterwards, and on the flawed one-type missions they learn to skip them (0 takes in 13 by their 4th–6th such mission). But the **dip** in their use is not tied to the card. Counted simply by decision number, noticers take the cards on 80% of their 1st session-1 decision and 46–50% of decisions 3–5, then 83% at decision 6, while never-said people rise to 89–94% in the same stretch. People who played the tactical-heavy game first had *not yet met* an illogical card in decisions 3–5 (95% of those decisions came before it), and their noticers dip just as much (-21 points per person, against -32 for those who had met it; game order makes no difference, p = .79). In that stretch noticers look at the cards and turn them down, on three-type missions too. Pooled over everyone the dip disappears, because the two groups move in opposite directions. Noticers climb back later in session 1 (+24 points per person) and start session 2 lower again (49%), but never-said people also start session 2 a little lower, so a weaker carry-over for noticers is not something these data can show (difference at the break -6 points [-32, +20]). Our best reading: "noticed" marks people who scrutinise the cards, and the dip is that scrutiny at the first busy stretch of the study, not a reaction to the illogical card. What triggers it at that moment we cannot say: mission mix, load, reserve shortage and game do not account for it. Suggestive (15 vs 18 people, window read off the curve). This finding also holds, sidelined, the flaw-visibility comparison and the recovery material merged in from two other findings.

Strength of evidence (register rating): suggestive.

7.8.1 The event study, aligned on the first illogical card (every step, pairs or threes)

Figure 28. Each person's decisions are numbered *within* the kind of mission plotted: their 1st, 2nd, … decision after the first illogical card. With every step shown, each point is then about 15 single decisions and its interval spans roughly 30 points either way. Pairs keep the shape with intervals about a third narrower, so pairs is the default. On **all missions**, noticers go from 76% before the card to 43–52% for their next ten decisions, recovering to 67–68% only from the 11th on. Never-said people stay at 67–75% throughout. Part of the noticers' low on all missions is the flawed one-type missions themselves, where they increasingly refuse the card. On **three-type missions**, noticers go 74% → 43% (1st–2nd) → 59–63% → 75%: a short dip and recovery. On one- and two-type missions almost nobody has a decision before the card, because the card only ever appears there, so only the after-card trend can be read. **The steps before the card** add little: most people have only one decision before it, so -2 and -3 are 8 decisions each (± 35 points). The last step before the card (-1) is lower than $-3/-2$ in *both* groups (64% and 56%), so it is not a noticer warning sign. **Read with the next section:** the aligned dip is not where it seems.

Key numbers.

- All missions, noticers (15): 76% before (n 53) → card 47% (n 15) → pairs +1–2 to +9–10: 43%, 47%, 43%, 46%, 52% → 67% (+11–12) → 68% (13th+, n 31)
- All missions, never said (18): 78% before (n 54) → 56% at the card → 67–75% for every later pair
- All missions, before → after, adjusted: noticers -14 points, p = .08; never said +9, p = .14; difference -14 [-30, +1], p = .06
- Three-type, noticers by pair: 74% before (n 51), 43% (n 30), 59% (n 29), 62% (n 24), 63% (n 19), 75% (n 12)
- Why expected is lower for noticers: their own-rate term (take rate on all their other decisions) is 56% vs 70%; 79% of a noticer's decisions come at or after the card, where they took 50% per person vs 72% (p = .007); before the card the groups were alike (68% vs 63%, p = .71)
- Mission-only expected (own-rate term set to the average): after the card 68% noticers vs 69% never-said; before 80% vs 81%
- Steps before the card, all missions: noticers ≤ -4 87% (n 23), -3 88% (n 8), -2 50% (n 8), -1 64% (n 14); never-said 82%, 100%, 88%, 56% (n 22, 8, 8, 16)

- -1 against $-3/-2$, the 16 people with three steps (all tactical-heavy-first): adjusted -13 points [-40, +13], p = .29
- Matched view (8 noticers with 3+ steps, 6 of whom met the card at the start of session 2): 87%, 88%, 50%, 63% before $\rightarrow$ 38% at the card $\rightarrow$ 31–44% for the next ten decisions

Method. Rows: strategic decisions of the 33 people, split into those who said they noticed (debrief or interview code C12, or a strict in-play remark; 15 people) and those who never said so (18). 'Before' and 'after' are relative to the first illogical card. That decision itself is excluded from both windows and shown as its own point ('the card') in the all-missions view. Change = the 'after' coefficient in a linear model of card uptake, with practice (decision number) held fixed and people clustered for the uncertainty. Raw version: uptake ~ after + practice + game. Adjusted version: (observed $-$ expected) ~ after + practice. The interaction asks whether the after-step differs between the groups. Label shuffle: the said/never-said labels were shuffled across people 5,000 times, to see how often a random split of the same people shows as large a difference in their own after-minus-before. Each person counts once in that test, so it is exact. Steps before the card: -1 is each person's last decision before it, -2 the one before that, and '≤ -4' pools everything earlier; the matched view keeps only the 16 people (8 per group) with at least three such decisions. Mission-only expected: the same model's prediction with each person's own-rate term set to everyone's average.

What participants said. 12 of 33 said in the debrief or interview that the strategy cards over-allocated or sent unneeded types (C12), and 12 commented on it during play (7 strictly about a card sending an unneeded type). The first sight is often puzzlement rather than a verdict ('unless I'm missing something', 'do fast drones have a higher chance of failing?'), and some took the card anyway (P04) or noticed only after taking it (P20, P29). Dissent: P10 decided the redundancy was sensible; P27 was not sure it was a fault.

Quotes selected for this section.

"Oh, this just sends a bunch of extra of these even though it doesn't need it?" — P19 (think-aloud s2 0:36)

"Why is it that the green ones, there's no green or red, unless I'm missing something." — P17 (think-aloud s1 5:25)

"And then when it came to assigning them, I noticed, oh, okay, it's made a mistake." — P20 (debrief 0:26:11)

"That seems like more drones than perhaps it needed, but I've had drones that are literally not needed for that one, so we'll do the aggressive again." — P04 (think-aloud s1 0:44)

"where we can go conservative, but we don't need any, in this one we don't need any lifter or cameras." — P30 (think-aloud s1 1:01)

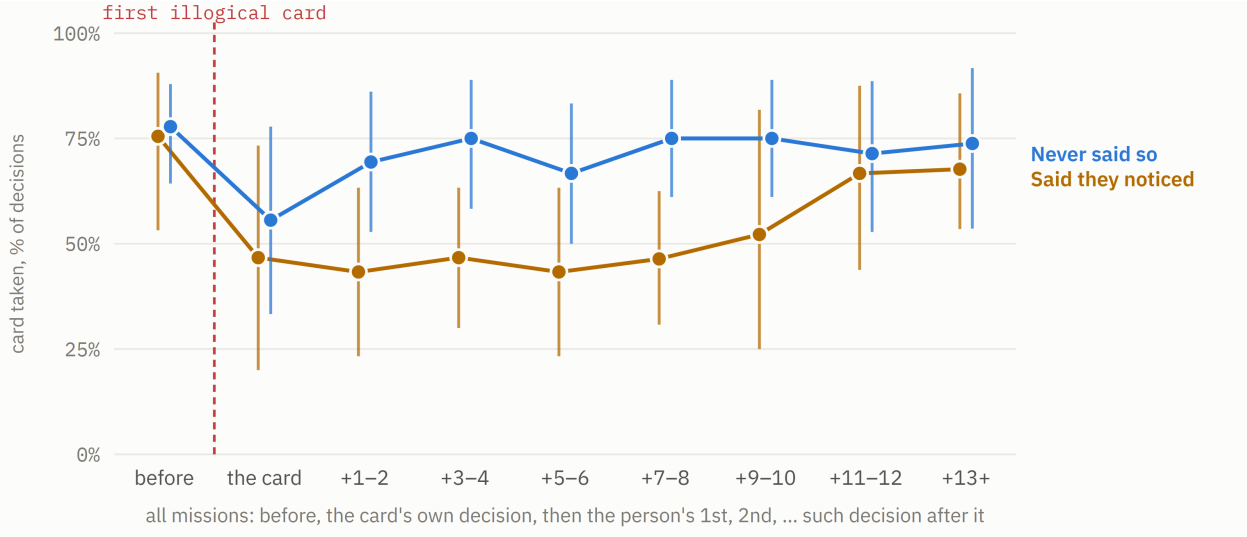
"But I find that the first task it allocated too much Lifter and Camera-1. So I think I'm less confident for it." — P35 (think-aloud s1 1:22)

"there was one example in the first round where it was required all fast drones. And I think the AI assistant recommended I add some red and green drones." — P20 (debrief 0:25:51)

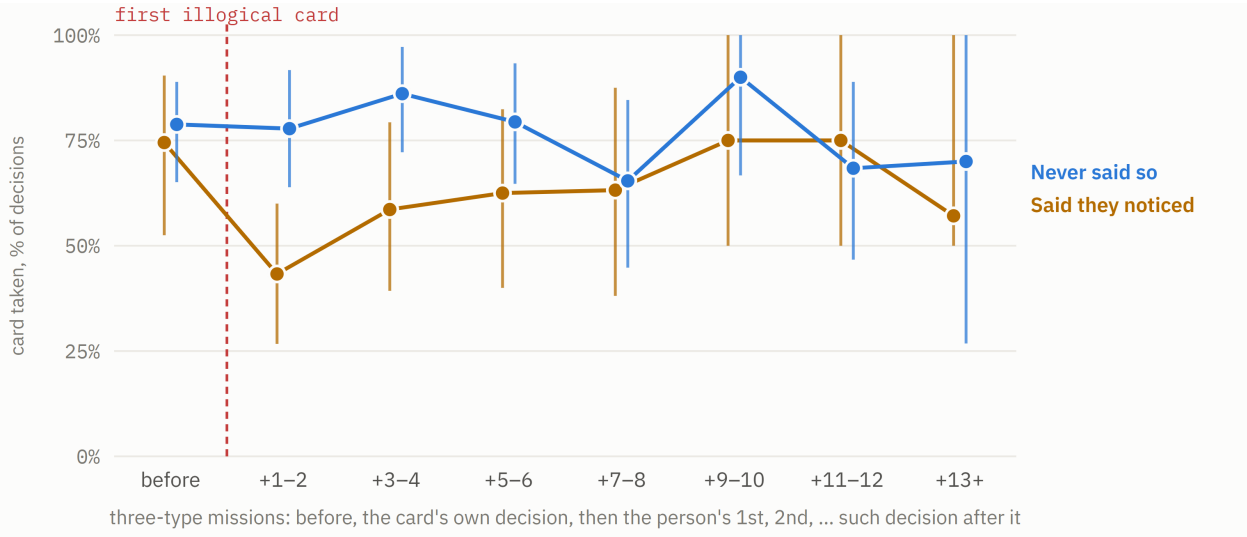
"I mean, it was giving me the wrong type of drones. It required only blue drones and kept giving me red and green ones." — P21 (debrief 0:35:03)

"But in conservator, you don't need camera or lifters, yet it adds it as reserve. Mm-hmm. Even for tasks it doesn't do. That doesn't make sense." — P13 (debrief 0:29:55)

"deploy a conservative strategy for that one doesn't make sense. I don't think because I don't need any red or blue chair drones." — P20 (think-aloud s1 3:41)



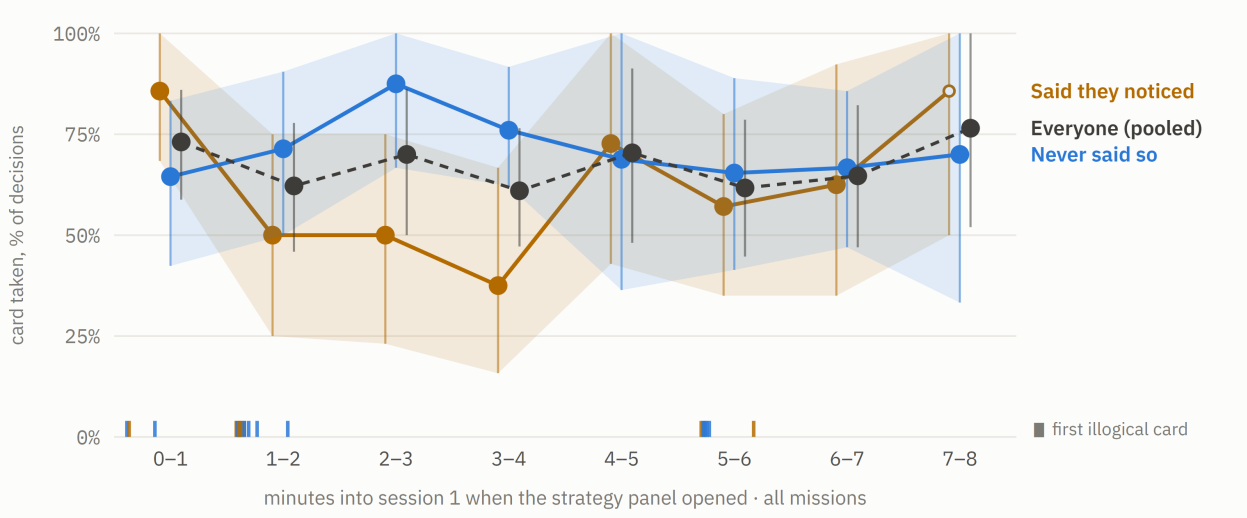
(a) All missions.



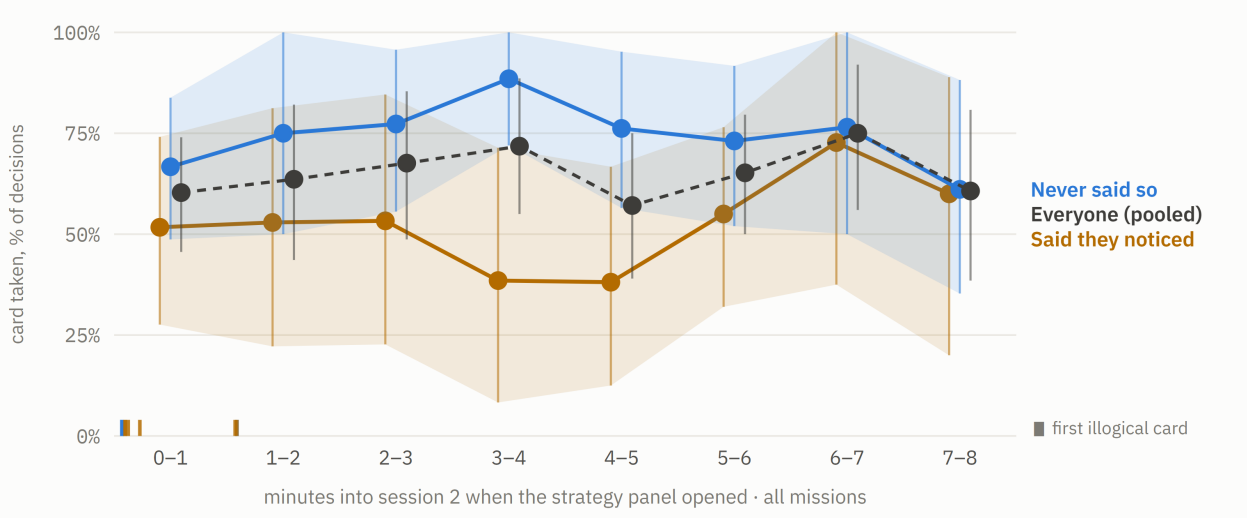
(b) Three-type missions only.

Figure 28: The event study, aligned on the first illogical card (every step, pairs or threes)

7.8.2 Card use through the session, noticers and never-said



(a) Session 1.



(b) Session 2.

Figure 29: Card use through the session, noticers and never-said

Figure 29. What is plotted. Every strategic decision is placed in the minute of the session when its strategy panel opened (0–1 is the first minute). Each point is the share of the decisions in that minute where the person took one of the two strategy cards instead of allocating by hand, with a 95% band. Nothing is lined up on the illogical card; the small ticks on the axis show when each person first met it. By default it shows **each group and everyone pooled** (dashed line). **Session 1**: noticers start at 86% in the first minute, fall to 50%, 50% and 38% over minutes 1–4, and come back to 73% in minute 4–5. Never-said people rise over the same minutes, 65% → 88%. Everyone pooled sits between the two and stays at 61–77%: the two groups move in opposite directions, so lumping them together hides the dip. That is the point of this chart. **Session 2**: noticers are lower throughout (52–53% in the first three minutes, 38% in minutes 3–5, then 55–73%); never-said people sit between 61% and 89%. Each minute holds 10–30 decisions per group, so single minutes have wide bands: the shape, not any one point, is the result.

Key numbers.

- Session 1, noticers by minute 0–1 to 4–5: 86% (n 21), 50% (n 16), 50% (n 14), 38% (n 16), 73% (n 11)

- Session 1, never said: 65% (n 31), 71%, 88%, 76%, 69%
- Session 2, noticers: 52% (n 29), 53%, 53%, 38% (n 13), 38% (n 21), 55%, 73%, 60%; never said 61–89%

Method. Rows: every strategic decision, binned by the minute of the session at which its panel opened (the last bin, 7–8 min, is thin). Rate = share of decisions where a card was taken; 95% bands from resampling people. Adjusted = the group's overall rate plus the mean of (observed − expected) from the adjustment model, i.e. relative to each person's own habit given the mission in front of them. Game order: strategic-heavy-first players met the illogical card at about 67 s of session 1; 7 tactical-heavy-first players at about 315 s, the other 9 at the very start of session 2.

What participants said. Nobody describes a dip-and-recover in their own words, but the pieces are there: several noticed early in session 1 (P35 in the first two minutes, P13 at minute 3, P20 at minute 4) and went manual for a while, and those who say why they kept using the cards give speed — it is less reliable 'but at least it saves time' (P28). P20, asked if their trust was destroyed: 'No. Definitely not… most of the time it seemed accurate'.

Quotes selected for this section.

> *"I think it's less reliable than before, but I still use its suggestion, and that's the case. Yeah, because at least it saves time." — P28 (debrief 0:29:56)*

7.8.3 Card use by decision number, session 1 then session 2, not aligned to the card

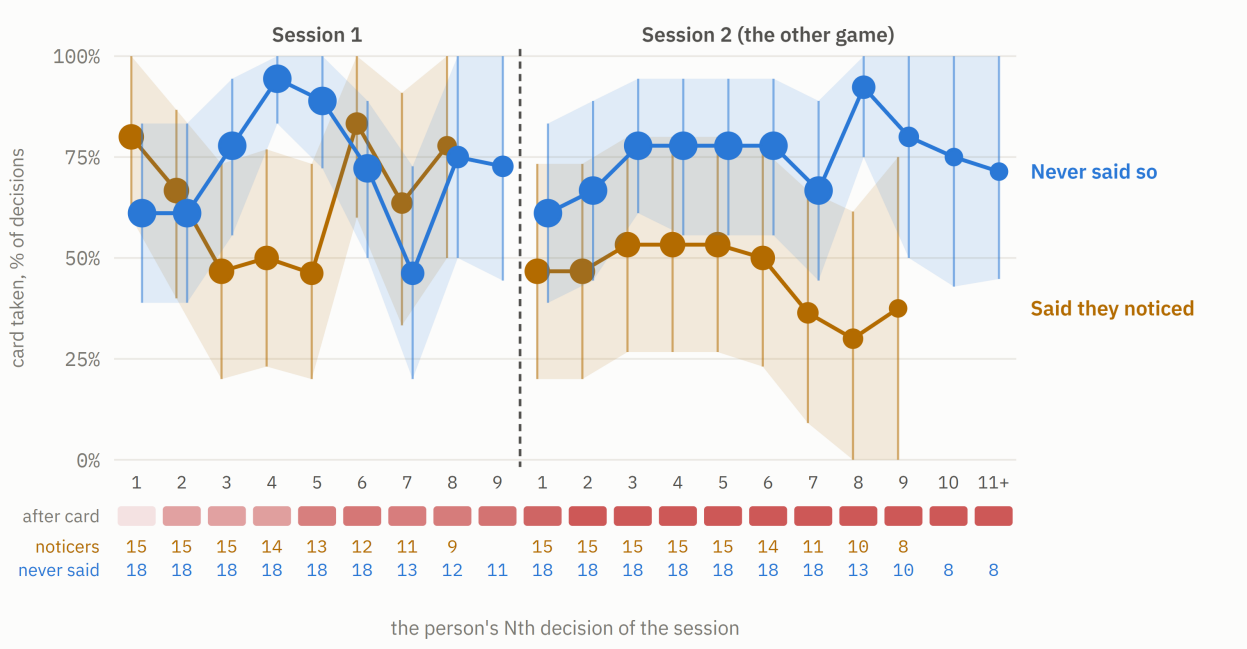


Figure 30: Card use by decision number, session 1 then session 2, not aligned to the card

Figure 30. Each person's decisions are numbered in the order they made them within each session (or counting only three-type, one/two-type, or missions with no illogical card). **Session 1 and session 2 sit on one axis**, separated by a dashed line, with the two sessions' lines deliberately not joined (session 2 is a different game, after a break and a survey). **Later numbers are different people**: only those who made that many decisions reach them, and they are mostly players of the strategic-heavy game, which has more missions. So a point is drawn only if at least 8 people of its group reach it; dot size and the numbers under the axis show how many people each point rests on. **Session 1**: noticers 80%, 67%, 47%, 50%, 46%, then 83%, 64%, 78% for decisions 6–8; never-said 61%, 61%, 78%, 94%, 89%, 72%. Noticers dip at 3–5 and climb back (+24 points per person). The same shape holds in a fixed panel of the 9 noticers and 12 never-said people who made at least 8 decisions (noticers 44% at decisions 4–5, 78–89% at 6

and 8), so it is not a change of who is counted. **Session 2**: noticers 47–53% for six decisions, then 30–38% at 7–9; never-said people 61–92%. The late low is the mission mix: around decisions 7–8 of the strategic-heavy game half the missions show the illogical card, and noticers took it on 17% of those against 62% of the rest. Leaving those missions out, noticers and never-said people meet at decisions 7–8 (71% against 70–73%). Noticers start session 2 below where they ended session 1 (64% → 49%), but never-said people also start a little lower (76% → 69%); the difference at the break is small and uncertain (−6 points [−32, +20]), and −9 [−30, +12] for noticers after allowing for the mission mix.

Key numbers.

- Everyone, session 1, decisions 1–6: 70%, 64%, 64%, 75%, 71%, 77% (n 33 → 30)
- Noticers, session 1: 80%, 67%, 47%, 50%, 46%, 83%, 64%, 78% (decisions 1–8, 15 → 9 people); never said: 61%, 61%, 78%, 94%, 89%, 72%, 46%, 75%, 73% (decisions 1–9, 18 → 11 people)
- Session 1 fixed panel (9 noticers, 12 never-said, decisions 1–8): noticers 78, 67, 56, 44, 44, 89, 56, 78%; never said 50, 42, 67, 92, 83, 67, 50, 75%
- Session 1 tail (hidden): decision 9, 6 noticers 83%; decision 10, 5 noticers 1 of 5 took (20% [4, 62]), 7 never-said 43% [16, 75]; 11+, 2 noticers 3 of 7 (one took 3 of 4, the other 0 of 3), 5 never-said 7 of 12. All strategic-heavy-game players; 57–67% of the 11+ decisions in the last 90 s
- Session 2 tail (hidden for noticers): decision 10, 5 noticers 60%; 11+, 4 noticers (all tactical-heavy first, so playing the strategic-heavy game) 12 of 13 = 92% [67, 99], against 8 never-said 20 of 28 = 71%. These 4 took cards on 42% of their decisions 1–6, below their group's 51%: not keen takers surviving
- Strategic-heavy game: share of decisions with an illogical card at decisions 7, 8, 9, 10, 11+: 48%, 55%, 30%, 4%, 5%. At decisions 7+, noticers took 17% of illogical-card missions (n 18, 9 people) and 62% of the rest (n 53); never-said 68% and 69%
- Session 2, no illogical card: noticers 71% at decisions 7 and 8 (7 people) against never-said 73% and 70%; earlier the gap is there (decisions 2–3: 33–47% against 67–89%)
- Session 2 fixed panel (8 noticers, 10 never-said, decisions 1–9): gap over the last 3 decisions −47 points [−81, −12] against −19 [−50, +12] before; on missions with no illogical card (panel to decision 6) −17 [−45, +12] against −30 [−61, +1]
- At the session break (last 3 of session 1 → first 3 of session 2): noticers 64% → 49%, −16 points [−38, +4], p = .17; adjusted −9 [−30, +12]; never-said 76% → 69%, −7 [−24, +7]; difference −6 [−32, +20], p = .66
- Recovery within session 1 (decisions 3–5 → 6+): noticers 48% (n 42) → 66% (n 50), +24 points per person [+5, +43], p = .04 (window read off the curve, so descriptive)

Method. Rows: every strategic decision; position = its rank among that person's decisions of the chosen kind in that session (positions past 10 pooled as 11+). **Minimum**: a point is drawn by default only if at least 8 people of its group (both game orders together) reach that number, about half of the smaller group. **Fixed panel**: per session and mission kind, K = the largest number that at least 8 people in each group reach; only people with at least K decisions, decisions 1–K (all missions: session 1 K = 8, 9 noticers and 12 never-said; session 2 K = 9, 8 and 10). 'No illogical card' drops the decisions on missions whose Conservative card showed an unused drone type, then renumbers. 'After card' strip = share of the position's decisions made at or after the person's first illogical card. Carry-over checks, per person then averaged (intervals from resampling people, p from a t-test over people): (a) recovery = the rest of session 1 (decisions 6+) minus decisions 3–5; (b) break = the first three decisions of session 2 minus the last three of session 1; (c) start-to-start. Group differences = the interaction in a linear model with people clustered; adjusted versions use observed − expected.

What participants said. 4 of 33 said they caught the assistant's mistakes more, or were more careful, in the second session (C15: P15, P17, P19, P33), and P15 and P33 add that in the first session there was no time to check. One said the opposite: P20 saw no strategic issues in the second run. No one connects it to which game they played.

Quotes selected for this section.

> *"the second round side because i have more experience after the first round and now i can check the whether the uh in these screens that tour i can find that it makes mistakes" — P15 (debrief 0:33:05)*

7.8.4 Lined up on the moment they said a strategy card was wrong: card use falls (7 people, a direction only)

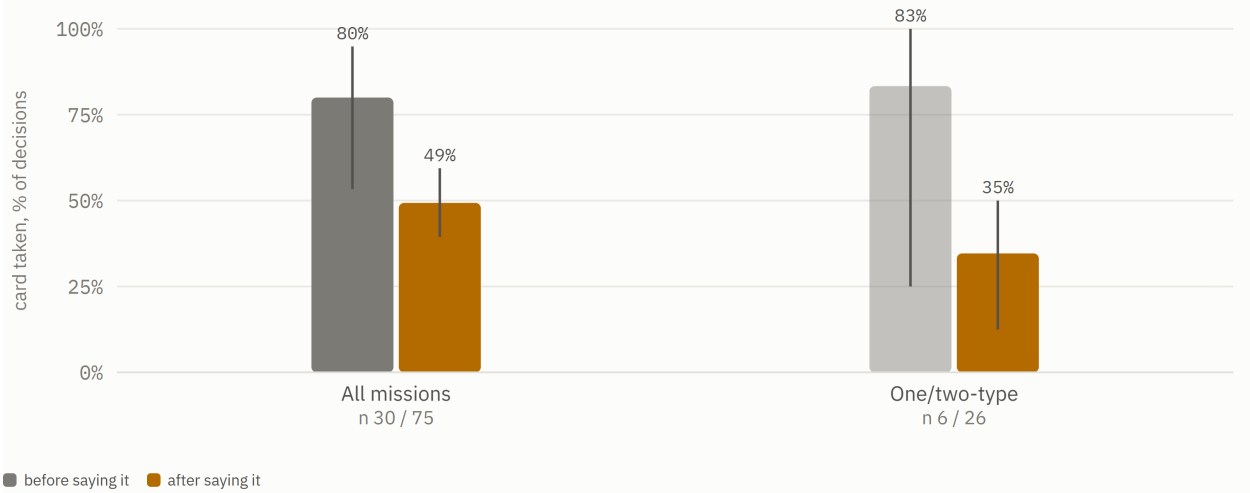


Figure 31: Lined up on the moment they said a strategy card was wrong: card use falls (7 people, a direction only)

Figure 31. *This section is about noticing the strategy card's flaw; the next section is a different event (the Tactical Assistant failing to offer a recovery plan), so the two are not in conflict.* Here each person is anchored on the moment they said, during play, that a card was wrong. Seven people made a strict remark (a card sends types the mission does not need, or a specific card is wrong); five more made broader 'too many drones' remarks. For four of the seven, an illogical card was on screen as they spoke. P30's remark was about an illogical card that opened 5 s later, so it was moved onto that card. P29 spoke 28 s after taking the illogical card. P04's remark was about too many drones on a three-type card. The chart pools each set of people, before vs after the remark, with **all missions and one- and two-type missions side by side**. On all missions the seven went from 80% to 49% (adjusted −21 points [−49, +7], p = .11, timing-matched placebo p = .51; five fell and two rose). On one- and two-type missions, the ones that carry the illogical card, the drop is larger, 83% → 35%, but that 'before' bar is only 6 decisions from 4 people, so it is a direction, not a measured size. Seven people cannot settle it; the direction matches the main result.

Key numbers.

- Strict remarks (7 people), all missions: 80% (n 30) → 49% (n 75); adjusted −21 points [−49, +7], p = .11; 5 fell, 2 rose; placebo p = .51
- Strict, one/two-type: 83% (n 6, 4 people) → 35% (n 26, 7 people); three-type: 79% (n 24) → 57% (n 49)
- Any remark (12 people), all missions: 74% (n 74) → 55% (n 124), adjusted −4 points, p = .60; one/two-type 67% (n 21) → 40% (n 48)

Method. The spoken moments were hand-coded from the in-play narration before any behaviour around them was computed (extras.SPOKEN_STRAT, session time). A decision whose card was open at that moment is the anchor decision. Otherwise, 'before' and 'after' are split at the remark. The same models and placebo as above are used.

What participants said. The in-play remarks are short and pointed ('It's just wrong', 'You don't need all of that') and are usually followed by going manual (P19, P29, P35, P13), sometimes by taking the card anyway (P04). Four people make further in-play remarks that the anchor list leaves out, late in session 2: P19, P21, P35, and P32. P32 has

no C12 code, so they are counted as never having said it, yet they said 'It's just wrong' about an unneeded Camera on a card (s2 5:45).

7.8.5 A different event: after the Tactical Assistant's first 'No workable plan', nothing changes on either tier

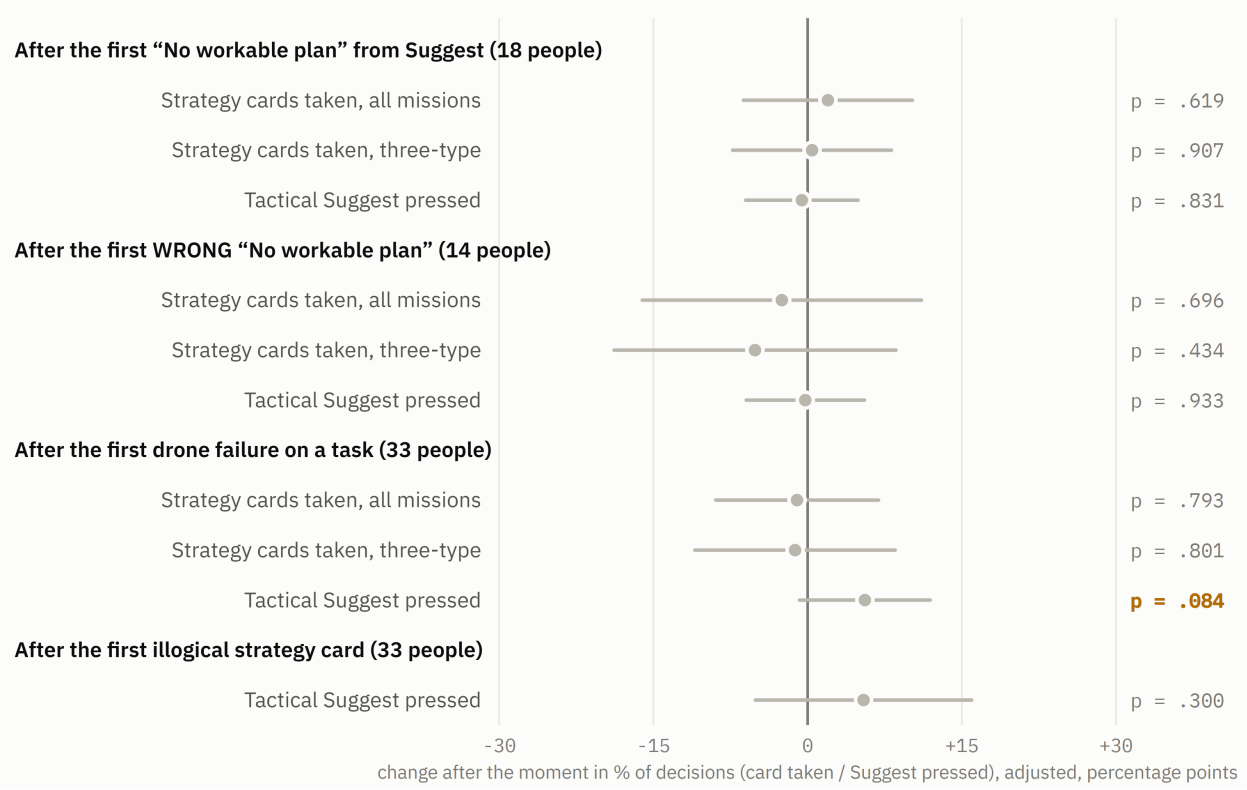


Figure 32: A different event: after the Tactical Assistant's first 'No workable plan', nothing changes on either tier

Figure 32. *Not the same moment as the previous section.* Here each person is lined up on the first time the Tactical Assistant's Suggest, asked for a plan to recover from a drone failure, answered 'No workable plan'. That is a failure of the other assistant, and nothing to do with noticing the illogical strategy card. Does *strategic* use go down after the first refusal, with the problem cascading up the chain? **It does not.** After their first 'No workable plan' (18 people), card uptake actually went *up* in raw terms (68% → 76%). The expected value went up just as much, because these refusals tend to come later, when people have settled in. Adjusted, the change is +2 points [−6, +10], with 10 people falling and 8 rising. On three-type missions it is 0 points. Tactical Suggest use also rose raw (78% → 87%) and is flat adjusted (−1 point). After the first *wrong* refusal (14 people) the picture is the same: strategic −2.5 points [−16, +11], tactical 0. The first drone failure (33 people) changes nothing either. These are absences of evidence, with intervals around ±10 points. They are not proof of no effect, but no cascade is visible.

Key numbers.

- First refusal (18): strategy cards, all missions, adjusted +2.0 points [−6, +10], p = .62; three-type +0.4, p = .91; tactical Suggest −0.6 [−6, +5], p = .83
- First WRONG refusal (14): strategy cards −2.5 [−16, +11], p = .70; three-type −5.1, p = .43; tactical −0.2, p = .93
- First drone failure (33): strategy cards −1.0, p = .79; tactical +5.6, p = .08
- After the first illogical card, tactical Suggest +5.4 points, p = .30 (no spill-down either)

Method. Same rebuilt method as Section 7.8. Each person's strategic decisions (or tactical plans) are numbered within the sample, after versus before the moment of their first refusal, with practice held fixed and people clustered. Adjusted = observed minus the expected value. The placebo moves each anchor to a random point drawn from the real anchor positions. All p-values are for the adjusted change.

What participants said. Five people reacted aloud to Suggest's 'No workable plan', and every one took it at face value and moved on to abandoning ('Oh well. We'll have to abandon it then') or doing it by hand. Nobody said a refusal made them doubt either assistant. That fits the null. The only counter-voice is P05's note that they would want to check a refusal themself, and P27 asked for the abort button to be highlighted when no plan exists.

7.8.6 Why the pre-registered test (H8) was dropped

H8 asked whether uptake falls after an operator 'realises' the card can be illogical. Its marker for that realisation was the first time a person went manual on an illogical-card mission and left out the unneeded type. The trouble is that, by that definition, every illogical-card decision *before* the marker was not such a manual choice. So the 'before' window was mostly forced to be card takes, near 100%. Any later manual choice then shows up as a drop, even if the person never changed at all. A simulation confirmed it. In simulated people who keep the same habits all study, the same marker produced a large 'drop' (average odds ratio 0.37, against the real 0.12). Measured against that, the real drop was ordinary (calibrated p = .30, about −10 points after correction). Defining the marker as the first manual choice on a *clean* card produced the same kind of drop. H8 was replaced by tests whose anchor is not defined by the outcome (the first illogical card shown; the moment someone said it), each checked against a placebo. Those are the tests in this finding.

Key numbers.

- Pre-registered estimate OR 0.12; the same test on no-change simulations averages OR 0.37; calibrated p = .30 (risk difference)
- Placebo marker on clean cards: OR 0.22, p = .03, the same kind of 'drop'

Method. From docs/analysis/round1/h8_illogical_card.md. The null calibration redrew each person's choices 1,000 times from their own constant rate, reapplied the marker and refitted the same model.

What participants said. The move the H8 marker counts — going manual and leaving out the unneeded type — is described by several (P19, P13, P16, P17, P29). But noticing and going manual are not the same event. Some saw the excess and still took the card (P04, P28 'I still use its suggestion'). Others went manual only to trim an extra of a needed type (P10, P01). So a first manual 'fix' is a poor stand-in for the moment someone realised.

7.9 The Tactical Assistant is trusted more, and most people say so too

Summary. On the post-session questionnaire, operators trusted the Tactical Assistant more than the Strategic Assistant: 5.44 against 4.61 on a 1–7 scale, 26 of 33 people higher for tactical (p = .001). Most of the gap opens in the second session and in Strategic Heavy games, and people who said in the debrief that they preferred the tactical one also scored it higher. Tactical plans get only a few seconds of checking whatever their size, which fits "checking is costly, so it is skipped", but that mechanism test is fragile. Stated trust and actual use line up when people are ranked against each other, but not in size and not over time.

Strength of evidence (register rating): moderate.

7.9.1 Trusted more: the basic result

Figure 33. After every session operators rated each assistant on the same six statements ("is reliable", "I trust its recommendations", . . .) from 1 to 7. Averaged over their two sessions, **26 of 33** rated the Tactical Assistant higher, 6 the Strategic Assistant, and 1 the same. The average gap is **+0.83 points** on the 7-point scale. That is a medium-sized effect (dz = .62), and every one of the six items points the same way.

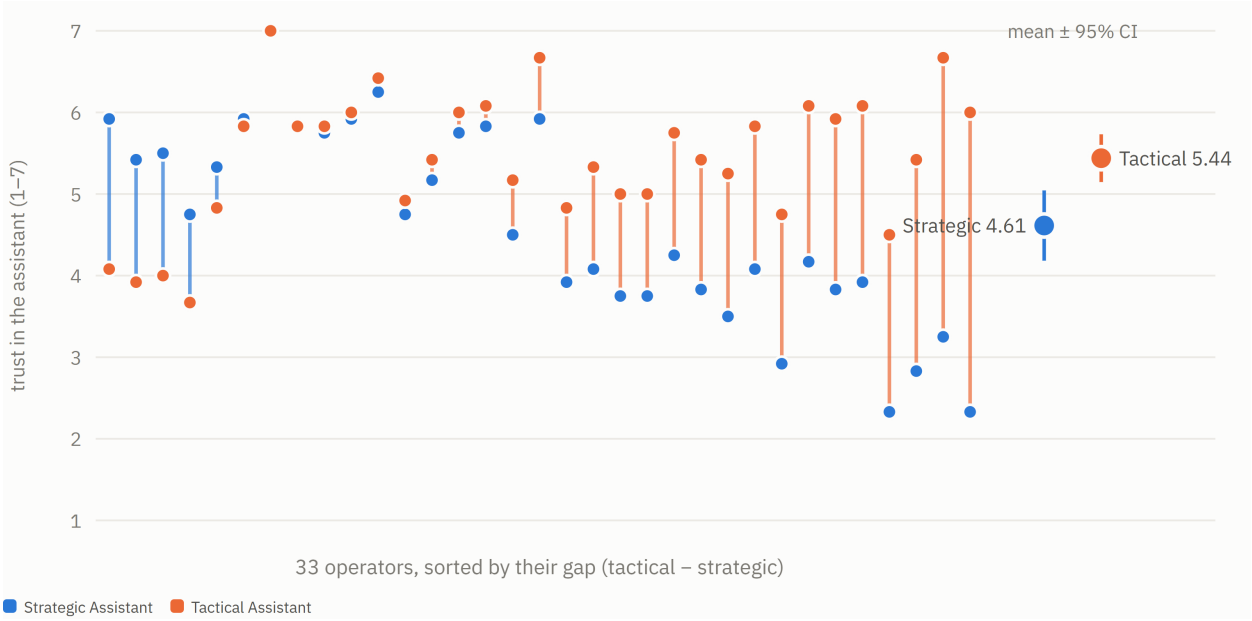


Figure 33: Trusted more: the basic result

Key numbers.

- Strategic 4.61 [4.18, 5.04], Tactical 5.44 [5.15, 5.73] (1–7, person means)
- Gap +0.83 [+0.35, +1.30]; paired t p = .0012; Wilcoxon p = .0010; dz = .62; n = 33
- 26 higher for tactical, 6 for strategic, 1 tied
- Item gaps: follow its recommendations +1.11, I trust it +1.01, reliable +0.94, performs well +0.74, useful +0.67 (all $p < .01$); confident in it +0.48 ($p = .13$)

Method. One value per person per assistant: the mean of the six items, averaged over that person's two sessions (33 people, P09 excluded). The test is a paired t-test on the difference tactical minus strategic, with a Wilcoxon signed-rank test as the distribution-free check. dz is the mean difference divided by the SD of the differences: .62 means the average person's gap is about six-tenths of the person-to-person spread in gaps. The 95% CI says the true average gap is very probably between +0.35 and +1.30 points. Items are tested the same way, one at a time.

What participants said. In the round-2 preference coding, 17 of 33 favour the Tactical Assistant, 7 the Strategic, 3 both, 4 'it depends' and 2 no view. The usual reason is that the strategic cards visibly waste drones while the tactical plans 'seemed to be doing what I was doing'. Dissent comes from P10, P06 and P07 (trusted strategic more), P31 ('the tactical layer required a little bit more human intervention'), P12 (strategic is the 'more trivial calculation') and P21/P31, who saw tactical flaws (no backup drone, illogical task order).

Quotes selected for this section.

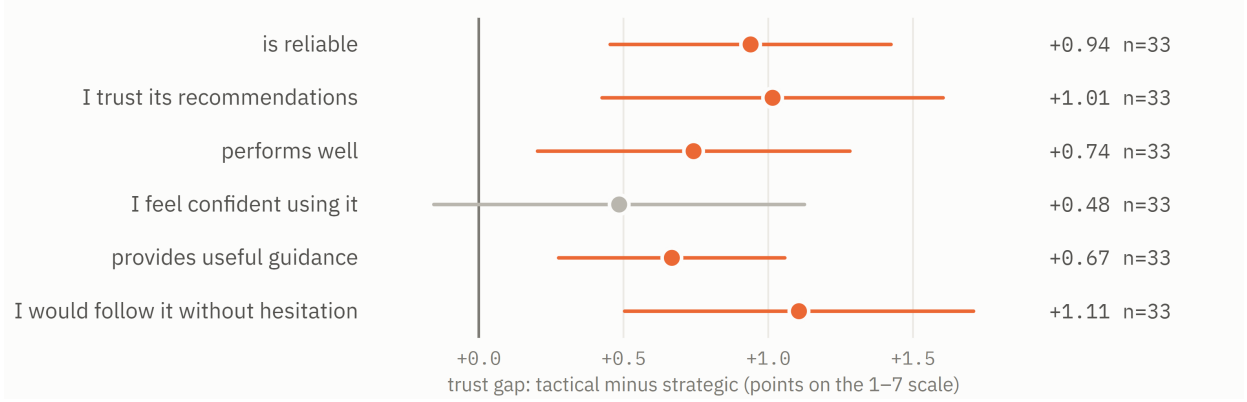
> *"But I think in the low level, the... Tactical level. Tactical level, it works very well." — P33 (debrief 0:28:24)*

> *"You can't blame the tactical for drone failures but you can blame the strategic for not having redundancy" — P03 (written interview)*

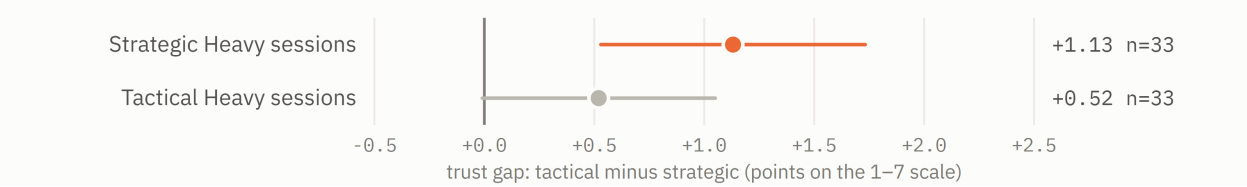
> *"So the tactical one seemed to be doing what I was doing. So quite reliably. And obviously the strategic one wasn't." — P32 (debrief 0:23:52)*

> *"But tactical was pretty good." — P13 (debrief 0:23:08)*

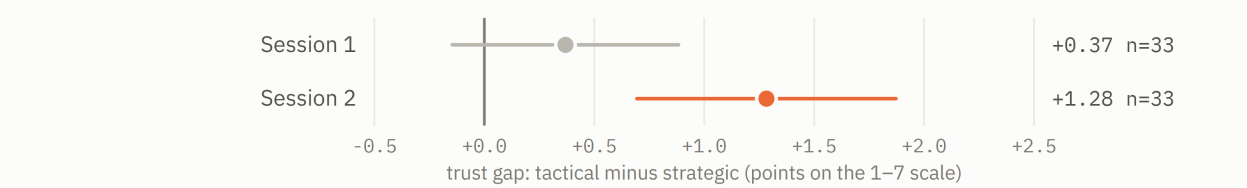
7.9.2 Where the gap comes from: the second session and the Strategic Heavy scenario



(a) By questionnaire item.



(b) By scenario played.



(c) By session order as well.

Figure 34: Where the gap comes from: the second session and the Strategic Heavy scenario

Figure 34. The gap is larger in Strategic Heavy sessions (+1.13) than in Tactical Heavy ones (+0.52, CI just reaches zero). It is also small after the first session (+0.37, CI crosses zero) and large after the second (+1.28). Both matter when taken together: holding session order fixed, Strategic Heavy adds about +0.6 to the gap and being the second session about +0.9. Tactical trust hardly moves (5.36 → 5.51 across sessions, and only +0.15 between scenarios); it is **strategic** trust that falls, and it is lowest in Strategic Heavy sessions. So "trusted more, especially in Strategic Heavy" largely means "the strategic one loses trust, especially in the scenario where its cards are used most". Four people said their trust *grew* with experience (C16). The chart opens on the six questionnaire items (every one favours tactical); **Scenario played** shows the two scenarios; **Also: first / second session** shows session order.

Key numbers.

- Session 1 gap +0.37 [−0.15, +0.88]; session 2 +1.28 [+0.69, +1.87] (n = 33 each)
- Strategic Heavy sessions +1.13 [+0.53, +1.73]; Tactical Heavy sessions +0.52 [−0.01, +1.05] (n = 33 each)
- Same people, Strategic Heavy minus Tactical Heavy: +0.61 [−0.01, +1.23], paired t p = .054, Wilcoxon p = .038; strategic trust −0.46 (p = .087), tactical +0.15 (p = .25)
- Both in one model: Strategic Heavy +0.64 [+0.11, +1.17], p = .019; second session +0.93 [+0.40, +1.47], p < .001
- Strategic trust 5.00 → 4.23 from session 1 to 2; tactical 5.36 → 5.51

Method. The tactical-minus-strategic difference for each session (66 sessions). By session: averaged over the 33 first and the 33 second sessions. By scenario: both games of each scenario pooled (33 sessions each). Every person played one session of each scenario, so the scenario comparison is also done within person (paired t and Wilcoxon on each person's two gaps). Session order and scenario are partly tangled (about half played Strategic Heavy first), so a model with both (GEE, gap ~ Strategic Heavy + second session, clustered by person) separates them. Means have t-based 95% CIs.

What participants said. Those who name a session say it was the strategic tier that slipped in the second session: P19 ('you dropped the ball a little'), P33, P15 and P17. Tactical trust was steady or up (P19, P26). Against the pattern, P06 says strategic trust went up over time and P20 saw strategic issues only in the first run. Nobody tied the change to the Strategic Heavy scenario as such.

7.9.3 Why the tactical plan is kept: checking it is brief and does not grow with the plan

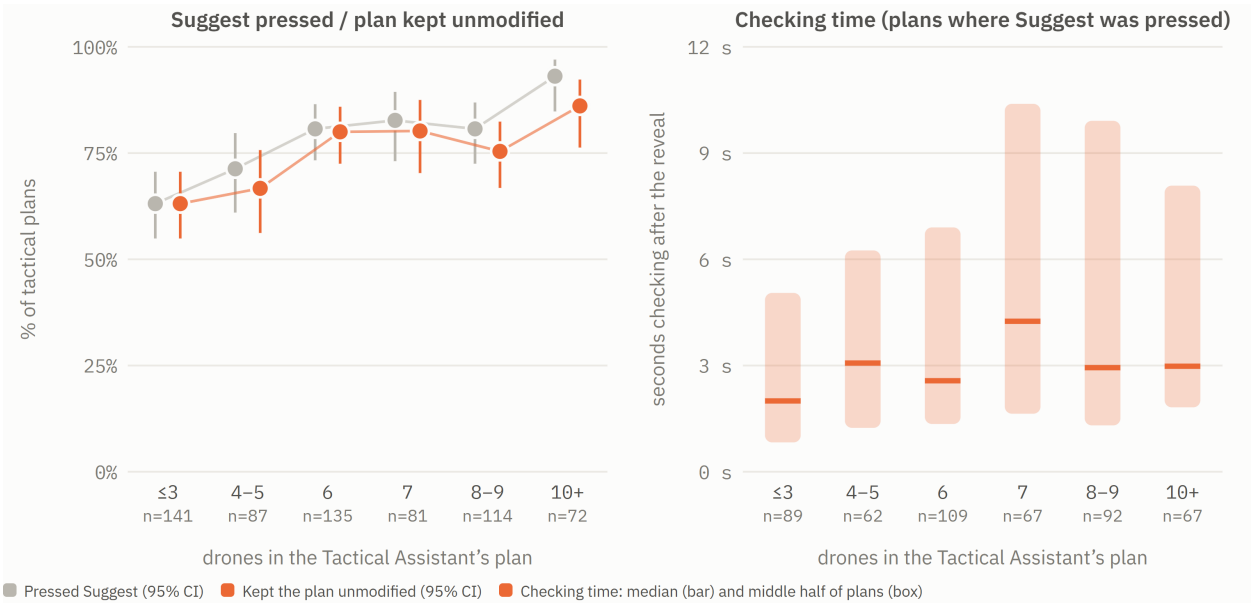


Figure 35: Why the tactical plan is kept: checking it is brief and does not grow with the plan

Figure 35. Bigger missions mean bigger tactical plans. As plans grow, operators press Suggest more often (63% of plans with 3 or fewer drones → 93% with 10 or more) and keep the plan unmodified more often (63% → 86%). Once Deploy unlocks, they spend a median of about **2–4 seconds** on the plan whatever its size. Per drone, checking therefore falls steeply as plans grow (0.7 s → 0.2 s). A strategy card gets about 9 s. Almost every consulted plan is deployed as it comes: 468 of 486. The more there is to check, the less of it is checked proportionally, and the plan is taken. Operators describe exactly this. Nine say they check when it is cheap and skip it under pressure (C17). Three say the plan was too complex or too fast to check (C14). Five say they checked nearly everything, and they do decide more slowly (C18). Only 3 of 33 ever mention noticing a tactical flaw (C13).

But the pre-registered mechanism test (H2) is fragile. All three predicted parts point the right way (p = .019 uncorrected), but it does not survive correction for the round's several tests (.065). Among consulted plans, larger ones are actually modified a little *more*. Keeping goes up with size mainly because people consult more on big plans, not because they check consulted plans more lightly. The logs also cannot tell "too costly to check" apart from "nothing ever looked wrong". The tactical plan's flaws are hard to see, while the cards' over-allocation is visible. That half of the explanation belongs to *tier-flaw-visibility* in Section 7.8 and is not repeated here.

Key numbers.

- Kept unmodified rises with plan size: OR 1.76 per SD [1.25, 2.47], p = .001 (all 630 plans)

- Consulted rises: OR 2.52 [1.87, 3.38]; kept once consulted FALLS: OR 0.68 [0.50, 0.94], p = .017
- By bin ($\leq$3, 4–5, 6, 7, 8–9, 10+ drones; n = 141, 87, 135, 81, 114, 72 plans): Suggest pressed 63%, 71%, 81%, 83%, 81%, 93%; kept unmodified 63%, 67%, 80%, 80%, 75%, 86%
- 10+ drones vs $\leq$3: Suggest pressed 93% vs 63%, OR 10.8 [3.2, 36.0]; kept 86% vs 63%, OR 5.2 [2.1, 12.7], both p < .001
- Median checking time by bin: 2.0, 3.1, 2.6, 4.2, 2.9, 3.0 s; per SD +0.60 s [$-$0.15, +1.35], p = .12; per drone 0.68 s at $\leq$3 drones $\rightarrow$ 0.23 s at 10+
- 468 of 486 consulted plans deployed unmodified (96%)
- Person-median check: tactical 2.85 s vs strategy card 8.80 s (n = 33, Wilcoxon p < .0001)
- H2 primary p = .019 uncorrected, .065 after Holm correction across round 1

Method. 630 confirmed tactical plans from 33 people (h2_verification_cost.load). They are grouped by the number of drones in the Tactical Assistant's plan into six bins with no gaps: $\leq$3, 4–5, 6, 7, 8–9 and 10+ drones. The bins were chosen so every bin has enough plans for a stable rate: at least 72 plans and at least 62 with a checking time in each (the smallest bins are 4–5 drones, 87 plans, and 10+, 72 plans). Every plan is used; none is dropped. "Consulted" means Suggest was pressed at least once. "Kept unmodified" means consulted and deployed with no change. "Checking time" runs from the last Suggest press to Deploy, minus the forced reveal: the planner adds one drone$\rightarrow$task pair per second and Deploy is locked until it finishes, so that wait is not checking. Effects per SD of plan size (drones $\times$ tasks) come from GEE models on all 630 plans, clustered by person, adjusted for scenario and session; they do not depend on the bins. An odds ratio of 1.76 means each step up in plan size multiplies the odds of keeping the plan by 1.76. The two ends (10+ vs $\leq$3 drones) are also compared directly. The card comparison takes each person's median net decision time on strategic decisions: strategic_choice latency minus the card reveal, over all strategic decisions, manual ones included. The H2 numbers are from docs/analysis/round1/h2_verification_cost.md.

What participants said. Speaking to this point: 3 of 33 say the tactical plan was too complex or fast to check (C14), 9 that checking depended on time (C17), and 5 that they always checked (C18). The reasons match the finding: the plan is hard to check, so it is taken on trust ('I just took the AI's word for it'). Several say it is not more trust but the time to check (P04). P14 is the clear dissent: they checked tactical plans more, drone by drone, because cards can be read 'in one glance'.

Quotes selected for this section.

> *"as soon as it got to any sort of non-linear, difficult, high dimensional problem, I just took the AI's word for it, essentially, probably because I couldn't assess in the time" — P22 (debrief 0:30:06)*

> *"Quite extensively. [...] Yeah, because it's an emergent scenario, so I don't have enough time to just check the details. So I think the AI suggestions is quite useful." — P27 (think-aloud s1 3:33)*

> *"My reason is there are so many drones, so I don't have enough energy and attention to just the check zone." — P27 (think-aloud s1 0:56)*

7.9.4 Stated vs revealed preference: they line up in rank, not in size or over time

Figure 36. Does the person who *says* (rates) the Tactical Assistant higher also *lean* on it more? Ranked against each other, yes, moderately. The person-level trust gap correlates ρ = .47 with the usage gap (share of tactical plans kept unmodified minus share of strategic decisions where a card was taken). 16 people favour tactical on both measures and 3 favour strategic on both. Three things do **not** line up:

(1) **Size.** Stated trust favours tactical strongly (26 of 33), but use barely does. Cards are taken on 64% of strategic decisions and tactical plans kept on 72%, a gap of +8 points whose CI includes zero. Nine people rate tactical higher yet use the strategic cards more.

(2) **Change.** Within a person, a rise or fall in trust from session 1 to 2 does not go with a matching change in use (ρ = .16 strategic, .13 tactical).

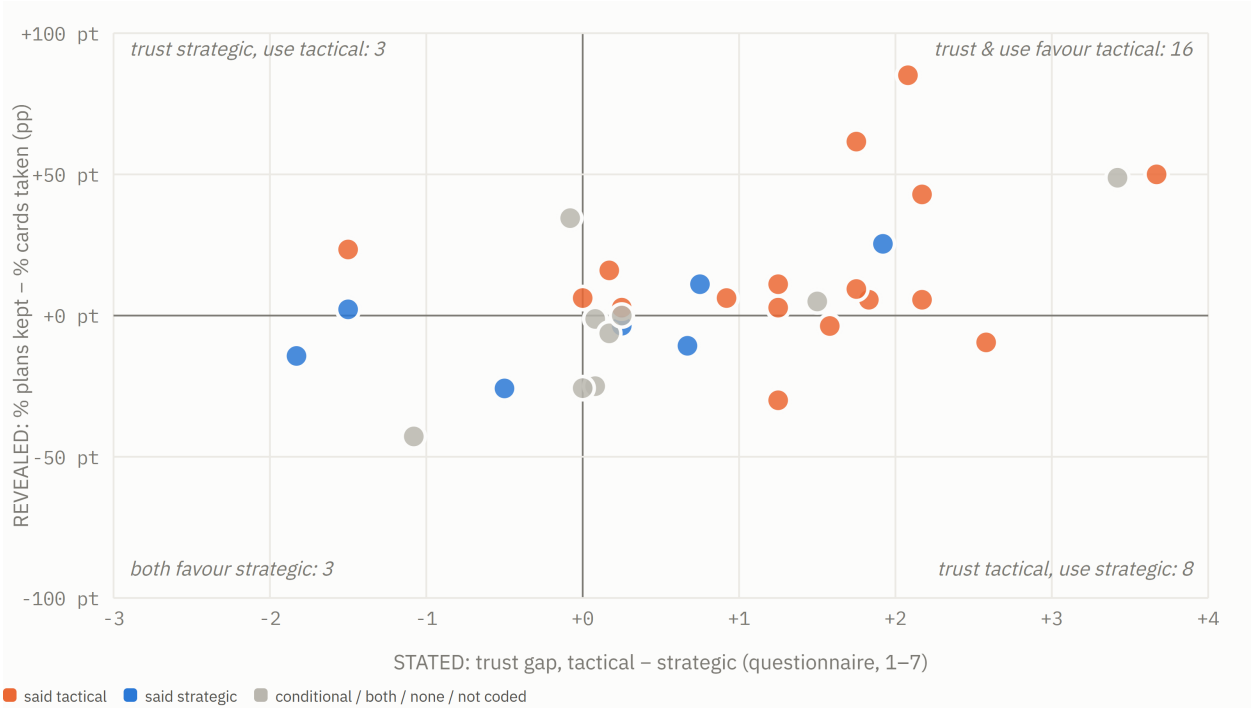


Figure 36: Stated vs revealed preference: they line up in rank, not in size or over time

(3) **Lead.** Session-1 trust does not predict how card use changes in session 2 ($\rho = .07$). Trust ratings read like a stable opinion of the assistant, while use follows the moment: mission size, queue, time. The ratings describe the person; they do not forecast their next move.

Key numbers.

- Trust gap vs usage gap (kept − took): $\rho = .47$ [.10, .73], p = .006, n = 33
- Quadrants: both favour tactical 16, both strategic 3, trust tactical/use strategic 9, trust strategic/use tactical 3 (2 on a zero line)
- Group use: took a card 64%, consulted the plan 76%, kept it 72%; usage gap +0.08 [−0.02, +0.17]
- Within person, change in strategic trust vs change in card use: $\rho = .16$ [−.27, .52], p = .38
- Session-1 strategic trust vs change in card use s1→s2: $\rho = .07$ [−.27, .40], p = .71; tactical .01
- Between people, strategic trust vs cards taken $\rho = .60$ [.30, .78]; tactical trust vs plan kept $\rho = .31$, p = .08

Method. One dot per person, both sessions pooled. Stated = mean tactical trust minus mean strategic trust (1–7). Revealed = share of their confirmed tactical plans kept unmodified minus share of their strategic decisions where they took a card (the alternative "pressed Suggest at all" gives the same picture, $\rho = .49$). Spearman ρ uses 4000-draw seeded bootstrap CIs. A caveat on (1): taking a card and keeping a plan are different acts with different costs, so the *level* of the usage gap is not a pure preference; the ranking is the cleaner comparison. "Change" correlates the session-2 minus session-1 differences within each person (33 people). "Lead" correlates session-1 trust with the later change in use. Session-level links are also shown (66 sessions): strategic trust ~ cards taken $\rho = .49$, tactical trust ~ plan kept $\rho = .27$. Note that the trust survey comes *after* the session it follows, so a same-session link can be use shaping trust as easily as the reverse.

What participants said. Several people say outright that how much they used an assistant was not the same thing as how much they trusted it. They took cards they rated lower because cards save time ('less reliable… but I still

use its suggestion', P28). P20 lost trust in the strategic tier yet found it 'a bit more useful', and P11 would 'miss both equally' but use the strategic one more. This matches the gap between trust and use found in the data.

7.9.5 All the trust–use correlations in one view

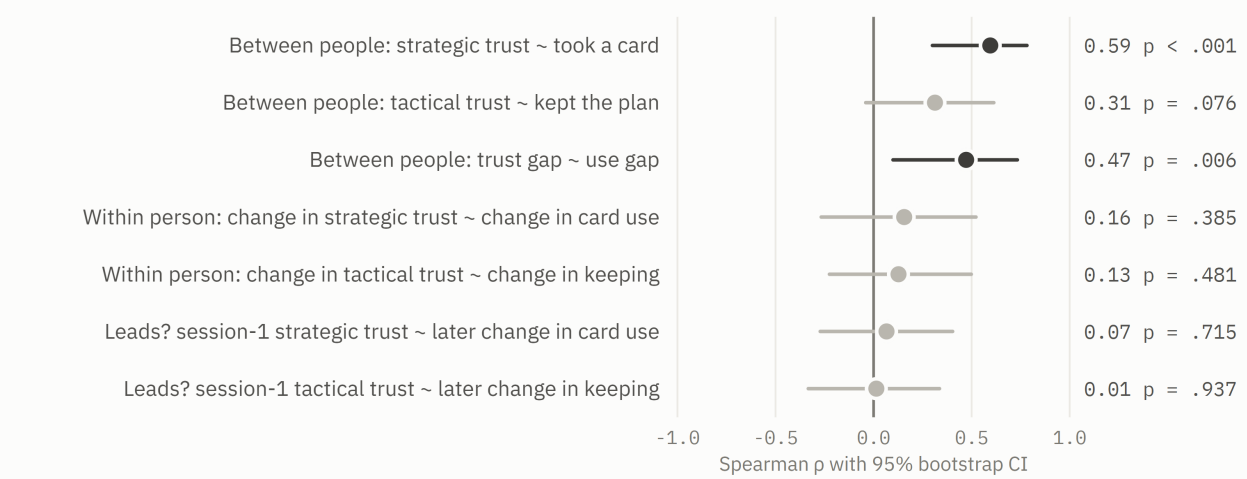


Figure 37: All the trust–use correlations in one view

Figure 37. Between people, trust and use move together, most clearly for the strategic tier. Within a person over time, and as a predictor of the next session, the correlations are near zero.

Method. Each row is a Spearman ρ with a 4000-draw seeded bootstrap 95% CI, at the person level (n = 33). A dark dot means $p < .05$. These are the same numbers as the section above, in one picture.

What participants said. 4 of 33 say their trust grew with experience or success (C16: P06, P10, P26, P32), and a few more describe a checking-then-trusting arc in play (P07, P06, P29). These accounts are all retrospective and smooth, while the data show no within-person link from trust to later use. Nobody describes trust falling and use following it within a session.

7.10 Beyond the findings: what operators said they want, believe and feel

Summary. Asked about real search and rescue, nobody rejected assistants like these, and 9 of 33 said a human must keep the final decision. What they asked for was mostly information: how the assistants decide (about 12) and failure risk or feedback on how they were doing (11). Some wanted help with a decision neither tier covers, choosing which mission to open next (P27, P32). How they pictured the assistants ranged from "a dozen lines of code" to something that learns or predicts, and several credited them with more than they had (learning from the operator, prediction, one shared plan across the tiers). Suggestive: one coder, and much of it unprompted talk during play.

Strength of evidence (register rating): suggestive.

7.10.1 What they asked for

Figure 38. The common thread is control plus information: keep the human in the decision and give them more to decide with (why the assistant suggested it, how likely drones are to fail, options in between). Only four wanted the assistant to do more. These are design requirements more than findings — and the "reasoning" request links to the tactical blind spot, where an explained refusal could have been checked.

Key numbers.

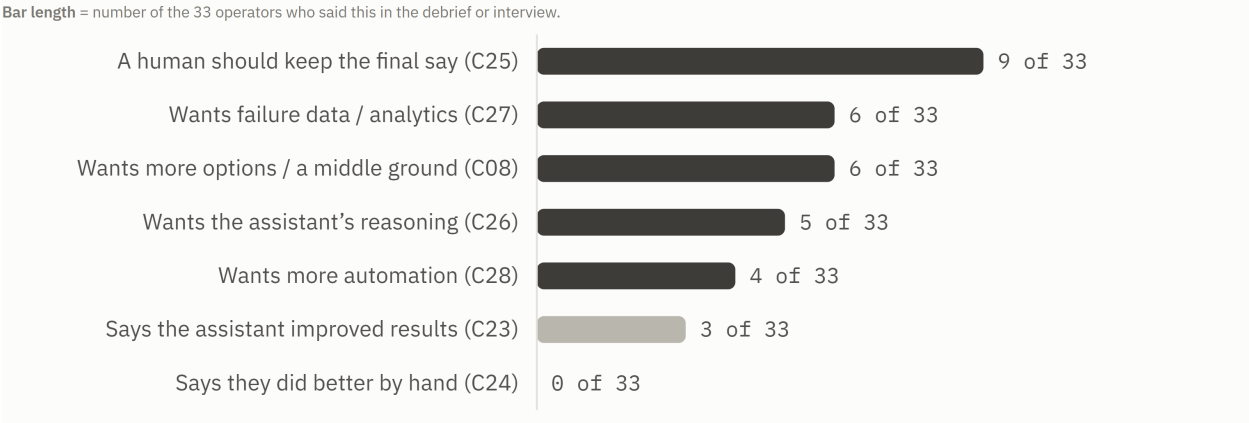


Figure 38: What they asked for

- Human final say 9 of 33 (8 unprompted); failure data 6; more options 6; reasoning 5; more automation 4 (all unprompted)

Method. Debrief transcripts and interview notes coded against the codebook (CODEBOOK.md): C25 human final say, C27 failure data, C08 middle ground / more options, C26 reasoning, C28 more automation. "Unprompted" = at least one piece of evidence not given as a bare yes to a leading question.

What participants said. Across the coded material, 9 of 33 said a human should keep the final say (8 unprompted), 6 wanted failure probabilities or performance data, 6 wanted options between the two cards, 5 wanted the assistant's reasoning and 4 wanted more automation. Reading everything (including the in-play talk) raises the information asks: about 11 asked or wondered how the assistants decided, and 8 asked for failure or performance numbers. The dissent is small but sharp: P11 does not want to be 'reduced to pressing allocate, suggest, deploy', and P36 warns that what you hand over you no longer have in your head when it fails. The mini sections below take each strand in turn.

7.10.2 Real search and rescue: welcome, with a human deciding

What people said when asked how they would feel about systems like this in real search and rescue. 16 of 33 are quoted here (25 answered the question); nobody rejected it, and the common condition was a human who keeps the final decision.

What participants said. Asked the closing 'five years from now' question, at least 14 welcomed the idea, nearly always for speed ('a bad approximation is better than no response at all'), and 9 said a human must keep the final decision (8 unprompted). Conditions came with it: well regulated (P20), only where the job is understood and stakes low (P03, P05), only if better than today's AI (P15). Four raised a deeper worry: who is accountable when an AI fails to save someone (P31, P22), a confident plan with a hole in it, repeated at scale (P36), and military use of the research (P31). When pressed on whether a busy human could really check, P24 conceded they could not and still wanted one there.

7.10.3 Wanting to know how the assistants decide

Requests for the assistants' reasoning, and moments where people could not work out why a card or plan looked the way it did. 12 of 33 said something on this.

What participants said. 5 of 33 asked outright for explanation, auditability or understandable tools (P02, P12, P16, P25, P36); 6 more asked mid-play why a card looked as it did or said they could not tell (P01, P05, P06, P13, P25 in play, P32). P01 named the bind: more information to handle means relying more and trusting less. The requests point both ways: P05 found the strategic cards the hard ones to understand, while P19 found the card simple enough that its

error was visible, and P22 said they trusted the tactical plan partly because they could not see how far from optimal it was. Nobody asked for an explanation of a single plan step; the asks were about how the assistant works in general.

7.10.4 Failure risk and feedback: wanting the numbers

Asks for drone-failure probabilities and for feedback on how well they (or the assistants) were doing, and misreadings of the random failures. 11 of 33 said something on this.

What participants said. 8 of 33 asked for failure probabilities or performance data (P10, P11, P12, P13, P25, P31, P32, P36); P10, P11 and P13 independently wanted the risk tied to place, terrain or drone type and fed into the allocation. Without it, 2 read patterns into failures that were in fact flat and random (P19: some tasks fail more; P01: the rate was 'cranked up'). 4 could not tell from the score how well they were doing (P10, P29, P31, P32). P36 adds the caution: a raw failure percentage needs context, or operators will refuse any risk, though 'the failure rate of doing nothing is 100%'.

7.10.5 More automation, and new jobs for an assistant

Wishes for the assistants to do more, or to do a job neither of them does now, and the few voices against. 8 of 33 said something on this.

What participants said. 4 of 33 asked in general for more automation (P13, P27, P30, P31) and 4 more proposed a specific new job: choose which mission to open next (P32, and P27's feasibility-then-criticality ordering), check the operator's own plan for slips (P11), red-team the assistants' outputs (P36), build back-up routes into the tactical plan (P21), re-route spares after a failure (P30) and look ahead to missions not yet arrived (P30). The common thread is help with what neither assistant touches: choosing what to work on, and failures. Against: P11 does not want to be reduced to clicking, and P36 warns that handing over the plan means playing catch-up when it fails.

7.10.6 What they thought the assistants were

How people pictured the assistants: simple rules or 'AI', whether they learn, predict or see beyond one mission. 13 of 33 said something on this.

What participants said. Two pictures sit side by side. 6 saw through to simple rules and judged them on that: 'a dozen lines of code' (P31), 'a matching algorithm' (P19), 'a little bit stupid', 'old computing' (P15), works but not optimal (P04); P32 and P04 said they would trust a provable optimiser over machine learning. Others credited the assistants with more than they had: P03 hoped it would learn from their choices, P27 thought it predicted, P13 wondered if it knew fast drones fail more, P21 assumed the two tiers share a plan. 2 noticed the cards look at one mission only (P24 had to ask, P30). 2 more took the illogical card for a bug or a planted error (P17, P27).

7.10.7 How the two tiers fit together

Remarks on the hierarchy itself: which level a human should hold, and how a strategic choice carries into the tactical one. 10 of 33 said something on this.

What participants said. The shared view is that the strategic choice sets up everything below it: a good allocation makes the tactical stage hard to fail, and drone failures can only be absorbed by drones sent at the start (P03, P19, P36). Several said the human belongs at the strategic level and the assistant at the tactical one, 'give it resources and allow it to optimise' (P05, P30, P32, P33 when asked, P12 reasoning that the strategic sum is the easier one), and P10 found an Aggressive card leaves only one sensible tactical plan. The cross-current: P21 builds the tactical plan themself because their strategic choice already carries a plan in their head, and P36 frames it as human-in-the-loop now, with on-the-loop needing far more trust.

7.10.8 Me or the assistant: who does it better?

Brought back from the sidelined 'who does better' finding: what people said about whether the assistant or they themselves got better results. 13 of 33 said something on this.

What participants said. Coded strictly, 3 of 33 said the assistant improved their results and none said they did better by hand. Read more widely, about 7 rated the assistant as good as or better than themselves (P06, P11, P24, P29, P31, P33, P35), nearly always because it was faster rather than wiser. 4 said they could do better given time or care (P04, P12, P17, P35) and P22 judged them about equal; P31 and P21 said they simply could not tell. Self-doubt shows through (P01 'I don't trust myself anymore', P35). This fits the data: the gain sits between people, not in what the assistant did for any one of them.

7.10.9 Interface and display asks

Concrete requests about the screens, plus the praise. 11 of 33 said something on this.

What participants said. 8 of 33 asked for something on screen: a total drone requirement per mission (P10, P13), a cue on one screen when the other needs attention (P10), drones-in-hub and feasibility in the queue (P27; the criticality sort exists but they had not found it), filtering by penalty (P19), confirmation before cancelling a task (P13), the missing green icon (P36), and keeping a Suggest plan when clicked away (P01). P06 said the task needs two monitors and P31 that the central map went unwatched. Praise came from 4 (P10, P14, P25, P07: easy, intuitive). Two people (P13, P14) were confused by the reversed NASA-TLX performance slider, which matters for that item's data.

7.10.10 It's a game: what real stakes, pace and training would change

How people saw the realism of the task: no lives at stake, the pace, what the simulation leaves out, and the training a real operator would need. 12 of 33 said something on this.

What participants said. 9 of 33 said real lives would change how they played: more stress (P01, P11, P15, P34), less decisiveness (P36), more care about efficiency (P07), while P01 expected to behave the same and saw that as the problem. Some rules read as unrealistic: abandoning a mission (P02, P13), partial rescue (P28: a drone that finds someone and leaves), task locations known in advance (P28, P36), the pace (P31, P36). 4 said real operators would need training or new skills (P29, P35, P36, P13), P36 putting the skill, not the trust, first.

7.10.11 Feelings: fun, stress, self-doubt, and talking to the assistant

The emotional side of the sessions: enjoyment, stress, doubts about their own ability, and treating the assistant like a person. 14 of 33 said something on this.

What participants said. 7 said it was fun or wanted to win (P01, P10, P11, P18, P24, P31, P36). Stress and drama ran alongside it, mostly around failures and empty reserves (P01 'I'm losing my job', P22 'everyone is going to die'). 5 voiced doubts about themselves (P01, P29, P33, P35, P27 not liking to decide fast), and for P27 and P35 it was a stated reason to rely more on the assistant. 4 talked to or personified the assistants: P19 scolded and apologised to the Strategic Assistant, P22 and P32 urged it on, P11 felt the pull to 'personify the computer and feel some sort of grudge', alongside pride when doing it themself.

8 Discussion

[To write. Material for it: the f-operator-wants finding (Section 7.10), the open questions recorded in the register, and the old draft's Step-4 benchmarking note: compare against `adams2025supervise` (workload at scale), `wischnewski2023measuring`/`wildman2024trust` (across-tier and multilevel trust), the variable-autonomy cluster including `oberacker2026mosaic`, and `ramchurn2021trustworthy` for error propagation across a pipeline.]

8.1 Limitations

[Draft list from the register: one sample of 33, almost all computing/engineering and strongly AI-positive (AIAS-4 mean 8.0/10); a game, not real stakes; four fixed games played in an order that depended on when a person took part; build changes during collection (Appendix B); single AI coder for the qualitative counts; the illogical card and the wrong refusal are side effects found after collection, not manipulations; trust measured only after each session, so a same-session trust–use link can run either way.]

9 Conclusion

[To write.]

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A Simulation Parameters

Table 4: Simulation parameters (scenario parameter set v2.1). Everything except the arrival rate and the category weights is identical in the two scenarios.

Parameter	**Description**	**Value**
Fleet		
Fast (blue)	count; speed	11; 11.0 u/s
Lifter (red)	count; speed	11; 10.0 u/s
Camera (green)	count; speed	11; 9.0 u/s
Missions		
Category penalty rate	A / B / C / D / E	0.05 / 0.10 / 0.15 / 0.25 / 0.40 pts/s
Archetype weights	Fast / Lifter / Camera / Mixed	38 / 26 / 21 / 15
Zone	radius; min. distance to hub; to other zones	80; 150; 200 units
Scenario-specific		
Strategic Heavy	mean inter-arrival; category weights A–E	37 s; 40/38/16/5/1
Tactical Heavy	mean inter-arrival; category weights A–E	75 s; 5/13/28/38/16
Session, assistants, failures		
Session length	per session	480 s (8 min)
Seed	all sessions	42 (four games; Section 6.1)
Strategy card reveal	per card, independent	≈4–5 s
Conservative card	spare per used type; top-up of remaining reserve	+1; 15% of every type
Suggest fill rate	tactical plan	1 drone→task entry per second
Drone failure hazard	per deployed drone per second	1/900 (`v1.4`+)
Post-failure grace	no further failure on that mission	30 s after recovery (`v1.5`+)
Penalty display step		15 s
Assistant error rates	ε_S, ε_T	0, 0 (all analysed sessions)

Table 5: Platform builds used during data collection, with the changes that bear on the data. The build is recorded in every session's log; the full history is in `docs/STUDY_BUILD.md`.

Build	Participants	Relevant to the data
`v1.0`	P01	Task points half the later values (rescaled in analysis); failures from a per-mission schedule.
`v1.2`	P02	As above; abandoned work re-queued as a residual mission from here on.
`v1.3`–`v1.4`	P03	Points doubled; pre-study AI-attitude scales added; live failure hazard (1/900 per drone-second) on a fixed grid from `v1.4`; substitute compositions executable.
`v1.5`–`v1.6`	P04	Recovery Suggest produces plans; 30 s post-failure grace; a half-built recovery plan survives a task completing.
`v1.7`–`v1.9`	P05–P07	Strategy-card score bars corrected (Speed = best ETA / ETA; Reserve = share of reserve left); allocating clears the tactical screen; deploying closes the planner.
`v1.10`–`v1.11`	P10, P11	Error-injection machinery added, switched off ($\varepsilon = 0$).
`v1.12`	P12–P30	NASA-TLX performance slider carries a "runs backwards" note.
`v1.13`	P31–P36	Games swapped: the same four mission streams played in the other session position.

B Build Versions

[P09 (`v1.11`, errors on) excluded. Check the table against docs/STUDY_BUILD.md before submission; the register's sensitivity analyses on the `v1.12`+ cohort are in the analysis scripts.]

Table 6: The same measures by the four fixed games (mean (SD) over sessions). SH = Strategic Heavy, TH = Tactical Heavy; the number is the mission stream (plan index), not the session order. Sessions per game: SH1 17, SH2 16, TH1 16, TH2 17.

Measure	SH1	SH2	TH1	TH2
Missions arrived (non-residual)	13.0 (0.0)	14.0 (0.0)	6.0 (0.0)	9.0 (0.0)
Strategic decisions made	10.8 (2.3)	12.1 (2.3)	6.2 (1.4)	7.6 (1.2)
Tactical plans confirmed	10.6 (2.5)	11.8 (2.4)	6.1 (1.5)	7.5 (1.1)
Tasks completed	25.1 (4.4)	28.0 (7.2)	21.2 (6.0)	32.8 (7.8)
Completion points[a]	1266 (248)	1288 (384)	1169 (331)	2058 (480)
Time penalty (points)	288 (52)	264 (66)	232 (79)	549 (92)
Score (points − penalty)[a]	978 (284)	1023 (443)	940 (398)	1509 (567)
Strategy card taken (% of decisions)	58 (23)	61 (31)	78 (29)	68 (26)
Manual before cards loaded (%)	15 (19)	22 (27)	11 (26)	11 (14)
Card edited, then deployed (%)	10 (21)	0 (2)	1 (4)	10 (19)
Suggest pressed (% of plans)	75 (22)	68 (30)	75 (35)	84 (21)
Suggest plan kept unmodified (% of plans)	72 (24)	66 (32)	74 (35)	76 (29)
Drone failures	3.2 (1.7)	4.9 (3.1)	3.1 (2.0)	4.5 (2.9)
Missions abandoned	1.0 (1.1)	1.4 (2.0)	0.7 (0.9)	0.9 (0.9)
NASA-TLX, raw mean (0–20)	9.4 (3.1)	8.2 (3.1)	6.5 (2.9)	10.6 (2.8)
Mental demand	13.1 (3.8)	10.2 (5.5)	7.0 (5.2)	14.7 (4.2)
Temporal demand	13.5 (3.5)	13.7 (4.4)	11.2 (5.4)	13.4 (4.5)
Performance (0 = perfect)	8.1 (4.2)	6.9 (5.6)	7.8 (5.8)	9.8 (3.8)
Effort	10.2 (4.7)	8.9 (4.7)	5.9 (3.4)	11.7 (4.0)
Frustration	6.9 (4.8)	5.1 (5.3)	3.6 (3.9)	8.7 (6.3)
Trust, Strategic Assistant (1–7)	4.35 (1.29)	4.42 (1.77)	4.95 (1.43)	4.75 (1.26)
Trust, Tactical Assistant (1–7)	5.45 (1.05)	5.58 (0.67)	5.43 (0.95)	5.30 (0.94)

C Supplementary Results

Sidelined analyses from the analysis register: robustness checks, alternative specifications and closer looks that support the main results but are not needed to follow them.

C.1 Each operator has a personal style — individual, not a type (supplementary analyses)

Supplement to Section 7.2.

C.1.1 After one session we can recognise a person's style in the next

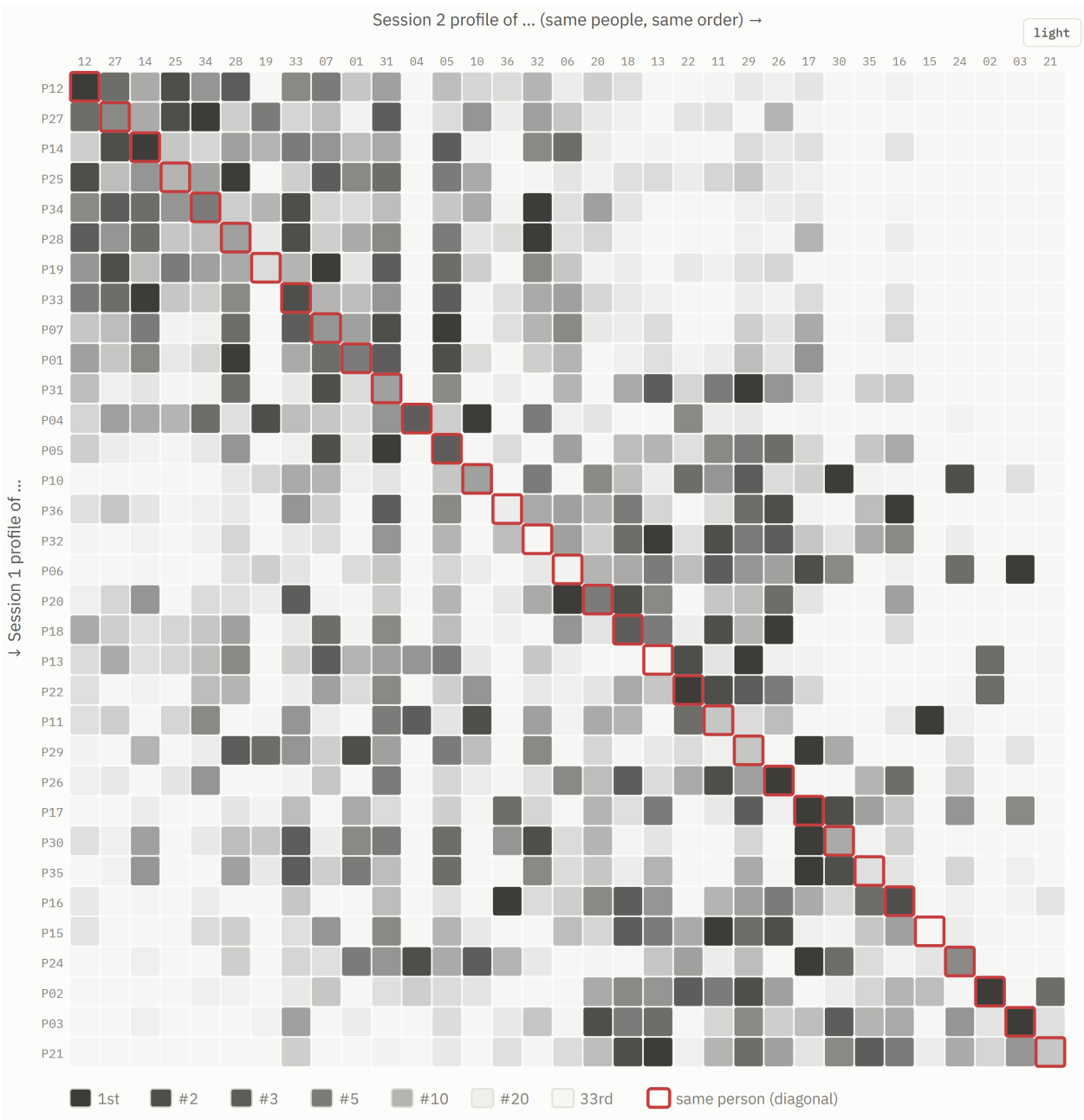


Figure 39: After one session we can recognise a person's style in the next

Figure 39. Each session is summarised as ten habits (below). Comparing everyone's session-1 profile with everyone's session-2 profile, the closest match is far more often the **same person** than chance allows. It is a "near neighbour" result, not exact identification: the own session is the single closest match for 7 of 33 people (chance: 1), but in the closest three for 41% (chance 9%), and on average ranks 8th of 33 (chance 17th). It survives taking the reliance axis out — so the style is more than "how much do you lean on the assistants". In the matrix, every row is one person's session 1, every column a session 2, darker = closer; the outlined diagonal is each person against themselves.

Key numbers.

- Own session nearest: 7 of 33 (session 1 $\rightarrow$ 2) and 5 of 33 (2 $\rightarrow$ 1); chance $\approx$ 1; permutation $p < .001$
- Own session in the closest 3: 41% vs 9% by chance; mean rank 8.2 of 33 vs 17.0 ($p < .001$)
- Reliance axis removed: 6 and 6 of 33, top-3 32%, mean rank 10.1 ($p < .001$) — the style is more than reliance
- Reliance axis alone: 4 and 4 of 33, top-3 27%, mean rank 11.1
- Profile correlation with own other session: median $r = .48$; with other people's: median $r = -.01$; own closer for 29 of 33 (sign test $p < .001$)

Method. Rows: 66 sessions (33 people $\times$ 2). Each session becomes a vector of ten habits: share of strategy cards taken; share of manual choices made before the cards loaded; share of cards edited; share of tactical plans where Suggest was pressed; share of Suggest plans kept unchanged; share of recovery episodes with Suggest pressed; time per strategic decision beyond the forced card delay (log); drones committed vs the Conservative card on single-type and on multi-type missions; card previews per decision. Each habit is turned into a z-score *within the game played* (so a hard game does not make everyone in it look alike; a habit with no data in a session, e.g. no recoveries, is set to that game's median). For each person's session-1 vector we measure the straight-line (Euclidean) distance to all 33 session-2 vectors and rank them; "rank 1" means their own session 2 is the nearest. Done both ways (session 1 looking among session 2s and vice versa). The p-value comes from shuffling who-is-who in session 2 10 000 times and counting how often chance does as well. "Reliance axis removed": the single strongest direction of variation (PC1, see below) is subtracted from every vector first. The profile correlation r is the Pearson correlation between a person's two ten-habit vectors (1 = same shape, 0 = unrelated). Exploratory and post hoc; the permutation test is clean. People are ordered by reliance in the matrix, so neighbours look alike — the claim is that the diagonal beats them.

C.1.2 What a personal style looks like

Figure 40. Every person's ten habits in session 1 (light) and session 2 (dark). The top rows are people whose two sessions have almost the same shape (r up to .89); at the bottom are the few whose style changed (P13, P29, P32). Most people repeat themselves.

Key numbers.

- 24 of 33 people have $r \geq .27$ between their two sessions; 4 have $r < 0$

Method. Each line joins the ten habit z-scores (within game, clipped at ± 2.5 for drawing). r = correlation of the two lines; #k = where their own session 2 ranked among 33 (1 = nearest).

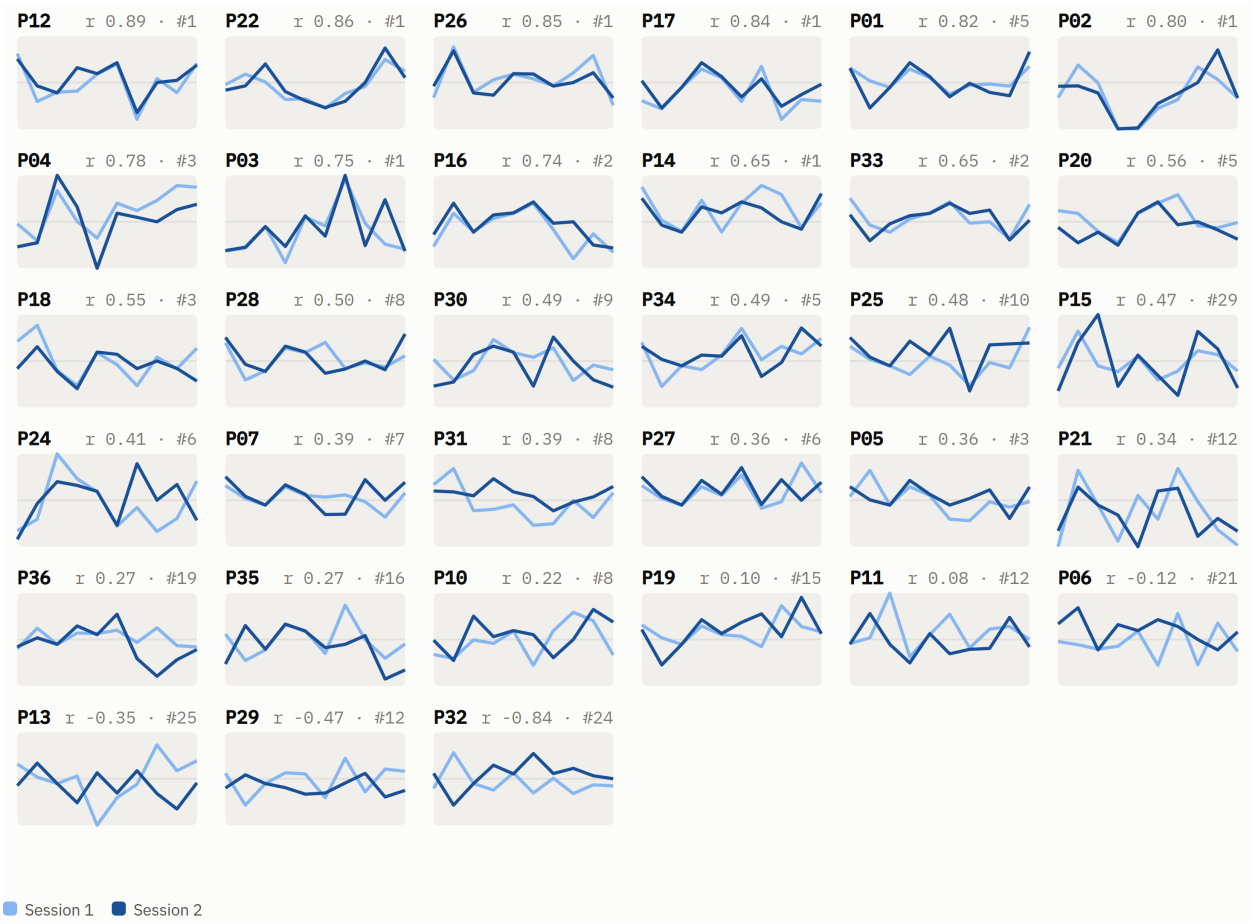


Figure 40: What a personal style looks like

C.1.3 Which habits carry over — and what each correlation means

Figure 41. Each small scatter is one habit: a dot is a person, x = that habit in session 1, y = the same habit in session 2. If people kept their place in the queue exactly, the dots would lie on the dashed line; if session 1 told us nothing, they would be a shapeless cloud. Six of the ten habits carry over clearly: keeping Suggest plans, pressing Suggest, taking cards, previewing cards, decision time, and Suggest in recovery. Topping up drones on single-type missions does not carry over at all.

Key numbers.

- Survive Holm: keeps Suggest plan ρ .65 [.34, .84]; presses Suggest .65 [.34, .84]; takes card .62 [.33, .78]; previews .56 [.25, .77]; decision time .52 [.20, .74]; Suggest in recovery .48 [.14, .74]
- Do not: edits a card .41; manual-before-cards (share of manual) .36; drones on multi-type .34; drones on single-type −.09
- Card uptake: ~2 distinct levels of people from one session, ~3 from two
- Score on the reliance axis, session 1 vs 2: ρ = .65 [.38, .82]

Method. ρ (Spearman) for habit X = the correlation, across the 33 people, between their rank on X in session 1 and their rank on X in session 2, each session first adjusted for the game played (z within game). ρ = .62 for "takes the strategy card" means people keep roughly the same order: high card-takers in session 1 are mostly high in session 2, though not exactly. 95% intervals come from resampling people 2000 times; Holm corrects for testing ten habits. "Distinct levels" (separation strata) says how many

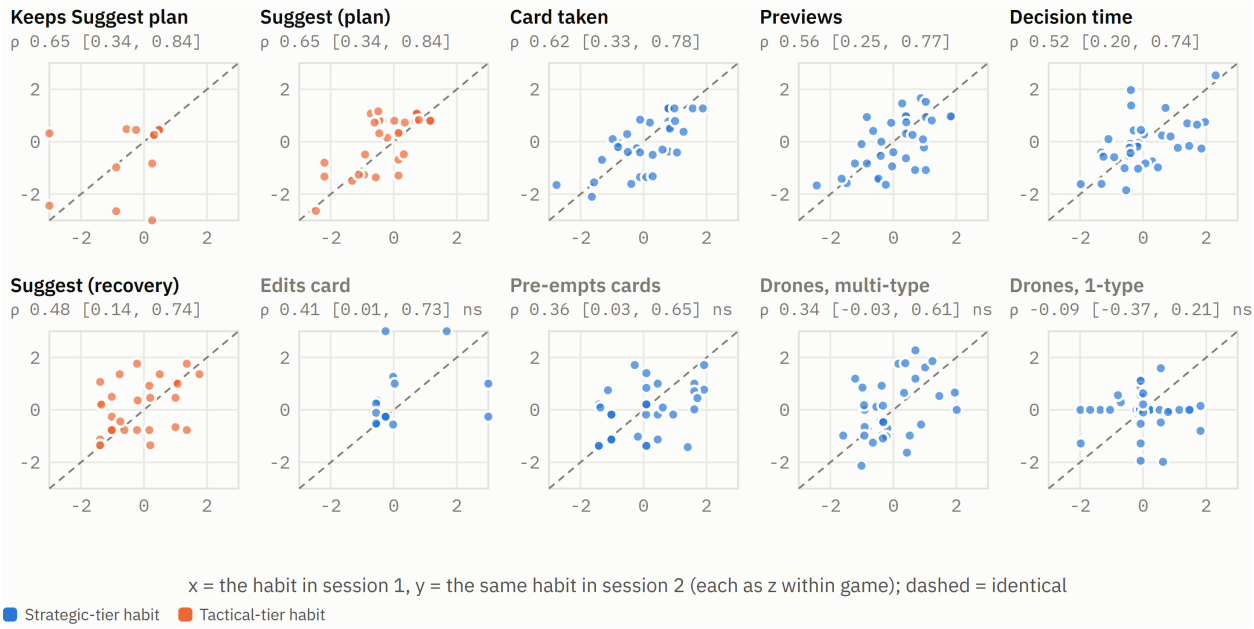


Figure 41: Which habits carry over — and what each correlation means

groups of people the habit can tell apart reliably: one session of card uptake separates about 2 (high vs low), the average of two sessions about 3. Caution on "keeps the Suggest plan": 24 of 33 people kept every Suggest plan unchanged in both sessions (a ceiling), so its high ρ mostly says that the nine who ever edit a plan tend to do so in both sessions. The pre-empting habit here is "share of *manual* choices made before the cards loaded" (undefined for 15 people who never chose manual in a session); the clearer per-decision version is in the next section.

C.1.4 Not waiting for the cards is one of those habits

Figure 42. Committing a manual allocation before the strategy cards have even loaded is something some people do and others never do. People who did it in session 1 did it on a quarter of their decisions in session 2; people who did not, on 8%. Fifteen people never did it in either session.

Key numbers.

- Session 1 vs 2: $\rho = .56$ [.25, .77], $p < .001$, $n = 33$
- Session-2 rate 25% for those who pre-empted in session 1 (n = 14) vs 8% for those who did not (n = 19)
- 86 of 564 decisions pre-empted; 46% of all manual choices; 15 of 33 never did it
- Round 2: person share of variance .24, mission .05

Method. Rows: 564 non-residual strategic decisions. Per person and session: share of decisions committed manually before the cards appeared (logged `manualBeforeCardsLoaded`). ρ = Spearman of the session-1 share against the session-2 share across 33 people. Variance shares (person ~24%, mission ~5%) are from round 2 (F4.1, crossed random intercepts for person and the exact mission, which repeats across people because games are fixed). The 15 "never" people sit together at the origin; ρ is driven both by them staying at zero and by the regular pre-empters staying high.

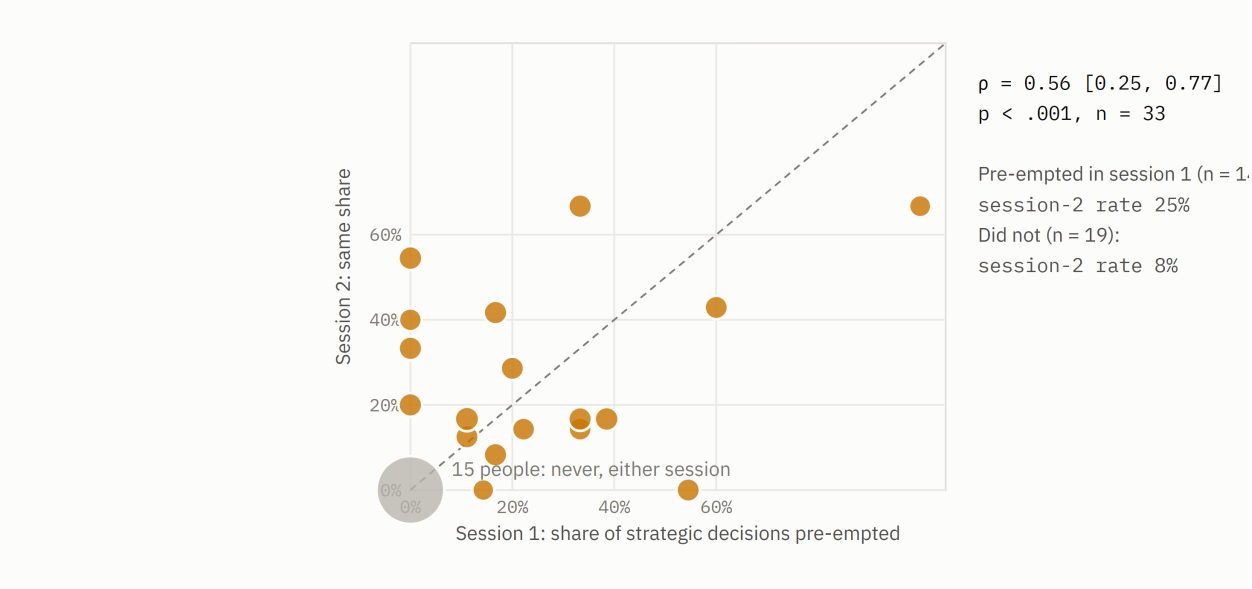


Figure 42: Not waiting for the cards is one of those habits

C.2 Small jobs by hand, big jobs to the assistant — on both tiers, and it is where the assistant saves time (supplementary analyses)

Supplement to Section 7.3.

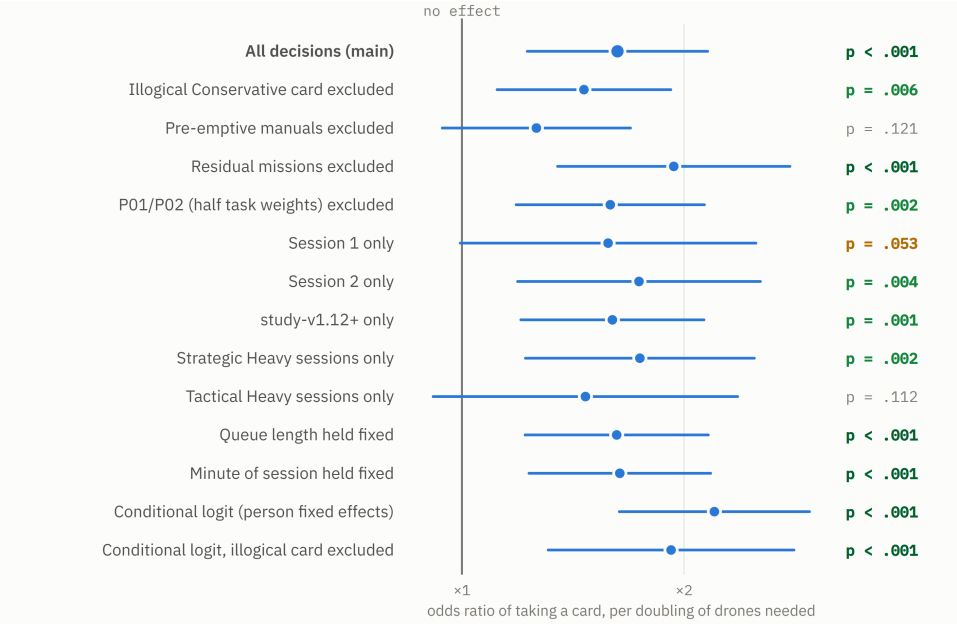
C.2.1 Is it an artefact? Checks, all reported

Figure 43. The worry: the illogical Conservative card (one that adds a drone of a type the mission does not need) appears only on one- and two-type missions — 71% of one-type decisions, none of the three-type ones — so "people avoid the card on small missions" could be "people avoid a visibly odd card". It is not: on one-type missions uptake is the same with an odd card (50%, 129 decisions) and without one (56%, 52 decisions), and with every odd-card decision removed the size effect is still there ($\times$1.46 per doubling; clean one-type vs three-type $\times$1.83). No ε-corrupted (deliberately wrong) cards were shown in any session, so card quality did not otherwise vary by size. The rule also holds with queue length held fixed, with the minute of session held fixed, per session, per scenario, without the half-weight participants and in a person-fixed-effects model. Two strategic rows do not reach $p < .05$: Tactical Heavy sessions alone (few one-type missions there) and cards-on-screen only (above). The tactical rule clears every row.

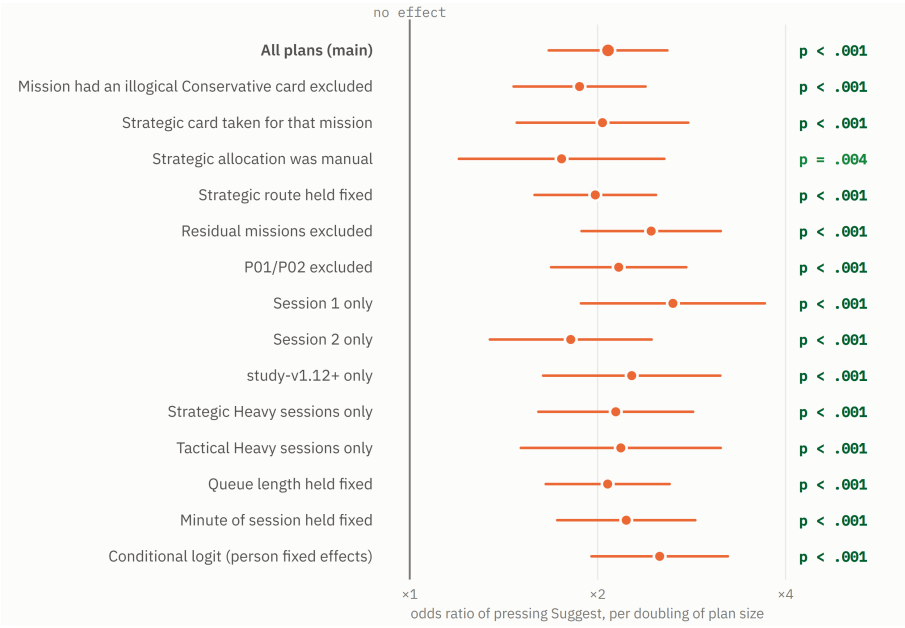
Key numbers.

- Strategic, illogical card excluded: $\times$1.46 [1.12, 1.92], p = .006 (474 decisions); conditional logit $\times$1.92 [1.31, 2.82]
- One-type missions, odd card vs clean card: odds $\times$0.77 [0.33, 1.78], p = .55 — no difference
- Strategic range across 14 specifications: $\times$1.26 to $\times$2.20; tactical range across 15: $\times$1.75 to $\times$2.64
- Tactical, illogical-card missions excluded: $\times$1.87 [1.47, 2.39]; strategic route held fixed $\times$1.98

Method. Each row refits the main within-person model on a subset or with one more control. Illogical card: the *odd* flag from build_findings_synthesis.load_decisions (Conservative card carries a type the mission needs none of). Queue held fixed: the number of missions waiting at the decision, split within/between person (the queue story itself is Section 7.6's). Conditional logit: a different model family that compares each person only with themselves by construction (people who always or never hand over drop out, hence fewer people). The odd-card story itself is Section 7.8's — here it is only checked as a confound.



(a) Strategic tier.



(b) Tactical tier.

Figure 43: Is it an artefact? Checks, all reported

C.2.2 Person by person

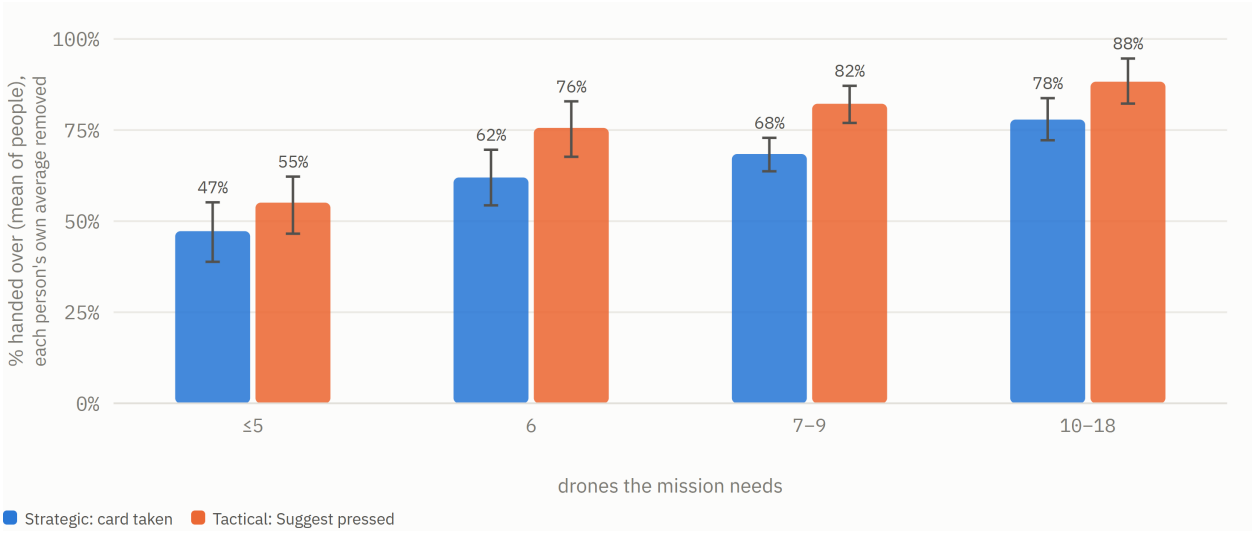


Figure 44: Person by person

Figure 44. The same rule with each person counted once: for each band of mission size, each person's own hand-over rate, averaged over people. The default view removes each person's overall level first, so it shows how the *same* people shift between small and big missions. Both tiers rise band by band. 22 of 33 people hand over more on their biggest band than their smallest, on each tier; only 4 (strategic) and 2 (tactical) do the reverse.

Key numbers.

- Strategic, own average removed: 47% ($\leq$5 drones, 32 people) $\rightarrow$ 62% $\rightarrow$ 68% $\rightarrow$ 78% (10–18, 33 people)
- Tactical, own average removed: 55% $\rightarrow$ 76% $\rightarrow$ 82% $\rightarrow$ 88%
- Biggest band above smallest: strategic 22 people, below 4, equal 7 (sign test $p < .001$); tactical 22, 2, 9 ($p < .001$)

Method. Bands of drones needed: $\leq$5, 6, 7–9, 10–18 (both tiers; drones needed is the measure both tiers share). A person contributes to a band if they made at least 2 decisions in it. "Own average removed" = the person's rate in the band minus their rate over all bands, plus the overall rate, so bars stay on the percentage scale. Whiskers: 95% intervals from resampling people. Sign test: for each person, their biggest band against their smallest; are "more" more common than "less" than a coin flip would give (ties dropped)?

C.2.3 The speed difference as estimates

Figure 45. The same comparison as model estimates: negative means the assistant route was faster for the same person.

Method. As in the section above. Each row is a within-person % change in decision time with its 95% interval.

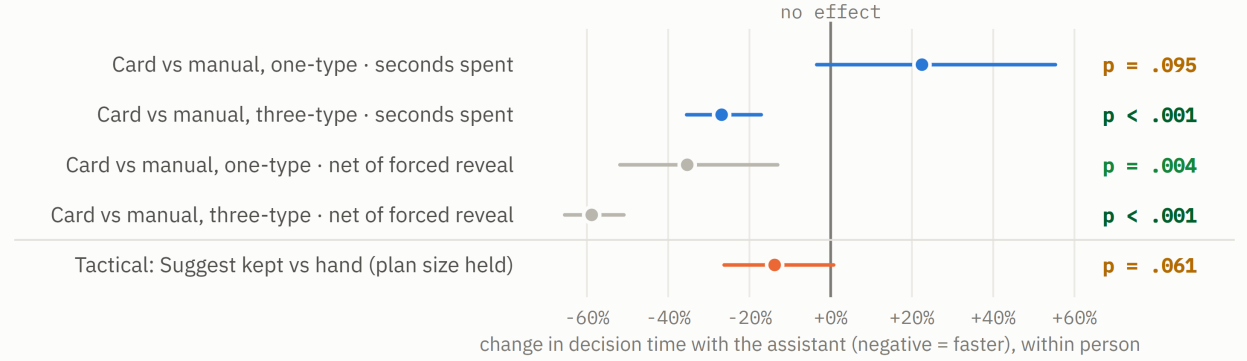


Figure 45: The speed difference as estimates

C.2.4 The exception that proves the rule: the people who call the assistant slow

Figure 46. Six people complain that the assistant is slow (C06: P06, P15, P16, P26, P27, P29). On every measure the two groups' intervals overlap: none of the ten differences is reliable with six people. The directions are worth a sentence, not a claim. They take cards and press Suggest at the same rate as everyone else (58% vs 66%; 74% vs 76%), and they lean *towards* the size rule: they decide before the cards load more often (30% vs 13%, p = .055) and their three-type minus one-type gap is larger (+40 vs +19 points, p = .09). Read that way, the complaint and the rule are the same observation (the fixed wait is not worth it on small jobs), and P29 says as much. But this is a reading of six people, not a finding.

Key numbers.

- Pre-emptive share 30% vs 13% (p = .055); three-type minus one-type card gap +40 vs +19 points (p = .09); decisions per session 10.2 vs 9.0 (p = .09)
- Tactical plan time 19.9 vs 25.8 s (p = .16); score +0.32 vs −0.07 SD (p = .34)
- Card uptake 58% vs 66% (p = .51); Suggest 74% vs 76% (p = .90) — reliance itself does not differ

Method. Per-person measures, averaged over the person's two sessions, for the six C06 people against the other 27. Bars are group means; whiskers are 95% intervals from resampling people within each group. Score = session score (P01/P02 rescaled for their half task weights) standardised within each of the four games. p = two-sided permutation test (group labels reshuffled 20,000 times); 10 comparisons, uncorrected, so expect about one p below .10 by chance.

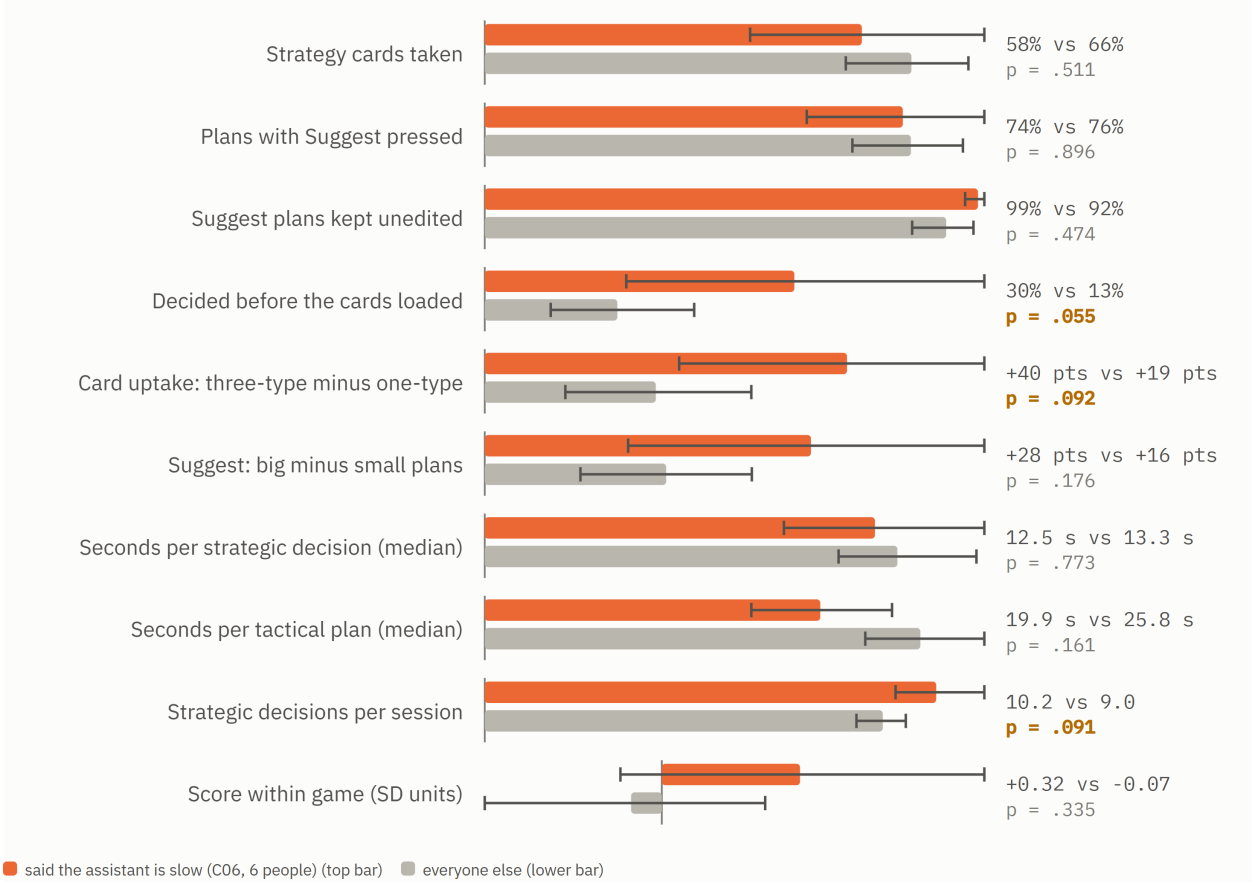


Figure 46: The exception that proves the rule: the people who call the assistant slow

C.3 A busier moment does not push people onto the assistants — even though they say it does (supplementary analyses)

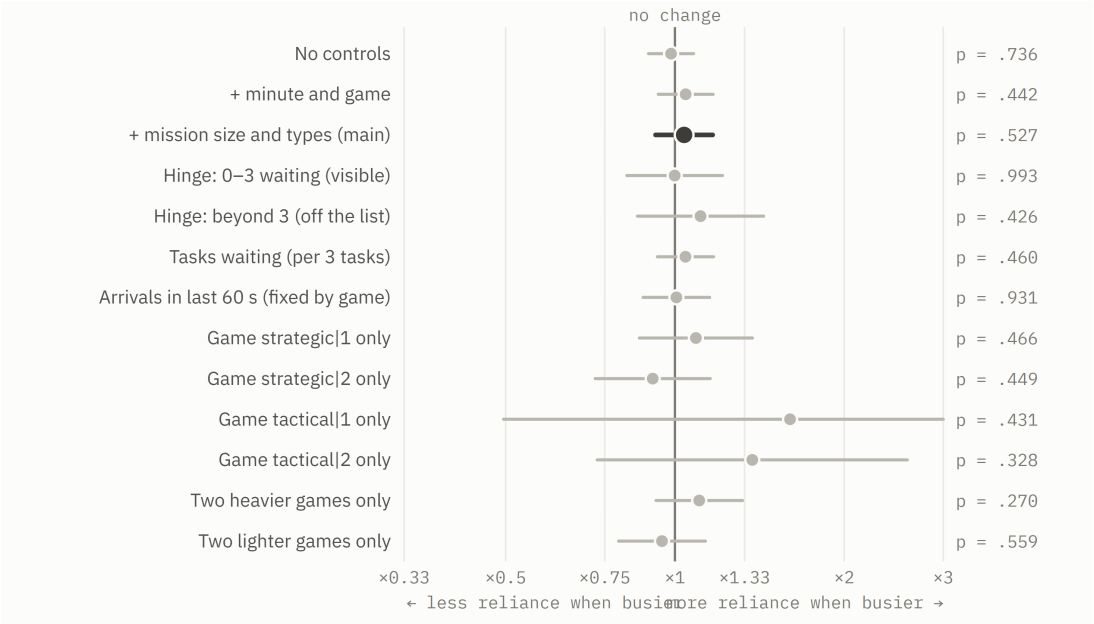
Supplement to Section 7.6.

C.3.1 Every way of asking it gives the same answer

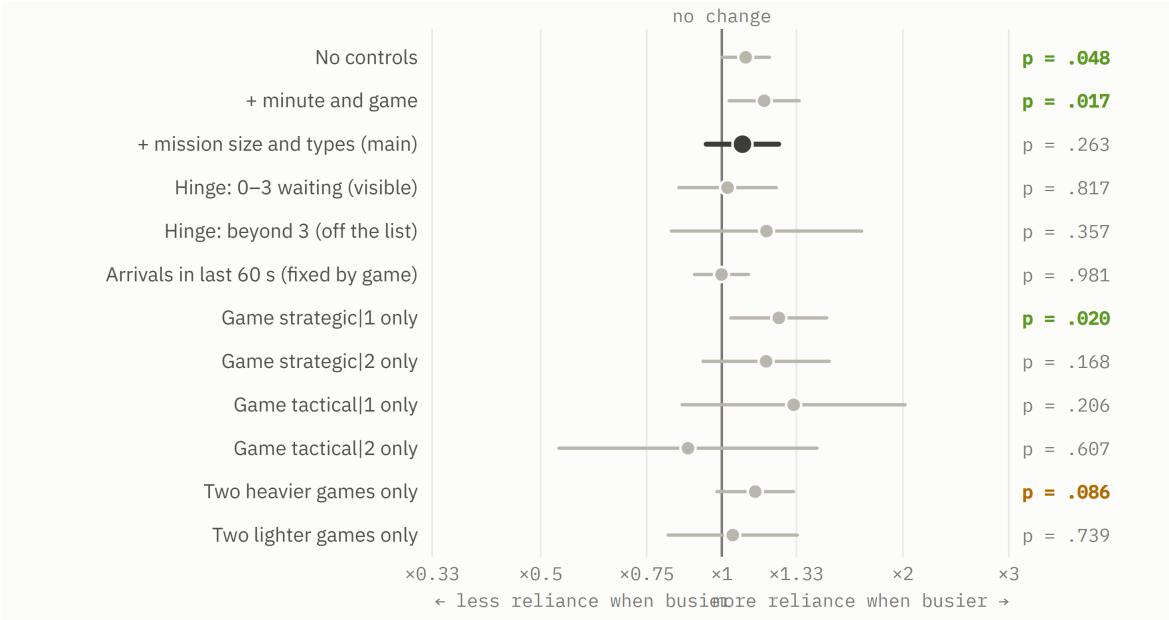
Figure 47. The within-person change in the odds of relying per extra mission waiting, under every specification tried — including each game on its own, the heavier vs the lighter pair of games, a 'hinge' that lets the effect differ beyond the visible list, tasks waiting instead of missions, and missions that had just arrived (which the operator cannot cause). Strategic: all 13 are compatible with no change. Tactical: the main model is flat, but two of twelve variants reach $p < .05$ (without mission size; Strategic Heavy game 1 alone) — a small upward lean that disappears once mission size is held fixed and is absent for the arrivals measure.

Key numbers.

- Strategic main: OR 1.04 [0.92, 1.17], p = .53 per extra mission waiting; range across 13 specs 0.91–1.60, none $p < .05$.
- Strategic hinge: 0–3 waiting OR 1.00 [0.82, 1.22]; beyond 3 OR 1.11 [0.86, 1.44] — no change at the visible limit.
- Tactical main: OR 1.08 [0.94, 1.24], p = .26; + minute and game only: 1.18 [1.03, 1.34], p = .017; arrivals in last 60 s: 1.00 [0.90, 1.11].
- Heavier games only (Strategic Heavy 1, Tactical Heavy 2): strategic 1.10 [0.93, 1.32], tactical 1.14 [0.98, 1.32].



(a) Strategic tier.

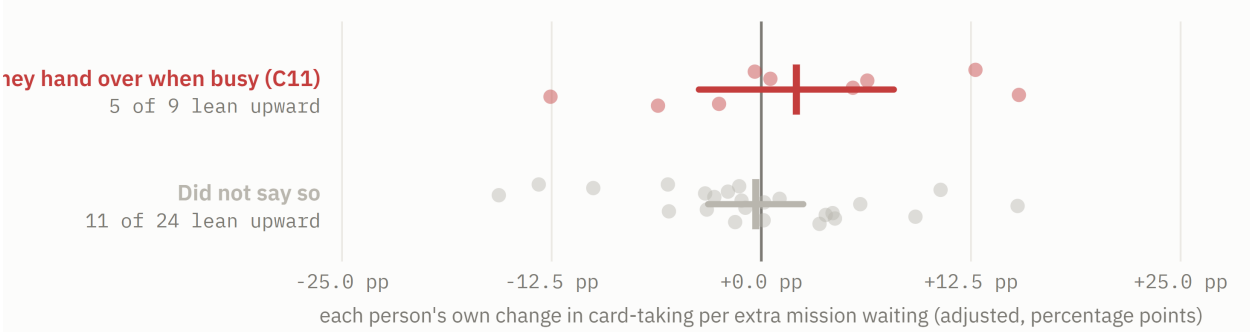


(b) Tactical tier.

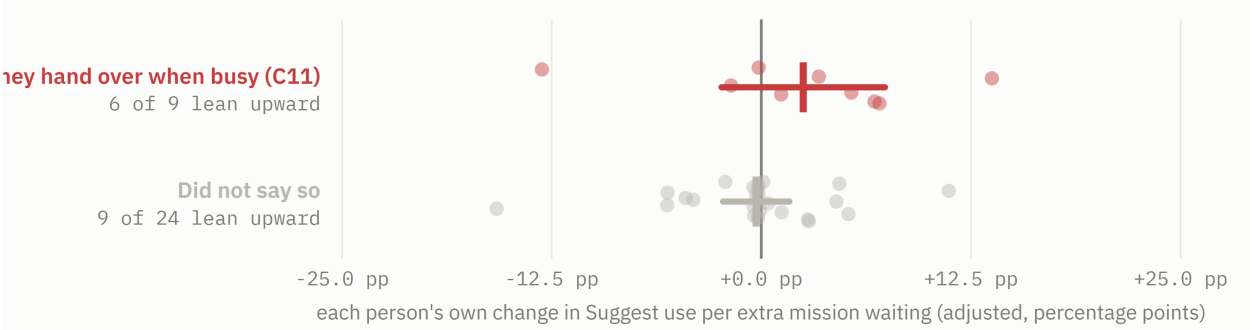
Figure 47: Every way of asking it gives the same answer

Method. GEE logistic regression (decisions clustered within person, bias-reduced standard errors). 'Within person' means the queue is centred on that person's own average, so the estimate compares a person when busier with the same person when less busy; their average queue is kept as a separate term. 'Main' adds mission size, tasks, drone types, minute and game. An odds ratio of 1.04 means the odds of taking a card are 4% higher per extra mission waiting. To make the interval concrete: at the main model's upper limit (1.17) uptake would rise from 65% at 0 waiting to 80% at 5; at its lower limit (0.92) it would fall to 56%. So the data rule out a large effect but not a modest one either way.

C.3.2 Said vs did: the people who said they hand over when busy



(a) Strategic tier, moment to moment.



(b) Tactical tier, moment to moment.

Figure 48: Said vs did: the people who said they hand over when busy

Figure 48. Nine operators (P05, P06, P11, P18, P20, P25, P32, P33, P35) said, unprompted, that they lean on the assistant more when busy, overwhelmed or rushed. Each dot is one person's own slope: how much their (adjusted) reliance changed per extra mission waiting. If what they said described what they did, the red group would sit clearly right of zero. It does not: on average they move about +2 points per extra mission, not distinguishable from zero or from the other 24 people. Their claim usually compared sessions ('the first one was so fast I used it a lot') — so the second view checks that version: reliance in their heavier game minus their lighter game. Again no difference.

Key numbers.

- Strategic slopes: said +2.1 pp per extra mission (5 of 9 upward) vs not said −0.3 pp (11 of 24 upward); p = .41.
- Tactical slopes: said +2.5 pp (6 of 9 upward) vs −0.3 pp (9 of 24); p = .09 — the closest thing to support, and not significant.
- Queue × said interaction: strategic OR 1.13 [0.83, 1.53] p = .44; tactical 1.18 [0.91, 1.53] p = .20. Within the 'said' group alone the queue OR is 1.03 (strategic) and 0.97 (tactical).
- Heavier minus lighter game, adjusted: said +3.8 pp strategic, +1.5 pp tactical; everyone −1.1 / +1.2 pp; all intervals span zero.

Method. **Moment to moment:** for each person with 6+ decisions and some spread in queue depth, a straight line through their (observed − expected) values against queue depth; the slope is in percentage points per extra mission waiting. Groups compared with a Mann–Whitney test; also tested as a queue × 'said' interaction in the GEE above. **Busier session:** the four games' load measured by total drones demanded per session (Tactical Heavy 2: 115; Strategic Heavy 1: 89; Strategic Heavy 2: 88; Tactical Heavy 1: 63); for each person, their mean (observed − expected) in the heavier of their two games minus the lighter. With 9 people in the 'said' group, only a large difference could have been detected — this is absence of evidence for the claim, and the point estimates are small.

C.3.3 In play: overload talk comes at slightly busier moments, reliance does not follow

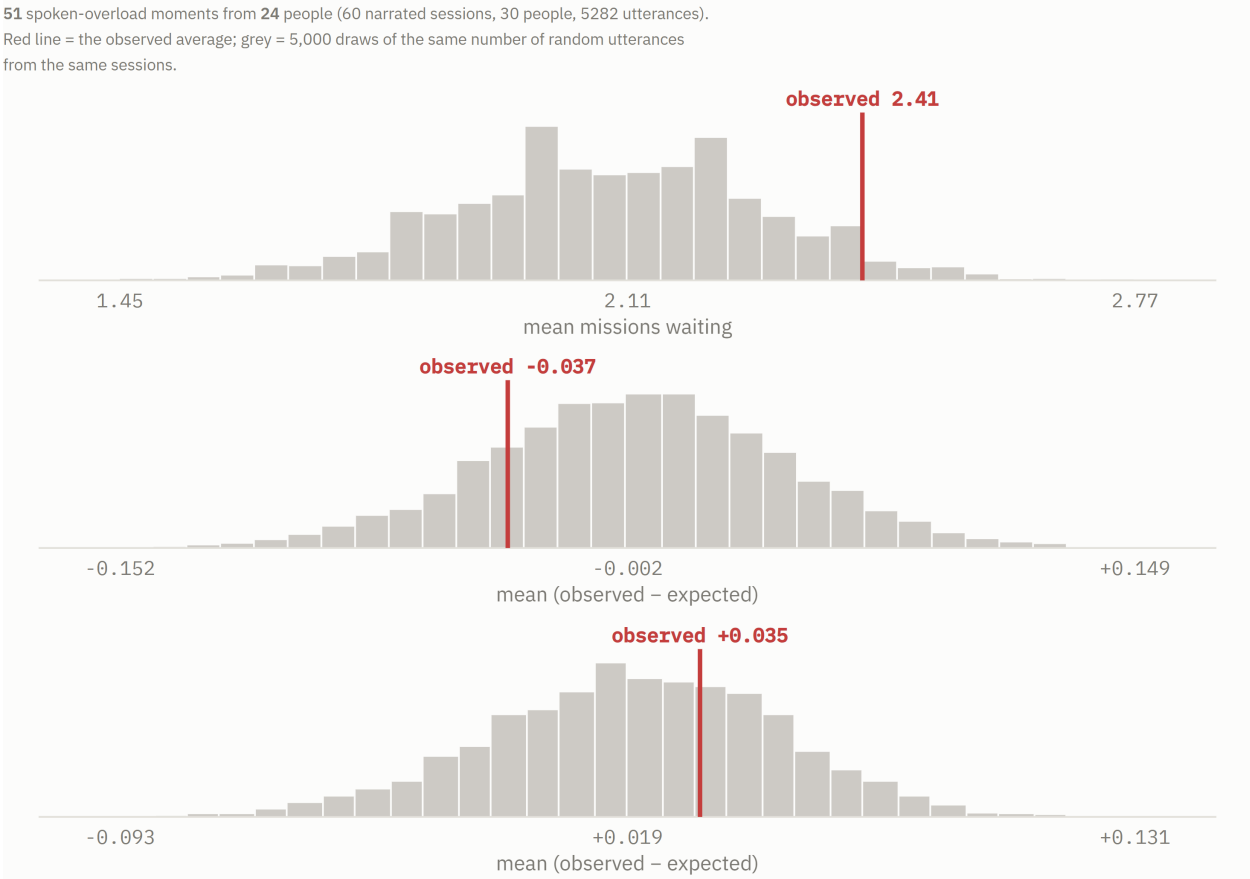


Figure 49: In play: overload talk comes at slightly busier moments, reliance does not follow Top: missions waiting at the overload moments; middle: strategy cards taken within ±60 s (observed − expected); bottom: Suggest pressed within ±60 s (observed − expected). Red line = observed mean; grey = 5,000 draws of the same number of random utterances from the same sessions.

Figure 49. Using the narration recorded during play (30 people, 60 sessions), we found every moment someone said an overload-type phrase ('too many', 'can't keep up', 'oh no', 'struggling'…): 51 moments from 24 people. All nine of the 'said' group produced at least one. At those moments slightly more missions were waiting than at a random utterance (2.4 vs 2.1; p = .09) — so what people feel as overload is weakly tied to the queue. But in the minute either side, their card-taking and Suggest use were exactly what their missions and habits predict. The felt pressure is real enough; it just does not turn into handing work over.

Key numbers.

- Missions waiting at overload moments 2.41 vs 2.09 at random utterances (95% of random draws 1.71–2.45), p = .09.

- Card-taking within ±60 s: −3.7 pp vs expected (random utterances +0.1), p = .38. Suggest: +3.5 pp (random +2.1), p = .65.
- 16 of the 51 moments came from the 9 people coded C11; 35 from others.

Method. Phrase list fixed in round 2 (before any behaviour was looked at), applied to the transcript of each session window (from segments.json; utterances 20 s apart merged into one moment). For each moment: missions waiting at that second (rebuilt from the log), and the mean (observed − expected) of strategic decisions and tactical plans started within ±60 s. Null: the same number of utterances per session drawn at random from ALL that session's speech, 5,000 times — so the comparison is 'overload talk' vs 'any talk', in the same people and sessions. Caveats: no speaker separation (researcher speech is in the transcripts); in round 2 about half of matched phrases were genuine stress on reading; alignment uses file times for 24 of 30 people (±2 s).

C.4 The dip in strategy-card use belongs to the people who later said they noticed the illogical card, early in the session; it shows up whether or not they had met that card yet (supplementary analyses)

Supplement to Section 7.8.

C.4.1 Strategic flaws are visible; tactical flaws are hidden

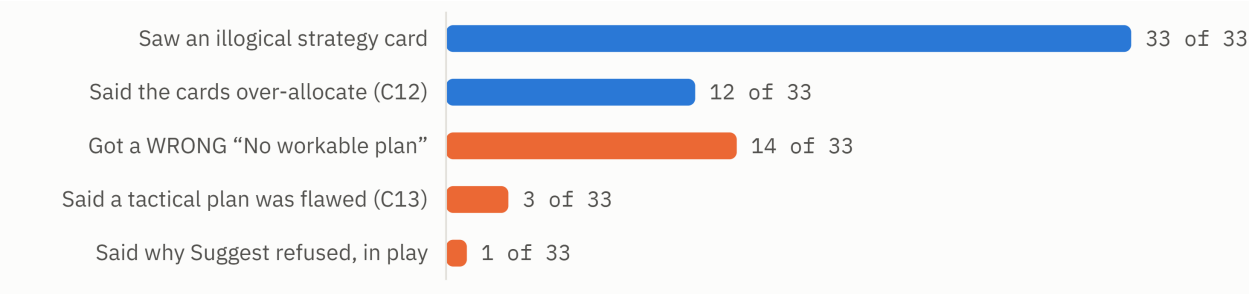


Figure 50: Strategic flaws are visible; tactical flaws are hidden

Figure 50. Neither assistant had errors injected for anyone analysed (the ε manipulation ran only for the excluded P09). Each tier still has a natural flaw. On the **strategic** side, it is the illogical extra drone type. It sits on the card face next to the mission's own requirements, everyone saw it, and 12 people mentioned it in debrief or interview (15 counting in-play remarks). On the **tactical** side, it is the wrong refusal: Suggest answers 'No workable plan' when a fix existed using a drone busy on another task. To the operator it looks exactly like a correct refusal. 14 people met one (23 wrong refusals out of 39). Four of them reacted aloud and accepted it, most saying they would have to abandon. Only P28 named the cause. Only 3 people ever described a tactical-plan flaw (C13), and just one of them had been wrongly refused. 'The strategic assistant is distrusted more' may partly be 'its mistakes are the ones you can see'.

Key numbers.

- Strategic: 33 of 33 saw the illogical card (132 of 606 decisions); 12 coded C12; 15 incl. strict in-play remarks
- Tactical: 14 of 33 got ≥1 wrong refusal (23 of 39 refusals); 3 coded C13, 1 of them among the 14; 4 reacted aloud to a wrong refusal, 1 named why

Method. Strategic exposure: people with ≥1 illogical-card decision. Tactical exposure: people with ≥1 Suggest press in a recovery that returned 'No workable plan' or a partial plan while a fix using a busy drone existed (the 'blind' flag in h3_recovery). Noticing: qualitative codes C12 and C13 (debrief or interview), plus the hand-coded in-play remarks (extras.SPOKEN_STRAT / SPOKEN_TAC), each matched to the nearest preceding Suggest press.

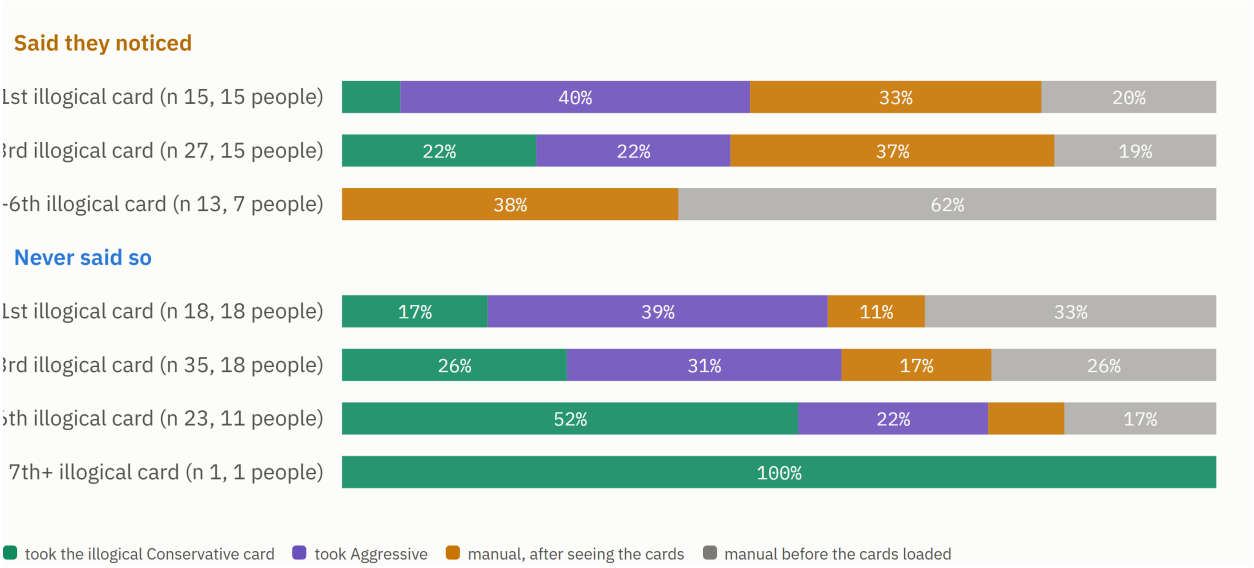


Figure 51: On the missions that carry the flaw, noticers stop using the cards

C.4.2 On the missions that carry the flaw, noticers stop using the cards

Figure 51. This is the cleanest behavioural trace, and it is where you would expect one. On their 1st illogical-card mission, noticers took a card 47% of the time and never-said people 56%. With each further illogical-card mission, noticers took a card about 14 points less often. By their 4th–6th they took one **0 times in 13** (7 people), going manual instead, mostly *before the cards had loaded* (62%). Non-noticers moved the other way: by their 4th–6th they took a card 74% of the time. Most of that was the illogical Conservative card itself (52%), which noticers took 0 times in 13. The handover analyst's check fits this: when people build a one-type mission by hand rather than take the Conservative card, nearly all of the 'trim' is dropping the unneeded-type spares (−1.2 drones overall, +0.05 on the types the mission needs). So hand-builders strip out exactly what is illogical; see Section 7.4. The manual-card-error card's version (people who mention a card error go manual more on one-type missions) holds in direction. But the same people also go manual more on three-type missions, where nothing is wrong (37% vs 22%). The one-type-*specific* part does not hold between people (interaction OR 1.12, p = .78). What is specific to the flawed missions is the *trend*: noticers learn to skip the cards there.

Key numbers.

- Took any card, per extra illogical card: noticers −13.9 points [−25, −3]; never-said +5.3 [−2, +12]; difference p < .001
- Took the illogical Conservative card: noticers −3.5 points per card; never-said +9.3 [+1, +18]; difference p = .001
- Manual before the cards loaded: noticers +15.4 points per card [+2, +29]; never-said −5.8; difference p = .012
- 4th–6th illogical card: noticers took a card 0 of 13 (7 people); never-said 74% of 23 (11 people), 52% the illogical card
- Manual on one-type missions: noticers 63% vs 37%, OR 2.41 [0.92, 6.32], p = .07; on three-type 37% vs 22%, OR 2.29, p = .06; one-type × noticed OR 1.12, p = .78

Method. Rows: the 132 illogical-card decisions, numbered per person (1st, 2nd, . . .). Each is coded by what happened: took the Conservative card (the illogical one), took Aggressive, manual after the cards loaded, or manual before they loaded (a pre-emptive choice that never saw the cards). Trend = a linear model of each outcome on the card's number (capped at 7) × group, with game held fixed and people clustered. The interaction p is whether the two groups' slopes differ. Between-person check (manual-card-error): manual ~ said + game on one-type missions, then on three-type, then the said × one-type interaction (logistic GEE, people

clustered).

C.4.3 What 'expected' and 'adjusted' mean

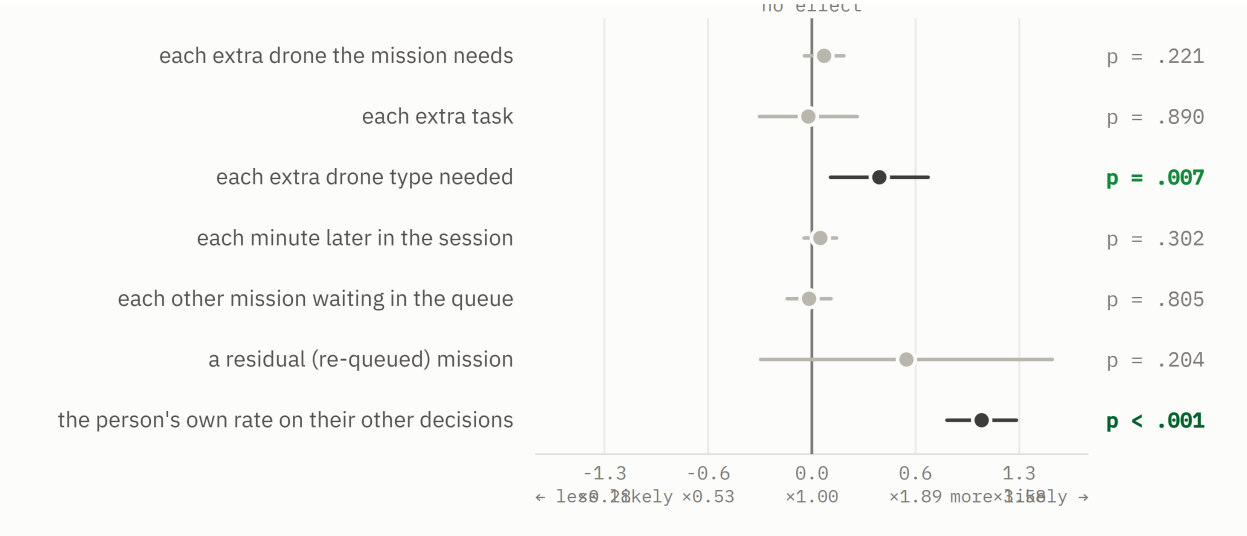


Figure 52: What 'expected' and 'adjusted' mean

Figure 52. **Expected** is the chance a model gives that this person takes a card on this decision. It knows only what the decision was (drones, tasks and drone types the mission needed), when it was (minute of the session, whether it was a re-queued mission, how many others were waiting), which of the four games was being played, and how often *this person* took cards on all their *other* decisions. **Adjusted** = the group's overall rate + the average of (observed − expected) at that position. In plain terms: 'how far above or below their usual self, given the mission in front of them, were people here?' The person term does most of the work (odds ×2.84 per unit; the model ranks takes above non-takes 81% of the time). So 'adjusted' mostly means 'compared with that person's normal habit'. Of the mission features, only the number of drone types matters (×1.51 per extra type). Because adjusted is a shifted average residual and not a probability, it can go above 100% at a small dot that sits far above its expectation. The old not-said +2 dot read 122% for this reason.

Key numbers.

- Person's own rate elsewhere: odds ×2.84 [2.29, 3.51] per logit unit
- Each extra drone type needed: ×1.51 [1.12, 2.04]; drones ×1.08, tasks ×0.98, minute ×1.05, queue ×0.98 (all n.s.)
- Model ranks takes above non-takes 81% of the time (AUC 0.81)

Method. Logistic regression over all 606 decisions: took ~ drones + tasks + types + minute + game + queue length + residual flag + logit(person's leave-one-out rate, shrunk with +1/+2). The same specification as extras2.add_expected; the script checks the fitted values are identical. The chart shows each term's odds ratio with a 95% interval.

C.4.4 Abandonment by what Suggest said

Figure 53. Every drone failure that needed the operator is one recovery episode. Grouped by what Suggest answered: a **wrong refusal** ended in abandonment 53% of the time (9 of 17). When Suggest gave a plan it was 6% (3 of 51), and when operators fixed things by hand without Suggest it was 13% (9 of 69). A **correct refusal**, where no fix existed with the drones on the mission, ended in abandonment every time (11 of 11), as it should. The striking row is the first one. These were saveable missions, and the same 14 people abandoned only 6 of their 61 other attempted fixable

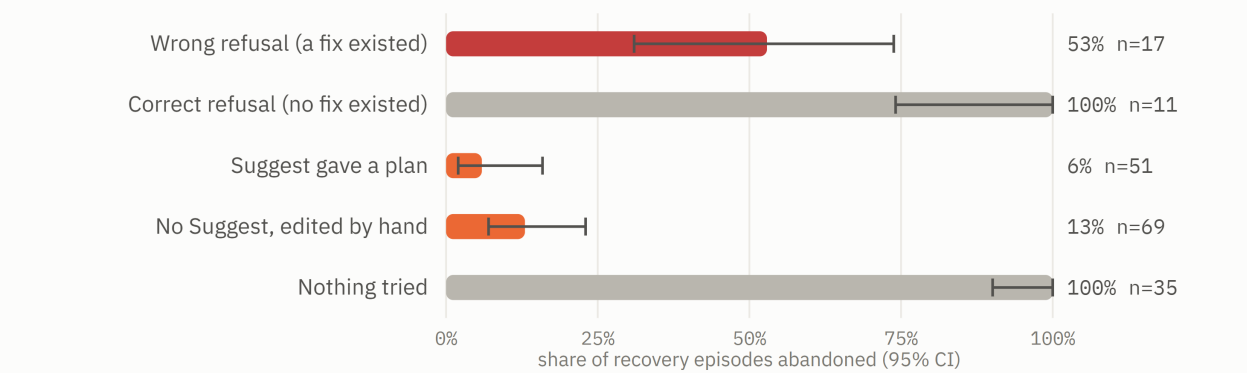


Figure 53: Abandonment by what Suggest said

failures. Abandonment was not their habit; the refusal is what is different. When nothing at all was tried, every failure was abandoned or left open (35 + 11).

Key numbers.

- Wrong refusal: 9 of 17 abandoned (53% [31%, 74%]), 14 people; all 17 fixable with drones on the mission
- Correct refusal: 11 of 11 abandoned (100% [74%, 100%]), 6 people
- Suggest gave a plan: 3 of 51 (6%); hand edits only: 9 of 69 (13%); nothing tried: 35 of 35 (+11 left open)
- Wrong refusal vs other attempted fixable failures: 9/17 vs 10/118; odds $\times$12.2, Fisher $p < .0001$; GEE $\times$12.3 [3.2, 47.6], $p = .0003$
- The same 14 people on their other attempted fixable failures: 6 of 61 abandoned

Method. Recovery episodes are rebuilt by h3_recovery.load_all (194 episodes, 33 people). It is a time-local rebuild of which drones were on the mission, idle or executing, at each Suggest press. A press is a **refusal** when Suggest could not cover every task the operator's live plan left short. A refusal is **wrong** when those tasks could have been covered by drones still on the mission, counting ones busy on another task (chaining them on). Suggest only offers drones that are not executing, which is its blind spot. An episode counts as a wrong refusal if it had at least one wrong-refusal press; as a correct refusal if it had refusals and none was wrong. Rates are abandoned ÷ (abandoned + recovered), with Wilson 95% CIs. The comparison counts wrong-refusal episodes against every other fixable episode where something was attempted (Suggest gave a plan, or hand edits): Fisher exact test, plus a GEE logit clustered by person (odds ratio 12 means the odds of abandoning were twelve times higher).

C.4.5 Do they check after a refusal? What the log shows between the refusal and the close

Figure 54. The timeline puts every refusal episode on one row, with 0 as the moment Suggest first said no. After a **wrong** refusal that ended in abandonment, not one operator dragged a drone in the planner. Three pressed Suggest again, got the same answer, and abandoned. None did anything on another mission in between, so they were not called away. 8 of the 9 abandoned within 10 s of the first refusal, at a median 3.8 s after the last one. That is time for a glance, not a re-plan. The 8 who **did not** take the refusal at its word look completely different: 7 of 8 started dragging drones, took a median 21 s, and saved the mission. After a correct refusal people behaved the same way as after a wrong one (abandoned in a median 4.4 s, no drags). That is right in those cases, and it shows operators get no cue to tell the two apart: the message is the same either way.

The verified case: P19, session 1, M003. Camera-2 failed. The Supply Drop needed a Camera. Two Lifters sat idle, and the only remaining Camera was busy on Search & Service, due to finish in about 45 s. Suggest said "No workable plan", and the operator abandoned 3 s later. Chaining that Camera on would have saved it (checked frame by frame on the recording). P19 later blamed themselves for failed missions (C19). No wrong-refusal abandoner is among the three who blamed the assistant (C20).

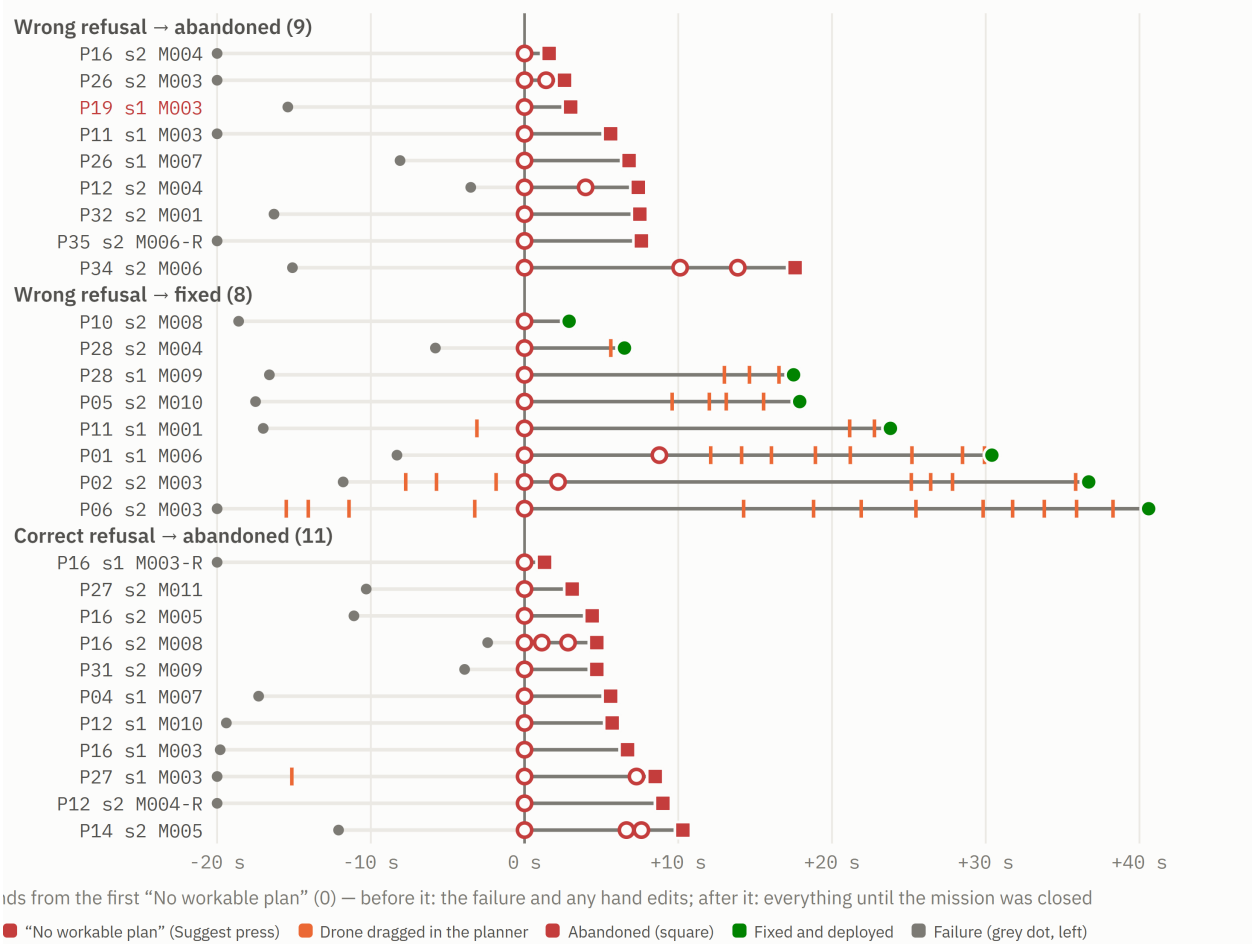


Figure 54: Do they check after a refusal? What the log shows between the refusal and the close

Key numbers.

- Wrong refusal → abandoned (9 episodes, 8 people): drags after the refusal 0 of 9; re-pressed Suggest 3 of 9; actions elsewhere 0 of 9; dragged before the refusal 0 of 9
- 8 of 9 abandoned within 10 s of the first refusal; median 6.8 s from the first refusal, 3.8 s from the last
- Wrong refusal → fixed (8 episodes, 7 people): 7 of 8 dragged drones after; median 20.9 s to deploy
- Correct refusal → abandoned (11, 6 people): 0 drags after, 3 retries, median 4.4 s from the last refusal
- Verified: P19 s1 M003, abandoned ~3 s after the refusal

Method. For each refusal episode the analysis takes the time of the first refusal and lists every logged operator action from then until the mission was abandoned or redeployed. That includes drags in the recovery planner (tactical_assignment_changed), further Suggest presses, and any action on another mission (strategic decisions, tactical confirms, recalls, other recoveries), from the raw event log. “Checked” here can only mean a *logged* action. Looking at the strategic map or the planner leaves no trace, so a fast abandonment could still hide a quick look, and the narration recordings are the place to check that. Seconds are session time; the session pauses if the primary window is hidden, so these are real play seconds.

C.5 The Tactical Assistant is trusted more, and most people say so too (supplementary analyses)

Supplement to Section 7.9.

C.5.1 Is the tactical plan given less attention in Strategic Heavy sessions?

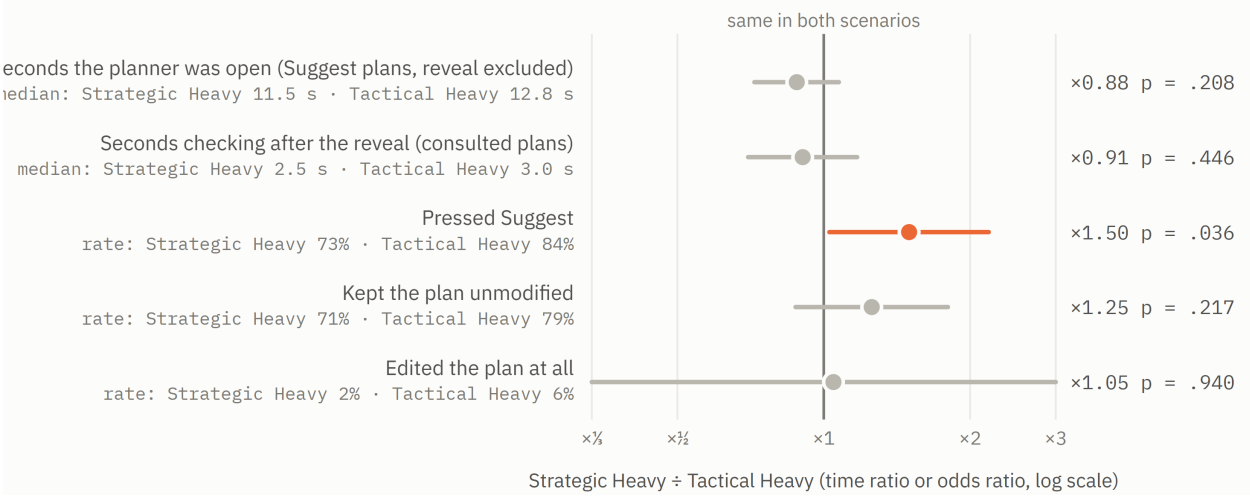


Figure 55: Is the tactical plan given less attention in Strategic Heavy sessions?

Figure 55. A check of the idea that tactical trust is higher in Strategic Heavy sessions because the tactical plan gets less attention there. Raw, it does: in Strategic Heavy sessions people press Suggest less, edit less and spend less time in the planner. But Strategic Heavy plans are much smaller (5.3 vs 8.1 drones). Compared at the same plan size and session order, the planner is open about 12% less long and the checking is about 9% shorter, but neither is distinguishable from no difference, and editing is the same. The one clear difference is the other way round: on same-size plans people press Suggest *more* in Strategic Heavy (odds $\times 1.5$). People whose planner time drops more do not clearly trust the Tactical Assistant more ($\rho = -.24$, p = .19). And tactical trust itself barely differs between the scenarios (+0.15, p = .25); the gap comes from strategic trust. So the attention account is not supported, though a small effect cannot be ruled out.

Key numbers.

- Planner time (Suggest plans): median 11.5 s Strategic Heavy vs 12.8 s Tactical Heavy; adjusted $\times 0.88$ [0.72, 1.07], p = .21
- Checking time after the reveal: median 2.5 s vs 3.0 s; adjusted $\times 0.91$ [0.70, 1.17], p = .45
- Suggest pressed: 73% vs 84% raw (OR 0.56, p = .005); same-size plans OR 1.50 [1.03, 2.18], p = .036
- Kept unmodified: 71% vs 79% raw (OR 0.67, p = .046); adjusted OR 1.26 [0.88, 1.80], p = .22
- Edited at all: 2% vs 6% raw (OR 0.44, p = .033); adjusted OR 1.05 [0.32, 3.47], p = .94
- Attention drop vs tactical-trust rise, person by person: $\rho = -.24$ [−.56, .12], p = .19, n = 31

Method. 630 tactical plans, 33 people (P09 excluded): 381 in Strategic Heavy sessions, 249 in Tactical Heavy. Five measures per plan. **Planner time**: from the planner opening to Deploy, minus the forced one-second-per-pair reveal, on plans where Suggest was pressed (486), so it is time spent with the assistant's plan. **Checking time**: last Suggest press to Deploy, minus the reveal (the "checking" measure of the section above). **Suggest pressed**, **kept unmodified** and **edited at all** on every plan. Each is compared Strategic Heavy vs Tactical Heavy with a GEE model clustered by person: times on the log scale (so the result is a ratio of typical seconds), yes/no measures as odds ratios. **Raw** has scenario only; **adjusted** adds plan size (drones × tasks) and session order, so it

compares like-for-like plans. The trust link: each person's average size-and-session-adjusted log planner time in each scenario, the Strategic Heavy minus Tactical Heavy difference, against their difference in tactical trust; Spearman ρ with a bootstrap 95% CI (31 people with Suggest plans in both scenarios).

C.5.2 A bad first session with the cards does not explain who loses strategic trust

If strategic trust fell because people had bad experiences with the cards, then those who met more illogical cards in session 1 should lose more trust. They do not. The raw correlation even points the other way ($\rho = +.40$), but that comes entirely from scenario order: people who played Strategic Heavy first see the odd cards early and lose the least trust (-0.29 against -1.26 for those who played it second). Once order is held constant, the correlation is +.14 (p = .42). Round 1's pre-registered version (missions that failed after a card) was also null: $\rho = -.05$ [$-.39$, .30]. Read this as "no evidence that how much went wrong drives it", not as proof that experience doesn't matter.

Key numbers.

- Illogical cards seen in session 1 vs change in strategic trust: $\rho = +.40$ [+.03, +.70], p = .02 — order artefact
- Same, holding scenario order fixed: $\rho = +.14$, p = .42
- Mean change in strategic trust -0.76; Strategic Heavy first -0.29, Tactical Heavy first -1.26
- H5 primary (card failures in session 1): $\rho = -.05$ [$-.39$, .30]

Method. One row per person (33). The predictor is the count of decisions in session 1 where an illogical strategy card was on screen (from load_decisions' odd flag). The outcome is session-2 minus session-1 strategic trust. Spearman ρ uses a 4000-draw seeded bootstrap CI. The partial version ranks both variables, removes the part explained by which scenario was played first, and correlates what is left. The $\rho = -.05$ is H5's primary test (docs/analysis/round1/h5_trust_drop.md).

C.5.3 What they said agrees with what they rated

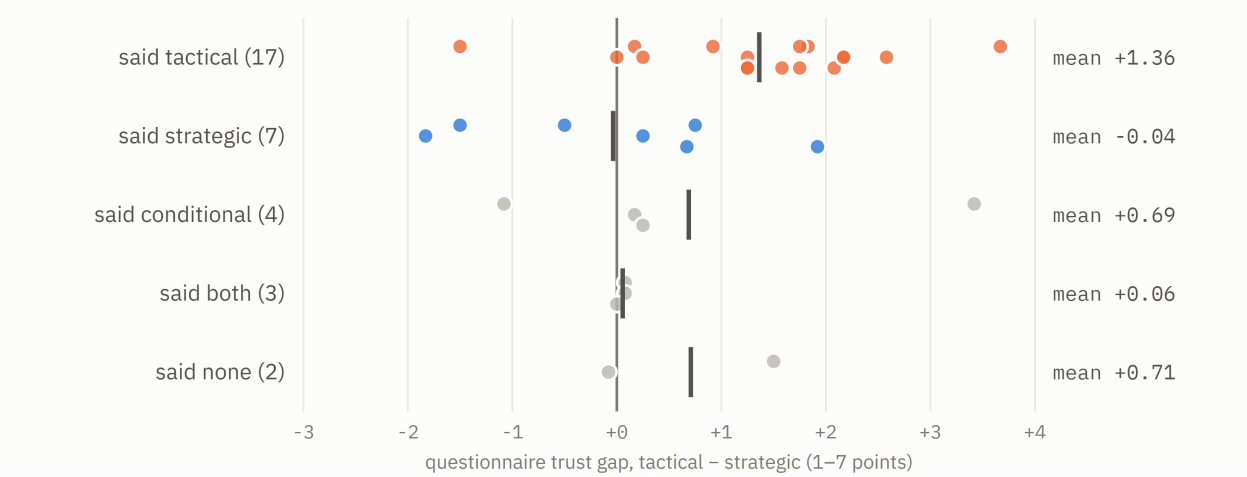


Figure 56: What they said agrees with what they rated

Figure 56. In the debriefs and interviews, 17 operators named the Tactical Assistant as the one they preferred or trusted, 7 the Strategic one, and 9 gave a conditional, "both" or no answer. Those who said tactical had a questionnaire gap of +1.36 on average; those who said strategic were at -0.04. For 18 of the 24 who named one side, the questionnaire points the same way. The coding is from one coder (an AI model) and has not been second-coded, so treat the counts as suggestive.

Key numbers.

- Said tactical 17, strategic 7, conditional 4, both 3, none 2
- Sign test 17 of 24: p = .064
- Questionnaire gap: tactical-sayers +1.36 [+0.76, +1.97] vs strategic-sayers −0.04 [−1.27, +1.19]; Mann–Whitney p = .033
- Questionnaire agrees with the stated side for 18 of 24

Method. Stated preference is the round-2 coding of each person's debrief and interview (docs/analysis/round2/wE_codes.csv, field pref). Its sign test asks whether 17 of 24 is more lopsided than a coin toss (exact binomial, two-sided, p = .064). The other test compares the questionnaire gap of tactical-sayers with strategic-sayers (Mann–Whitney, p = .033). Round 1's independent coding agreed poorly with it (κ = .27), and round 2's κ = .96 came from the same AI model. A human second coder is still needed (meth-qual-coder).

D Literature Suggestions

[Suggested additions keyed to the results, for the Related Work and Discussion. Not added to `refs.bib`: verify each reference before citing.]

- **Stable individual differences in reliance** (personal style, Section 7.2): Merritt & Ilgen (2008), "Not all trust is created equal: dispositional and history-based trust in human–automation interactions", *Human Factors* 50(2); Hoff & Bashir (2015), "Trust in automation: integrating empirical evidence on factors that influence trust", *Human Factors* 57(3).
- **Attitude scales not predicting behaviour**: Grassini (2023), "Development and validation of the AI Attitude Scale (AIAS-4)", *Frontiers in Psychology* 14 (the scale used); Schaefer et al. (2016), "A meta-analysis of factors influencing the development of trust in automation", *Human Factors* 58(3).
- **Task complexity and delegation** (mission size, Section 7.3): Lee & Moray (1994), "Trust, self-confidence, and operators' adaptation to automation", *International Journal of Human-Computer Studies* 40(1); Dzindolet et al. (2003), "The role of trust in automation reliance", *IJHCS* 58(6).
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